MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2003-2004

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Kamla Park, (water supply intake) East side.

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		03.04.03	-	-	14.07.03	-	-	21.10.03	-	-	19.01.04	-	-	-	-	-
2	Appearance		S.T	-	-	Turbid	-	-	S, Turbid	-	-	S, Turbid	-	-	-	-	-
3	Temperature	oC	32	-	-	22	-	-	22	-	-	-	-	-	25.3333	32	22
4	Turbidity	NTU	35	-	-	38	-	-	34	-	-	38	-	-	36.25	38	34
5	Colour	PCS	C.L.	-	-	Muddy	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O. L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	7.54	-	-	8.2	-	-	7.49	-	-	8.6	-	-	7.96	8.6	7.49
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	-	-	-	-	-	-	-	-			
9	T. Solids	mg/1	336	-	-	288	-	-	352	-	-	284	-	-	315	352	284
10	D. Solids	mg/1	272	-	-	192	-	-	288	-	-	202	-	-	238.5	288	192
11	S. Solids	mg/1	64	-	-	96	-	-	64	-	-	82	-	-	76.5	96	64
12	Amm. Nitrogen	mg/1	0.06	-	-	1.18	-	-	0.02	-	-	0.04	-	-	0.33	1.18	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.44	-	-	4.5	-	-	0.11	-	-	0.22	-	-	1.32	4.5	0.11
15	Phosphate (PO_4)	mg/1	0.25	-	-	0.223	-	-	0.275	-	-	0.132	-	-	0.22	0.275	0.132
16	Chloride	mg/1	24	-	-	24	-	-	-	-	-	10	-	-	19.3333	24	10
17	Sulphate (SO_4)	mg/1	14.41	-	-	14.41	-	-	-	-	-	12.64	-	-	13.82	14.41	12.64
18	T. Alkalinity	mg/1	174	-	-	74	-	-	104	-	-	130	-	-	120.5	174	74
19	T. Hardness	mg/1	178	-	-	52	-	-	110	-	-	114	-	-	113.5	178	52
20	Cal.Hardness	mg/1	136	-	-	38	-	-	84	-	-	68	-	-	81.5	136	38
21	Magnesium H.	mg/1	42	-	-	14	-	-	26	-	-	46	-	-	32	46	14
22	D. Oxygen	mg/1	7.8	-	-	8.8	-	-	7.8	-	-	10.5	-	-	8.725	10.5	7.8
23	B.O.D.	mg/1	3.5	-	-	3.4	-	-	2.1	-	-	2.8	-	-	2.95	3.5	2.1
24	C.O.D.	mg/1	36	-	-	36	-	-	28	-	-	30	-	-	32.5	36	28
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	2400	-	-	2400	-	-	2400	2400	2400	2400	-	-	2400	2400	2400
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	2.24	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2003-2004

Natural Water Resource : Upper Lake water , Bhopal.

Description of Sampling Station : Yatch Club. (South side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		03.04.03	-	-	14.07.03	-	-	21.10.03	-	-	19.01.04	-	-	-	-	-
2	Appearance		S.T.	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	32	-	-	34	-	-	28	-	-	36	-	-	32.5	36	28
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	7.42	-	-	7.88	-	-	7.6	-	-	8.87	-	-	7.94	8.87	7.42
8	Sp. Conductivity	µMhos/cm.	Disorde.	-	-	-	-	-	-	-	-	-	-	-			
9	T. Solids	mg/1	310	-	-	276	-	-	284	-	-	274	-	-	286	310	274
10	D. Solids	mg/1	252	-	-	208	-	-	226	-	-	196	-	-	220.5	252	196
11	S. Solids	mg/1	58	-	-	68	-	-	58	-	-	78	-	-	65.5	78	58
12	Amm. Nitrogen	mg/1	0.03	-	-	0.08	-	-	Nil	-	-	0.03	-	-	0.05	0.08	0.03
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.24	-	-	0.85	-	-	0.05	-	-	0.18	-	-	0.33	0.85	0.05
15	Phosphate (PO_4)	mg/1	0.175	-	-	0.112	-	-	0.08	-	-	0.112	-	-	0.12	0.175	0.08
16	Chloride	mg/1	21	-	-	20	-	-	12	-	-	13	-	-	16.5	21	12
17	Sulphate (SO_4)	mg/1	12.64	-	-	10.8	-	-	7.48	-	-	10.81	-	-	10.4325	12.64	7.48
18	T. Alkalinity	mg/1	156	-	-	72	-	-	98	-	-	130	-	-	114	156	72
19	T. Hardness	mg/1	164	-	-	48	-	-	102	-	-	110	-	-	106	164	48
20	CalciumHardness	mg/1	126	-	-	36	-	-	82	-	-	74	-	-	79.5	126	36
21	Magnesium H.	mg/1	38	-	-	12	-	-	20	-	-	36	-	-	26.5	38	12
22	D. Oxygen	mg/1	8.2	-	-	8.1	-	-	7.1	-	-	11	-	-	8.6	11	7.1
23	B.O.D.	mg/1	2.4	-	-	2.3	-	-	1.9	-	-	2.6	-	-	2.3	2.6	1.9
24	C.O.D.	mg/1	25	-	-	23.3	-	-	20	-	-	28	-	-	24.075	28	20
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	2400	-	-	2400	2400	2400
28	F. Coliform	MPN/100ml	-	-	-	-'	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2003-2004

Natural Water Resource : Upper Lake water , Bhopal.

Description of Sampling Station : Bairagarh West side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		03.04.03	-	-	14.07.03	-	-	21.10.03	-	-	19.1.04	-	-	-	-	-
2	Appearance		S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	28	-	-	32	-	-	32	-	-	42	-	-	33.5	42	28
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O. L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	7.44	-	-	7.89	-	-	7.44	-	-	8.58	-	-	7.84	8.58	7.44
8	Sp. Conductivity	µMhos/cm.	Disorde.	-	-	-	-	-	-	-	-	-	-	-			
9	T. Solids	mg/1	288	-	-	282	-	-	272	-	-	318	-	-	290	318	272
10	D. Solids	mg/1	234	-	-	224	-	-	210	-	-	236	-	-	226	236	210
11	S. Solids	mg/1	54	-	-	58	-	-	62	-	-	82	-	-	64	82	54
12	Amm. Nitrogen	mg/1	0.02	-	-	0.75	-	-	0	-	-	-	-	-	0.26	0.75	0.00
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	0.05	-	-			
14	Nitrate Nitrogen	mg/1	0.28	-	-	1.8	-	-	0.06	-	-	0.24	-		0.60	1.8	0.06
15	Phosphate (PO_4)	mg/1	0.144	-	-	0.104	-	-	0.08	-	-	0.144	-	-	0.12	0.144	0.08
16	Chloride	mg/1	19	-	-	23	-	-	14	-	-	16	-	-	18	23	14
17	Sulphate (SO_4)	mg/1	7.48	-	-	13.31	-	-	5.48	-	-	14.4	-	-	10.1675	14.4	5.48
18	T. Alkalinity	mg/1	158	-	-	76	-	-	94	-	-	160	-	-	122	160	76
19	T. Hardness	mg/1	154	-	-	48	-	-	100	-	-	130	-	-	108	154	48
20	Cal.Hardness	mg/1	124	-	-	36	-	-	90	-	-	84	-	-	83.5	124	36
21	Magnesium H.	mg/1	30	-	-	12	-	-	10	-	-	48	-	-	25	48	10
22	D. Oxygen	mg/1	7.8	-	-	8.4	-	-	7.2	-	-	10.5	-	-	8.475	10.5	7.2
23	B.O.D.	mg/1	2.4	-	-	2.4	-	-	2.1	-	-	2.2	-	-	2.275	2.4	2.1
24	C.O.D.	mg/1	25	-	-	25	-	-	24	-	-	26	-	-	25	26	24
25	Sodium	mg/1	-	-	-	-	-	-		-	-		-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2003-2004

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Karbala (North side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		3.4.03	-	-	14.07.03	-	-	21.10.03	-	-	19.1.04	-	-	-	-	-
2	Appearance		S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	34	-	-	34	-	-	36	-	-	24	-	-	32	36	24
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	7.88	-	-	8.01	-	-	7.6	-	-	8.66	-	-	8.04	8.66	7.6
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	-	-	-	-	-	-	-	-			
9	T. Solids	mg/1	358	-	-	278	-	-	318	-	-	244	-	-	299.5	358	244
10	D. Solids	mg/1	274	-	-	216	-	-	240	-	-	186	-	-	229	274	186
11	S. Solids	mg/1	64	-	-	62	-	-	78	-	-	58	-	-	65.5	78	58
12	Amm. Nitrogen	mg/1	0.08	-	-	0.8	-	-	0.01	-	-	0.02	-	-	0.23	0.80	0.01
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.32	-	-	2.5	-	-	0.12	-	-	0.12	-	-	0.77	2.5	0.12
15	Phosphate (PO_4)	mg/1	0.225	-	-	0.132	-	-	0.144	-	-	0.087	-	-	0.15	0.225	0.087
16	Chloride	mg/1	28	-	-	22	-	-	16	-	-	14	-	-	20	28	14
17	Sulphate (SO_4)	mg/1	12.48	-	-	12.64	-	-	7.48	-	-	9.15	-	-	10.4375	12.64	7.48
18	T. Alkalinity	mg/1	154	-	-	78	-	-	110	-	-	130	-	-	118	154	78
19	T. Hardness	mg/1	160	-	-	50	-	-	74	-	-	116	-	-	100	160	50
20	Cal.Hardness	mg/1	124	-	-	36	-	-	62	-	-	78	-	-	75	124	36
21	Magnesium H.	mg/1	36	-	-	14	-	-	12	-	-	38	-	-	25	38	12
22	D. Oxygen	mg/1	7.2	-	-	8.5	-	-	7.6	-	-	10.7	-	-	8.5	10.7	7.2
23	B.O.D.	mg/1	2.8	-	-	2.4	-	-	2.2	-	-	1.8	-	-	2.3	2.8	1.8
24	C.O.D.	mg/1	28	-	-	25	-	-	27	-	-	20	-	-	25	28	20
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-			-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2004-2005

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Kamla Park, (water supply intake) East side.

S.No.	Characteristic	Unit	April	May	June		Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date		2.4.04			19.7.04			28.10.04			4.1.05			-	-	-
2	Appearance		S.T.	-	-	S.Turbid	-	-	S.T.	-	-	S.Turbid	-	-	-	-	-
3	Temperature	oC	29	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	32	-	-	34	-	-	32	-	-	19	-	-	29.25	34	19
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	#DIV/0!	#DIV/0!	#DIV/0!
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	#DIV/0!	#DIV/0!	#DIV/0!
7	pH	pH Unit	7.74	-	-	8.48	-	-	7.66	-	-	7.74	-	-	7.905	8.48	7.66
8	Sp. Conductivity	µMhos/cm.	*	-	-	*	-	-	*	-	-	*	-	-	#DIV/0!	#DIV/0!	#DIV/0!
9	T. Solids	mg/1	310	-	-	286	-	-	290	-	-	266	-	-	288	310	266
10	D. Solids	mg/1	258	-	-	208	-	-	202	-	-	212	-	-	220	258	202
11	S. Solids	mg/1	52	-	-	78	-	-	88	-	-	54	-	-	68	88	52
12	Amm. Nitrogen	mg/1	0.02	-	-	0.04	-	-	0.06	-	-	Nil	-	-	0.04	0.06	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	#DIV/0!	#DIV/0!	#DIV/0!
14	Nitrate Nitrogen	mg/1	0.04	-	-	0.22	-	-	0.08	-	-	0.04	-	-	0.095	0.22	0.04
15	Phosphate (PO_4)	mg/1	0.114	-	-	0.144	-	-	0.275	-	-	0.132	-	-	0.16625	0.275	0.114
16	Chloride	mg/1	20	-	-	18	-	-	16	-	-	16	-	-	17.5	20	16
17	Sulphate (SO_4)	mg/1	13.31	-	-	12.64	-	-	14.4	-	-	12.6	-	-	13.2375	14.4	12.6
18	T. Alkalinity	mg/1	162	-	-	130	-	-	96	-	-	102	-	-	122.5	162	96
19	T. Hardness	mg/1	166	-	-	118	-	-	76	-	-	104	-	-	116	166	76
20	Cal.Hardness	mg/1	128	-	-	76	-	-	54	-	-	60	-	-	79.5	128	54
21	Magnesium H.	mg/1	38	-	-	42	-	-	22	-	-	44	-	-	36.5	44	22
22	D. Oxygen	mg/1	7.7	-	-	8.6	-	-	7.6	-	-	7.4	-	-	7.825	8.6	7.4
23	B.O.D.	mg/1	3.2	-	-	2.8	-	-	3.1	-	-	1.1	-	-	2.55	3.2	1.1
24	C.O.D.	mg/1	34	-	-	36.9	-	-	33.3	-	-	22	-	-	31.55	36.9	22
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	2400	-	-	2400	-	-	960			390	-	-	1537.5	2400	390
28	F. Coliform	MPN/100ml		-	-		-	-		-	-		-	-	-	-	-
29	T.K.N.	mg/1		-	-		-	-		-	-		-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2004-2005

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Yatch Club. (South side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		2.4.04	-	-	19.7.04	-	-	28.4.04	-	-	12.1.05	-	-	-	-	-
2	Appearance		S.T.	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-			
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-			
4	Turbidity	NTU	30	-	-	30	-	-	30	-	-	-	-	-	30	30	30
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-			
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L	-	-	O.L.	-	-			
7	pH	pH Unit	7.53	-	-	8.42	-	-	7.43	-	-	7.44	-	-	7.705	8.42	7.43
8	Sp. Conductivity	µMhos/cm.	-	-	-	*	-	-	-	-	-	*	-	-	#DIV/0!	#DIV/0!	#DIV/0!
9	T. Solids	mg/1	284	-	-	280	-	-	288	-	-	230	-	-	270.5	288	230
10	D. Solids	mg/1	238	-	-	212	-	-	214	-	-	190	-	-	213.5	238	190
11	S. Solids	mg/1	46	-	-	68	-	-	74	-	-	40	-	-	57	74	40
12	Amm. Nitrogen	mg/1	0.08	-	-	0.02	-	-	0.04	-	-	0.02	-	-	0.04	0.08	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	0.18	-	-	-	-	-	-	-	-	0.18	0.18	0.18
14	Nitrate Nitrogen	mg/1	0.85	-	-	0.132	-	-	0.24	-	-	0.15	-	-	0.343	0.85	0.132
15	Phosphate (PO_4)	mg/1	0.112	-	-	16	-	-	0.144	-	-	0.4	-	-	4.164	16	0.112
16	Chloride	mg/1	18	-	-	10.8	-	-	14	-	-	11	-	-	13.45	18	10.8
17	Sulphate (SO_4)	mg/1	10.81	-	-	120	-	-	12.64	-	-	16.6	-	-	40.0	120	10.81
18	T. Alkalinity	mg/1	176	-	-	110	-	-	98	-	-	136	-	-	130.0	176	98
19	T. Hardness	mg/1	158	-	-	74	-	-	84	-	-	122	-	-	109.5	158	74
20	CalciumHardness	mg/1	124	-	-	36	-	-	64	-	-	96	-	-	80.0	124	36
21	Magnesium H.	mg/1	34	-	-	8.5	-	-	20	-	-	26	-	-	22.1	34	8.5
22	D. Oxygen	mg/1	8	-	-	2.5	-	-	7.8	-	-	8	-	-	6.6	8	2.5
23	B.O.D.	mg/1	2.6	-	-	57.7	-	-	2.8	-	-	1.9	-	-	16.3	57.7	1.9
24	C.O.D.	mg/1	26	-	-	-	-	-	29	-	-	18	-	-	24.3333	29	18
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	2400	-	-	-	-	-	-	-	-	2400	-	-	2400	2400	2400
28	F. Coliform	MPN/100ml	-	-	-	-'	-	-	-	-	-	-	-	-			
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			

Cat. C

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2004-2005

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Karbala (North side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		2.4.04	-	-	19.7.04	-	-	28.10.04	-	-	12.1.05	-	-	-	-	-
2	Appearance		S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-			
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-			
4	Turbidity	NTU	30	-	-	28	-	-	34	-	-	22	-	-	28.5	34	22
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-			
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-			
7	pH	pH Unit	7.72	-	-	8.4	-	-	7.84	-	-	7.38	-	-	7.835	8.4	7.38
8	Sp. Conductivity	µMhos/cm.	*	-	-	*	-	-	*	-	-	*	-	-			
9	T. Solids	mg/1	316	-	-	282	-	-	318	-	-	224	-	-	285	318	224
10	D. Solids	mg/1	258	-	-	218	-	-	254	-	-	188	-	-	229.5	258	188
11	S. Solids	mg/1	58	-	-	64	-	-	64	-	-	36	-	-	55.5	64	36
12	Amm. Nitrogen	mg/1	0.8	-	-	0.03	-	-	0.04	-	-	0.02	-	-	0.2	0.8	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.26	-	-	-	-	-	0.3	0.26	0.26
14	Nitrate Nitrogen	mg/1	0.05	-	-	0.16	-	-	0.144	-	-	0.2	-	-	0.1	0.2	0.05
15	Phosphate (PO_4)	mg/1	0.225	-	-	0.13	-	-	24	-	-	0.245	-	-	6.2	24	0.13
16	Chloride	mg/1	22	-	-	14	-	-	16.64	-	-	22	-	-	18.7	22	14
17	Sulphate (SO_4)	mg/1	12.64	-	-	10.81	-	-	102	-	-	15.8	-	-	35.3	102	10.81
18	T. Alkalinity	mg/1	146	-	-	122	-	-	98	-	-	132	-	-	124.5	146	98
19	T. Hardness	mg/1	154	-	-	116	-	-	74	-	-	140	-	-	121	154	74
20	Cal.Hardness	mg/1	122	-	-	84	-	-	24	-	-	112	-	-	85.5	122	24
21	Magnesium H.	mg/1	32	-	-	32	-	-	7.4	-	-	28	-	-	24.85	32	7.4
22	D. Oxygen	mg/1	32	-	-	8.4	-	-	2.4	-	-	7.8	-	-	12.65	32	2.4
23	B.O.D.	mg/1	7.3	-	-	2.4	-	-	27	-	-	2.8	-	-	9.875	27	2.4
24	C.O.D.	mg/1	2.5	-	-	2.4	-	-	-	-	-	22	-	-	8.967	22	2.4
25	Sodium	mg/1	26	-	-	27.7	-	-	-	-	-	-	-	-	26.85	27.7	26
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-		-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. D

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2004-2005

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Bairagarh West side of the Lake

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		2.4.04	-	-	19.7.04	-	-	28.10.04	-	-	12.1.05	-	-	-	-	-
2	Appearance		S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	26	-	-	24	-	-	30	-	-	26	-	-	26.5	30	24
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-			
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O. L.	-	-	O.L.	-	-			
7	pH	pH Unit	7.39	-	-	8.38	-	-	7.44	-	-	7.53	-	-	7.685	8.38	7.39
8	Sp. Conductivity	µMhos/cm.	*	-	-	*	-	-	*	-	-	*	-	-			
9	T. Solids	mg/1	292	-	-	278	-	-	284	-	-	238	-	-	273	292	238
10	D. Solids	mg/1	242	-	-	216	-	-	226	-	-	194	-	-	219.5	242	194
11	S. Solids	mg/1	50	-	-	62	-	-	58	-	-	44	-	-	53.5	62	44
12	Amm. Nitrogen	mg/1	0.75	-	-	0.02	-	-	0.02	-	-	0.02	-	-	0.2025	0.75	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.06	-	-	0.08	-	-	0.18	-	-	0.25	-		0.1425	0.25	0.06
15	Phosphate (PO_4)	mg/1	0.144	-	-	0.118	-	-	0.23	-	-	0.375	-	-	0.21675	0.375	0.118
16	Chloride	mg/1	18	-	-	12	-	-	18	-	-	14	-	-	15.5	18	12
17	Sulphate (SO_4)	mg/1	9.98	-	-	7.48	-	-	10.81	-	-	14.4	-	-	10.6675	14.4	7.48
18	T. Alkalinity	mg/1	152	-	-	120	-	-	96	-	-	138	-	-	126.5	152	96
19	T. Hardness	mg/1	148	-	-	108	-	-	10.88	-	-	126	-	-	98.22	148	10.88
20	Cal.Hardness	mg/1	120	-	-	80	-	-	96	-	-	94	-	-	97.5	120	80
21	Magnesium H.	mg/1	28	-	-	28	-	-	88	-	-	32	-	-	44	88	28
22	D. Oxygen	mg/1	7.9	-	-	8.2	-	-	70	-	-	7.7	-	-	23.45	70	7.7
23	B.O.D.	mg/1	2.6	-	-	2	-	-	18	-	-	2.3	-	-	6.225	18	2
24	C.O.D.	mg/1	28	-	-	18.48	-	-	8.1	-	-	26	-	-	20.145	28	8.1
25	Sodium	mg/1	-	-	-	-	-	-	2.1	-	-		-	-	2.1	2.1	2.1
26	Potassium	mg/1	-	-	-	-	-	-	24	-	-	-	-	-	24	24	24
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. D

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2005-2006

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Kamla Park, (water supply intake) East side.

S.No.	Characteristic	Unit	April	May	June		Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date		-		22.6.05	-	-	27.9.05	-		5.12.05	-		8.3.06	-	-	-
2	Appearance		-	-	S.Turbid	-	-	Tub.	-	-	S.Turbid	-	-	S.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	24	-	-	23	-	-	24.5	23.833	24.5	23
4	Turbidity	NTU	-	-	30	-	-	48	-		36	-	-	28	35.5	48	28
5	Colour	PCS	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.			
6	Odour	T. No	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.			
7	pH	pH Unit	-	-	7.86	-	-	8.09	-	-	7.56	-	-	8.38	7.9725	8.38	7.56
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	-	324	-	-	410	-	-	284	339.33	410	284
9	T. Solids	mg/1	-	-	156	-	-	282	-	-	462	-	-	268	292	462	156
10	D. Solids	mg/1	-	-	132	-	-	208	-	-	388	-	-	224	238	388	132
11	S. Solids	mg/1	-	-	24	-	-	74	-	-	462	-	-	44	151	462	24
12	Amm. Nitrogen	mg/1	-	-	Nil	-	-	0.02	-	-	0.06	-	-	0.03	0.03667	0.06	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	-	-	0.22	-	-	0.3	-	-	0.58	-	-	0.24	0.335	0.58	0.22
15	Phosphate (PO_4)	mg/1	-	-	0.132	-	-	0.325	-	-	0.275	-	-	0.144	0.219	0.325	0.132
16	Chloride	mg/1	-	-	26	-	-	17	-	-	28	-	-	18	22.25	28	17
17	Sulphate (SO_4)	mg/1	-	-	12.64	-	-	10.81	-	-	14.41	-	-	10.81	12.1675	14.41	10.81
18	T. Alkalinity	mg/1	-	-	152	-	-	98	-	-	184	-	-	152	146.5	184	98
19	T. Hardness	mg/1	-	-	60	-	-	104	-	-	166	-	-	140	117.5	166	60
20	Cal.Hardness	mg/1	-	-	40	-	-	62	-	-	124	-	-	104	82.5	124	40
21	Magnesium H.	mg/1	-	-	20	-	-	42	-	-	42	-	-	36	35	42	20
22	D. Oxygen	mg/1	-	-	9.2	-	-	7.5	-	-	7.5	-	-	7.1	7.825	9.2	7.1
23	B.O.D.	mg/1	-	-	3.6	-	-	1.9	-	-	3.4	-	-	1.8	2.675	3.6	1.8
24	C.O.D.	mg/1	-	-	45	-	-	28	-	-	35	-	-	18.1	31.525	45	18.1
25	Sodium	mg/1	-	-	-	-	-			-	-	-	-	-			
26	Potassium	mg/1		-	-	-	-			-	-	-	-	-			
27	T. Coliform	MPN/100ml		-	1640	-	-	2400		-	2400	-	-	1600	2010	2400	1600
28	F. Coliform	MPN/100ml		-	-	-	-	-		-	-	-	-	-	-	-	-
29	T.K.N.	mg/1		-	-	-	-	-		-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2005-2006

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Yatch Club. (South side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	27.6.05			27.9.05	-		5.12.05	-		8.3.06	-	-	-
2	Appearance		-	-	S.Turbid	-	-	Tub.	-	-	S.Turbid	-	-	S.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	24	-	-	44	-		28	-	-	26	30.5	44	24
5	Colour	PCS	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-
6	Odour	T. No	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-
7	pH	pH Unit	-	-	7.68	-	-	7.93	-	-	7.44	-	-	8.34	7.85	8.34	7.44
8	Sp. Conductivity	µMhos/cm.	-	-	310	-	-	306	-	-	348	-	-	324			
9	T. Solids	mg/1	-	-	268	-	-	262	-	-	392	-	-	314	309	392	262
10	D. Solids	mg/1	-	-	220	-	-	194	-	-	328	-	-	268	252.5	328	194
11	S. Solids	mg/1	-	-	48	-	-	68	-	-	64	-	-	46	56.5	68	46
12	Amm. Nitrogen	mg/1	-	-	0.03	-	-	Nil	-	-	0.04	-	-	0.06	0.04	0.06	0.03
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-		-	-	-			
14	Nitrate Nitrogen	mg/1	-	-	0.16	-	-	0.08	-	-	0.35	-	-	0.18	0.19	0.35	0.08
15	Phosphate (PO_4)	mg/1	-	-	0.225	-	-	0.237	-	-	0.144	-	-	0.225	0.21	0.237	0.144
16	Chloride	mg/1	-	-	14	-	-	15	-	-	22	-	-	22	18.25	22	14
17	Sulphate (SO_4)	mg/1	-	-	15.8	-	-	9.15	-	-	10.81	-	-	12.64	12.1	15.8	9.15
18	T. Alkalinity	mg/1	-	-	138	-	-	90	-	-	174	-	-	158	140	174	90
19	T. Hardness	mg/1	-	-	124	-	-	92	-	-	158	-	-	144	129.5	158	92
20	CalciumHardness	mg/1	-	-	96	-	-	64	-	-	124	-	-	106	97.5	124	64
21	Magnesium H.	mg/1	-	-	28	-	-	28	-	-	34	-	-	38	32	38	28
22	D. Oxygen	mg/1	-	-	7.8	-	-	7.7	-	-	7.8	-	-	7.9	7.8	7.9	7.7
23	B.O.D.	mg/1	-	-	2	-	-	1.6	-	-	2.5	-	-	2.1	2.05	2.5	1.6
24	C.O.D.	mg/1	-	-	21	-	-	24	-	-	26	-	-	22.2	23.3	26	21
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-'	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2005-2006

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Karbala (North side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling			-	27.6.05	-		27.9.05	-		5.12.05	-	-	8.3.06	-	-	-
2	Appearance		-	-	S.Turbid	-	-	Tub.	-	-	S.Turbid	-	-	S.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	22	-	-	56	-		30	-	-	22	32.5	56	22
5	Colour	PCS	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-
6	Odour	T. No	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-
7	pH	pH Unit	-	-	8.64	-	-	8.14	-	-	7.88	-	-	8.21	8.22	8.64	7.88
8	Sp. Conductivity	µMhos/cm.	-	-	330	-	-	342	-	-	352	-	-	322	336.50	352	322
9	T. Solids	mg/1	-	-	300	-	-	306	-	-	348	-	-	354	327	354	300
10	D. Solids	mg/1	-	-	244	-	-	224	-	-	284	-	-	298	262.5	298	224
11	S. Solids	mg/1	-	-	56	-	-	82	-	-	64	-	-	56	64.5	82	56
12	Amm. Nitrogen	mg/1	-	-	0.02	-	-	0.02	-	-	0.06	-	-	0.05	0.04	0.06	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-		-	-	-			
14	Nitrate Nitrogen	mg/1	-	-	0.12	-	-	0.25	-	-	0.48	-	-	0.33	0.30	0.48	0.12
15	Phosphate (PO_4)	mg/1	-	-	0.144	-	-	0.362	-	-	0.225	-	-	0.275	0.25	0.362	0.144
16	Chloride	mg/1	-	-	20	-	-	21	-	-	20	-	-	26	21.75	26	20
17	Sulphate (SO_4)	mg/1	-	-	10.81	-	-	9.98	-	-	64	-	-	12.64	24.3575	64	9.98
18	T. Alkalinity	mg/1	-	-	150	-	-	106	-	-	148	-	-	158	140.5	158	106
19	T. Hardness	mg/1	-	-	106	-	-	88	-	-	158	-	-	166	129.5	166	88
20	Cal.Hardness	mg/1	-	-	86	-	-	66	-	-	124	-	-	134	102.5	134	66
21	Magnesium H.	mg/1	-	-	20	-	-	22	-	-	34	-	-	32	27	34	20
22	D. Oxygen	mg/1	-	-	6.9	-	-	7.4	-	-	8.6	-	-	7.3	7.55	8.6	6.9
23	B.O.D.	mg/1	-	-	2.1	-	-	2.2	-	-	2.4	-	-	2.2	2.225	2.4	2.1
24	C.O.D.	mg/1	-	-	23.3	-	-	30	-	-	27	-	-	22.2	25.625	30	22.2
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	T. Coliform	MPN/100ml	-	-	-	-	-	-		-	-	-	-	-	-	-	-
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2005-2006

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Bairagarh West side of the Lake

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	27.6.05	-	-	27.9.05	-	-	5.12.05	-	-	8.3.06	-	-	-
2	Appearance		-	-	S.Turbid	-	-	Tub.	-	-	S.Turbid	-	-	S.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	26	-	-	36	-	-	32	-	-	25	29.75	36	26
5	Colour	PCS	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-
6	Odour	T. No	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-
7	pH	pH Unit	-	-	8.79	-	-	7.58	-	-	7.46	-	-	8.08	7.98	8.79	7.46
8	Sp. Conductivity	µMhos/cm.	-	-	348	-	-	314	-	-	366	-	-	344	343	366	314
9	T. Solids	mg/1	-	-	312	-	-	264	-	-	360	-	-	368	326	368	264
10	D. Solids	mg/1	-	-	248	-	-	202	-	-	294	-	-	310	263.5	310	202
11	S. Solids	mg/1	-	-	64	-	-	62	-	-	66	-	-	58	62.5	66	58
12	Amm. Nitrogen	mg/1	-	-	0.05	-	-	Nil	-	-	0.08	-	-	0.06	0.063	0.08	0.05
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	-	-	0.22	-	-	0.04	-	-	0.58	-	-	0.35	0.2975	0.58	0.04
15	Phosphate (PO_4)	mg/1	-	-	0.275	-	-	0.2	-	-	0.275	-	-	0.225	0.24375	0.275	0.2
16	Chloride	mg/1	-	-	18	-	-	16	-	-	24	-	-	24	20.5	24	16
17	Sulphate (SO_4)	mg/1	-	-	17.84	-	-	7.48	-	-	15.8	-	-	14.41	13.88	17.84	7.48
18	T. Alkalinity	mg/1	-	-	146	-	-	94	-	-	154	-	-	0.06	98.52	154	0.06
19	T. Hardness	mg/1	-	-	116	-	-	86	-	-	160	-	-	152	128.5	160	86
20	Cal.Hardness	mg/1	-	-	92	-	-	66	-	-	124	-	-	122	101	124	66
21	Magnesium H.	mg/1	-	-	24	-	-	20	-	-	36	-	-	30	27.5	36	20
22	D. Oxygen	mg/1	-	-	7.6	-	-	7.8	-	-	8.4	-	-	7.6	7.85	8.4	7.6
23	B.O.D.	mg/1	-	-	1.9	-	-	1.4	-	-	2.6	-	-	1.8	1.925	2.6	1.4
24	C.O.D.	mg/1	-	-	19	-	-	22	-	-	28	-	-	18.3	21.83	28	18.3
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-A

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2006-2007

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Near w/s intake well at Kamla Park (East side of the lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date		3.4.06		-	24.7.06	-	-	9.10.06	-	-	31.1.07	-	-	-	-	-
2	Appearance		Sl.T.	-	-	S.T.	-	-	ST	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	25.5	-	-	22.5	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	27	-	-	36	-	-	38	-	-	40	-	-	35.25	40	27
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.74	-	-	8.29	-	-	7.62	-	-	8.17	-	-	8.21	8.74	7.62
8	Sp. Conductivity	µMhos/cm.	254	-	-	328	-	-	418	-	-	324	-	-	331.00	418	254
9	T. Solids	mg/1	284	-	-	272	-	-	394	-	-	276	-	-	306.5	394	272
10	D. Solids	mg/1	220	-	-	208	-	-	326	-	-	208	-	-	240.5	326	208
11	S. Solids	mg/1	284	-	-	64	-	-	68	-	-	68	-	-	121	284	64
12	Amm. Nitrogen	mg/1	0.04	-	-	0.06	-	-	0.06	-	-	0.02	-	-	0.05	0.06	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.14	-	-	0.28	-	-	0.62	-	-	0.45	-	-	0.37	0.62	0.14
15	Phosphate (PO_4)	mg/1	0.225	-	-	0.144	-	-	0.144	-	-	0.35	-	-	0.22	0.35	0.144
16	Chloride	mg/1	24	-	-	16	-	-	30	-	-	19	-	-	22.25	30	16
17	Sulphate (SO_4)	mg/1	-	-	-	10.81	-	-	14.41	-	-	14.14	-	-	13.12	14.41	10.81
18	T. Alkalinity	mg/1	144	-	-	136	-	-	190	-	-	146	-	-	154	190	136
19	T. Hardness	mg/1	136	-	-	122	-	-	176	-	-	138	-	-	143	176	122
20	Cal.Hardness	mg/1	12	-	-	78	-	-	132	-	-	110	-	-	83	132	12
21	Magnesium H.	mg/1	28	-	-	44	-	-	44	-	-	28	-	-	36	44	28
22	D. Oxygen	mg/1	7.6	-	-	8.4	-	-	7.6	-	-	7.6	-	-	7.8	8.4	7.6
23	B.O.D.	mg/1	2.1	-	-	2.7	-	-	3.5	-	-	2.9	-	-	2.8	3.5	2.1
24	C.O.D.	mg/1	22	-	-	36	-	-	36	-	-	34	-	-	32	36	22
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1		-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	1600	-	-	812	-	-	-	-	-	836	-	-	1082.67	1600	812
28	F. Coliform	MPN/100ml		-	-	-	-	-		-	-	-	-	-	-	-	-
29	T.K.N.	mg/1		-	-	-	-	-		-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2006-2007

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		3.4.06		-	24.07.06	-	-	9.10.06	-	-	31.01.07	-	-	-	-	-
2	Appearance		S.T	-	-	S.T.	-	-	ST	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	26	-	-	34	-	-	32	-	-	32	-	-	31	34	26
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.54	-	-	8.38	-	-	7.56	-	-	8.12	-	-	8.15	8.54	7.56
8	Sp. Conductivity	µMhos/cm.	262	-	-	326	-	-	360	-	-	298	-	-	311.50	360	262
9	T. Solids	mg/1	320	-	-	284	-	-	398	-	-	242	-	-	311	398	242
10	D. Solids	mg/1	248	-	-	222	-	-	320	-	-	188	-	-	244.5	320	188
11	S. Solids	mg/1	72	-	-	62	-	-	78	-	-	54	-	-	66.5	78	54
12	Amm. Nitrogen	mg/1	0.05	-	-	0.04	-	-	0.06	-	-	Nil	-	-	0.05	0.06	0.04
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.28	-	-	0.2	-	-	0.35	-	-	0.2	-	-	0.26	0.35	0.2
15	Phosphate (PO_4)	mg/1	0.325	-	-	0.375	-	-	0.175	-	-	0.25	-	-	0.28	0.375	0.175
16	Chloride	mg/1	28	-	-	20	-	-	24	-	-	16	-	-	22	28	16
17	Sulphate (SO_4)	mg/1	10.81	-	-	9.12	-	-	9.98	-	-	10.81	-	-	10.18	10.81	9.12
18	T. Alkalinity	mg/1	140	-	-	130	-	-	186	-	-	124	-	-	145	186	124
19	T. Hardness	mg/1	134	-	-	122	-	-	164	-	-	116	-	-	134	164	116
20	CalciumHardness	mg/1	108	-	-	80	-	-	120	-	-	94	-	-	100.5	120	80
21	Magnesium H.	mg/1	26	-	-	42	-	-	44	-	-	22	-	-	33.5	44	22
22	D. Oxygen	mg/1	7.7	-	-	8.2	-	-	7.7	-	-	7.9	-	-	7.875	8.2	7.7
23	B.O.D.	mg/1	2.3	-	-	2.7	-	-	2.6	-	-	2.4	-	-	2.5	2.7	2.3
24	C.O.D.	mg/1	24	-	-	30	-	-	28	-	-	26	-	-	27	30	24
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1		-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml		-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml		-	-	-	-	-		-	-	-	-	-			
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2006-2007

Natural Water Resource : Upper Lake , Bhopal

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		3.4.06		-	24.07.06	-	-	9.10.06	-	-	31.1.07	-	-	-	-	-
2	Appearance		Sl.T.	-	-	S.T.	-	-	ST	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	20	-	-	32	-	-	32	-	-	36	-	-	30	36	20
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	O.L.	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.68	-	-	8.31	-	-	7.81	-	-	8.14	-	-	8.24	8.68	7.81
8	Sp. Conductivity	µMhos/cm.	264	-	-	318	-	-	364	-	-	316	-	-	315.50	364	264
9	T. Solids	mg/1	298	-	-	268	-	-	354	-	-	264	-	-	296	354	264
10	D. Solids	mg/1	230	-	-	200	-	-	280	-	-	202	-	-	228	280	200
11	S. Solids	mg/1	68	-	-	68	-	-	74	-	-	62	-	-	68	74	62
12	Amm. Nitrogen	mg/1	0.05	-	-	0.02	-	-	0.08	-	-	0.02	-	-	0.04	0.08	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.28	-	-	0.18	-	-	0.5	-	-	0.35	-	-	0.33	0.5	0.18
15	Phosphate (PO_4)	mg/1	0.375	-	-	0.112	-	-	0.275	-	-	0.325	-	-	0.27	0.375	0.112
16	Chloride	mg/1	26	-	-	15	-	-	22	-	-	18	-	-	20.25	26	15
17	Sulphate (SO_4)	mg/1	-	-	-	7.48	-	-	15.8	-	-	13.31	-	-	12.1967	15.8	7.48
18	T. Alkalinity	mg/1	148	-	-	128	-	-	160	-	-	134	-	-	142.5	160	128
19	T. Hardness	mg/1	140	-	-	126	-	-	172	-	-	128	-	-	141.5	172	126
20	Cal.Hardness	mg/1	112	-	-	90	-	-	130	-	-	102	-	-	108.5	130	90
21	Magnesium H.	mg/1	28	-	-	36	-	-	42	-	-	26	-	-	33	42	26
22	D. Oxygen	mg/1	7.5	-	-	8.1	-	-	8.8	-	-	7.7	-	-	8.025	8.8	7.5
23	B.O.D.	mg/1	2.2	-	-	2.2	-	-	2.5	-	-	2.7	-	-	2.4	2.7	2.2
24	C.O.D.	mg/1	23	-	-	29	-	-	29	-	-	32	-	-	28.25	32	23
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1		-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	812	-	-	-	-	-	-	-	-	812	812	812
28	F. Coliform	MPN/100ml		-	-	-	-	-		-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2006-2007

Natural Water Resource : Upper Lake , Bhopal.

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		3.4.06		-	24.07.06	-	-	9.10.06	-	-	31.1.07	-	-	-	-	-
2	Appearance		S.T	-	-	S.T.	-	-	ST	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	26	-	-	28	-	-	34	-	-	34	-	-	30.5	34	26
5	Colour	PCS	C.L.	-	-	C.L.	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L	-	-	O.L	-	-	O.L	-	-	O.L	-	-	-	-	-
7	pH	pH Unit	8.68	-	-	834	-	-	7.54	-	-	7.94	-	-	214.54	834	7.54
8	Sp. Conductivity	µMhos/cm.	280	-	-	316	-	-	376	-	-	308	-	-	320.00	376	280
9	T. Solids	mg/1	282	-	-	270	-	-	370	-	-	254	-	-	294	370	254
10	D. Solids	mg/1	208	-	-	220	-	-	300	-	-	196	-	-	231	300	196
11	S. Solids	mg/1	74	-	-	50	-	-	70	-	-	58	-	-	63	74	50
12	Amm. Nitrogen	mg/1	0.06	-	-	0.02	-	-	0.06	-	-	Nil	-	-	0.05	0.06	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.18	-	-	0.16	-	-	0.6	-	-	0.25	-	-	0.30	0.6	0.16
15	Phosphate (PO_4)	mg/1	0.275	-	-	0.132	-	-	0.325	-	-	0.3	-	-	0.26	0.325	0.132
16	Chloride	mg/1	26	-	-	14	-	-	26	-	-	17	-	-	20.75	26	14
17	Sulphate (SO_4)	mg/1	15.8	-	-	5.48	-	-	12.64	-	-	11.64	-	-	11.39	15.8	5.48
18	T. Alkalinity	mg/1	144	-	-	128	-	-	168	-	-	130	-	-	142.5	168	128
19	T. Hardness	mg/1	132	-	-	118	-	-	178	-	-	122	-	-	137.5	178	118
20	Cal.Hardness	mg/1	110	-	-	84	-	-	130	-	-	98	-	-	105.5	130	84
21	Magnesium H.	mg/1	22	-	-	34	-	-	48	-	-	24	-	-	32	48	22
22	D. Oxygen	mg/1	7.5	-	-	7.9	-	-	8.6	-	-	7.8	-	-	7.95	8.6	7.5
23	B.O.D.	mg/1	2.4	-	-	2.1	-	-	2.7	-	-	2.5	-	-	2.43	2.7	2.1
24	C.O.D.	mg/1	25	-	-	20	-	-	30	-	-	28	-	-	25.75	30	20
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1		-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml		-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml		-	-	-	-	-		-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT-B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2007-2008

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Kamla Park (East side of the lake)

S.No.	Characteristic	Unit	*April*	May	*June*	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date		23.4.07	-	12.6.07	25.7.07	-	-	5.10.07	-	-	17.1.08	-	-	-	-	-
2	Appearance		S.T.	-	S.T.	Turbid	-	-	S.T.	-	-	ST	-	-	-	-	-
3	Temperature	oC	28	-	-	28	-	-	28.8	-	-	24	-	-	27.2	28.8	24
4	Turbidity	NTU	28	-	28	34	-	-	34	-	-	32	-	-	31.2	34	28
5	Colour	PCS	C.L.	-	C.L.	S muddy	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	O.L.	O.L.	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.68	-	8.16	8.11	-	-	8.18	-	-	8.49	-	-	8.32	8.68	8.11
8	Sp. Conductivity	µMhos/cm.	284	-	320	316	-	-	328	-	-	334	-	-	316.4	334	284
9	T. Solids	mg/1	298	-	288	222	-	-	282	-	-	330	-	-	284	330	222
10	D. Solids	mg/1	242	-	234	198	-	-	216	-	-	266	-	-	231.2	266	198
11	S. Solids	mg/1	56	-	54	24	-	-	66	-	-	64	-	-	52.8	66	24
12	Amm. Nitrogen	mg/1	0.04	-	0.12	0.08	-	-	0.02	-	-	0.08	-	-	0.07	0.12	0.02
13	Nitrite Nitrogen	mg/1	-	-	0.055	-	-	-	0.08	-	-	0.11	-	-	0.08167	0.11	0.055
14	Nitrate Nitrogen	mg/1	0.11	-	0.14	0.3	-	-	0.16	-	-	0.18	-	-	0.18	0.3	0.11
15	Phosphate (PO_4)	mg/1	0.225	-	0.275	0.132	-	-	0.225	-	-	0.225	-	-	0.22	0.275	0.132
16	Chloride	mg/1	24	-	28	12.2	-	-	24	-	-	22	-	-	22.04	28	12.2
17	Sulphate (SO_4)	mg/1	12.64	-	14.45	10.81	-	-	12.64	-	-	15.51	-	-	13.21	15.51	10.81
18	T. Alkalinity	mg/1	148	-	154	100	-	-	140	-	-	124	-	-	133.2	154	100
19	T. Hardness	mg/1	134	-	144	94	-	-	138	-	-	112	-	-	124.4	144	94
20	Cal.Hardness	mg/1	106	-	112	48	-	-	108	-	-	84	-	-	91.6	112	48
21	Magnesium H.	mg/1	28	-	32	46	-	-	30	-	-	28	-	-	32.8	46	28
22	D. Oxygen	mg/1	7.5	-	7.6	8.4	-	-	7.5	-	-	8.2	-	-	7.84	8.4	7.5
23	B.O.D.	mg/1	2.1	-	2.1	2.1	-	-	2.5	-	-	2.6	-	-	2.28	2.6	2.1
24	C.O.D.	mg/1	19.04	-	27.24	19.36	-	-	29.4	-	-	28.32	-	-	24.672	29.4	19.04
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	1600	-	1600	64	-	-	1600	-	-	1600	-	-	1292.8	1600	64
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2007-2008

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.07	-	-	25.7.07	-	-	5.10.07	-	-	17.1.08	-	-	-	-	-
2	Appearance		S.T.	-	-	Turbid	-	-	S.T.	-	-	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	24	-	-	32	-	-	32	-	-	24	-	-	28	32	24
5	Colour	PCS	C.L.	-	-	S muddy	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	OL	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.48	-	-	8.27	-	-	8.42	-	-	8.51	-	-	8.42	8.51	8.27
8	Sp. Conductivity	µMhos/cm.	272	-	-	314	-	-	310	-	-	330	-	-	306.50	330	272
9	T. Solids	mg/1	318	-	-	244	-	-	248	-	-	320	-	-	282.50	320	244
10	D. Solids	mg/1	250	-	-	204	-	-	202	-	-	262	-	-	229.50	262	202
11	S. Solids	mg/1	68	-	-	40	-	-	46	-	-	58	-	-	53.00	68	40
12	Amm. Nitrogen	mg/1	0.05	-	-	0.06	-	-	0.04	-	-	0.04	-	-	0.05	0.06	0.04
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.06	-	-	0.08	-	-	0.07	0.08	0.06
14	Nitrate Nitrogen	mg/1	0.15	-	-	0.4	-	-	0.16	-	-	0.18	-	-	0.22	0.4	0.15
15	Phosphate (PO_4)	mg/1	0.275	-	-	0.325	-	-	0.25	-	-	0.144	-	-	0.25	0.325	0.144
16	Chloride	mg/1	26	-	-	11.9	-	-	18	-	-	16	-	-	17.975	26	11.9
17	Sulphate (SO_4)	mg/1	-	-	-	9.15	-	-	9.15	-	-	10.81	-	-	9.70	10.81	9.15
18	T. Alkalinity	mg/1	142	-	-	164	-	-	128	-	-	100	-	-	133.5	164	100
19	T. Hardness	mg/1	136	-	-	160	-	-	118	-	-	90	-	-	126	160	90
20	CalciumHardness	mg/1	110	-	-	72	-	-	94	-	-	64	-	-	85	110	64
21	Magnesium H.	mg/1	26	-	-	88	-	-	24	-	-	26	-	-	41	88	24
22	D. Oxygen	mg/1	7.6	-	-	8.1	-	-	7.8	-	-	8.4	-	-	7.98	8.4	7.6
23	B.O.D.	mg/1	2.2	-	-	2.1	-	-	2	-	-	1.8	-	-	2.03	2.2	1.8
24	C.O.D.	mg/1	19.04	-	-	19.36	-	-	19.6	-	-	18.88	-	-	19.22	19.6	18.88
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2007-2008

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.07	-	-	25.7.07	-	-	5.10.07	-	-	17.1.08	-	-	-	-	-
2	Appearance		S.T.	-	-	Turbid	-	-	S.T.	-	-	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	22	-	-	32	-	-	30	-	-	26	-	-	27.5	32	22
5	Colour	PCS	C.L.	-	-	S muddy	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	OL	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.64	-	-	8.22	-	-	8.14	-	-	8.48	-	-	8.37	8.64	8.14
8	Sp. Conductivity	µMhos/cm.	268	-	-	304	-	-	322	-	-	428	-	-	330.50	428	268
9	T. Solids	mg/1	284	-	-	640	-	-	278	-	-	346	-	-	387.00	640	278
10	D. Solids	mg/1	222	-	-	246	-	-	214	-	-	288	-	-	242.50	288	214
11	S. Solids	mg/1	62	-	-	394	-	-	64	-	-	58	-	-	144.50	394	58
12	Amm. Nitrogen	mg/1	0.06	-	-	0.02	-	-	0.02	-	-	0.08	-	-	0.05	0.08	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.055	-	-	0.014	-	-	0.03	0.055	0.014
14	Nitrate Nitrogen	mg/1	0.18	-	-	0.28	-	-	0.14	-	-	0.12	-	-	0.18	0.28	0.12
15	Phosphate (PO_4)	mg/1	0.325	-	-	0.275	-	-	0.112	-	-	0.275	-	-	0.25	0.325	0.112
16	Chloride	mg/1	28	-	-	18.4	-	-	22	-	-	16	-	-	21.1	28	16
17	Sulphate (SO_4)	mg/1	14.81	-	-	12.64	-	-	10.81	-	-	7.48	-	-	11.435	14.81	7.48
18	T. Alkalinity	mg/1	144	-	-	82	-	-	144	-	-	136	-	-	126.5	144	82
19	T. Hardness	mg/1	140	-	-	74	-	-	132	-	-	118	-	-	116	140	74
20	Cal.Hardness	mg/1	114	-	-	50	-	-	106	-	-	70	-	-	85	114	50
21	Magnesium H.	mg/1	26	-	-	24	-	-	26	-	-	48	-	-	31	48	24
22	D. Oxygen	mg/1	7.4	-	-	8.2	-	-	7.4	-	-	8.7	-	-	7.925	8.7	7.4
23	B.O.D.	mg/1	2.2	-	-	2.9	-	-	2.3	-	-	2.6	-	-	2.5	2.9	2.2
24	C.O.D.	mg/1	28.8	-	-	67.76	-	-	29.4	-	-	28.32	-	-	38.57	67.76	28.32
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B		**B**			**B**			**B**			**B**					

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2007-2008

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.07	-	-	25.07.07	-	-	4.10.07	-	-	16.1.08	-	-	-	-	-
2	Appearance		S.T.	-	-	Turbid	-	-	Turbid	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	24	-	-	30	-	-	28	-	-	34	-	-	29	34	24
5	Colour	PCS	C.L.	-	-	S muddy	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	OL	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.58	-	-	8.14	-	-	7.84	-	-	8.04	-	-	8.15	8.58	7.84
8	Sp. Conductivity	µMhos/cm.	254	-	-	306	-	-	328	-	-	428	-	-	329.00	428	254
9	T. Solids	mg/1	282	-	-	382	-	-	256	-	-	318	-	-	309.50	382	256
10	D. Solids	mg/1	210	-	-	318	-	-	202	-	-	262	-	-	248.00	318	202
11	S. Solids	mg/1	72	-	-	64	-	-	54	-	-	56	-	-	61.50	72	54
12	Amm. Nitrogen	mg/1	0.02	-	-	0.02	-	-	Nil	-	-	0.02	-	-	0.02	0.02	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.02	-	-	0.14	-	-	0.08	0.14	0.02
14	Nitrate Nitrogen	mg/1	0.11	-	-	0.18	-	-	0.06	-	-	0.3	-	-	0.16	0.3	0.06
15	Phosphate (PO_4)	mg/1	0.144	-	-	0.132	-	-	0.225	-	-	0.45	-	-	0.24	0.45	0.132
16	Chloride	mg/1	25	-	-	27	-	-	12	-	-	13	-	-	19.25	27	12
17	Sulphate (SO_4)	mg/1	9.48	-	-	5.48	-	-	10.81	-	-	10.81	-	-	9.145	10.81	5.48
18	T. Alkalinity	mg/1	140	-	-	176	-	-	146	-	-	204	-	-	166.5	204	140
19	T. Hardness	mg/1	132	-	-	170	-	-	134	-	-	210	-	-	161.5	210	132
20	Cal.Hardness	mg/1	108	-	-	112	-	-	106	-	-	162	-	-	122	162	106
21	Magnesium H.	mg/1	24	-	-	58	-	-	28	-	-	48	-	-	39.5	58	24
22	D. Oxygen	mg/1	7.3	-	-	5.1	-	-	8.3	-	-	7.8	-	-	7.13	8.3	5.1
23	B.O.D.	mg/1	2.4	-	-	2.7	-	-	2.6	-	-	3.2	-	-	2.73	3.2	2.4
24	C.O.D.	mg/1	28.56	-	-	77.44	-	-	29.64	-	-	39.52	-	-	43.79	77.44	28.56
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat.		**B**			**B**			**B**			**D**					

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2007-2008

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		16.4.07	-	-	26.7.07	-	-	4.10.07	5.11.07	31.12.07	17.1.08	-	-	-	-	-
2	Appearance		S.T.	-	-	S.T.	S.T.	-	Turbid	Turbid	Turbid	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	29	-	-	-	-	-	29.0	29.0	29.0
4	Turbidity	NTU	36	-	-	23	12	-	46	48	38	24	-	-	32.4	48.0	12.0
5	Colour	PCS	C.L.	-	-	C.L.	C.L.	-	C.L.	C.L.	C.L.	C.L.	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	OL	-	OL	OL	OL	OL	-	-	-	-	-
7	pH	pH Unit	8.58	-	-	7.78	-	-	8.17	8.23	8.16	8.49	-	-	8.2	8.6	7.8
8	Sp. Conductivity	µMhos/cm.	444	-	-	464	460	-	476	498	472	344	-	-	451.1	498.0	344.0
9	T. Solids	mg/1	494	-	-	472	472	-	370	398	358	346	-	-	415.7	494.0	346.0
10	D. Solids	mg/1	422	-	-	398	396	-	298	320	296	298	-	-	346.9	422.0	296.0
11	S. Solids	mg/1	72	-	-	74	76	-	72	78	62	48	-	-	68.9	78.0	48.0
12	Amm. Nitrogen	mg/1	0.14	-	-	0.06	Nil	-	0.02	0.02	0.02	0.12	-	-	0.1	0.1	0.0
13	Nitrite Nitrogen	mg/1	-	-	-	0.012	Nil	-	0.16	0.22	0.16	0.012	-	-	0.1	0.2	0.0
14	Nitrate Nitrogen	mg/1	0.18	-	-	0.22	0.04	-	0.3	0.4	0.35	0.11	-	-	0.2	0.4	0.0
15	Phosphate (PO_4)	mg/1	0.337	-	-	0.225	0.112	-	0.525	0.6	0.525	0.225	-	-	0.4	0.6	0.1
16	Chloride	mg/1	10	-	-	13.7	12.1	-	16	18	16	12	-	-	14.0	18.0	10.0
17	Sulphate (SO_4)	mg/1	4.48	-	-	15.81	3.48	-	14.14	15.8	14.97	9.41	-	-	11.2	15.8	3.5
18	T. Alkalinity	mg/1	110	-	-	114	100	-	218	232	214	122	-	-	158.6	232.0	100.0
19	T. Hardness	mg/1	120	-	-	122	80	-	234	244	226	114	-	-	162.9	244.0	80.0
20	CalciumHardness	mg/1	78	-	-	90	76	-	182	186	174	86	-	-	124.6	186.0	76.0
21	Magnesium H.	mg/1	42	-	-	32	4	-	52	58	52	28	-	-	38.3	58.0	4.0
22	D. Oxygen	mg/1	7.1	-	-	29.9	8.5	-	8.5	8.1	8.2	9.7	-	-	11.4	29.9	7.1
23	B.O.D.	mg/1	1.4	-	-	5.7	5.5	-	4.1	4.3	3.7	2.6	-	-	**3.9**	5.7	1.4
24	C.O.D.	mg/1	19.04	-	-	61.88	58.08	-	49.4	59.28	48	28.22	-	-	46.3	61.9	19.0
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT -C

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2007-2008

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.07	-	-	25.7.07	10.8.07	-	4.10.07	5.11.07	-	16.01.08	-	-	-	-	-
2	Appearance		S.T.	-	-	Turbid	S.T.	-	Turbid	Turbid	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	22	-	-	30	38	-	66	64	-	48	-	-	44.7	66	22
5	Colour	PCS	C.L.	-	-	S.muddy	C.L.	-	C.L.	C.L.	-	C.L.	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	OL	-	OL	OL	-	OL	-	-	-	-	-
7	pH	pH Unit	8.64	-	-	8.14	7.8	-	8.63	8.74	-	8.34	-	-	8.4	8.74	7.8
8	Sp. Conductivity	µMhos/cm.	268	-	-	306	544	-	524	568	-	508	-	-	453.0	568	268
9	T. Solids	mg/1	284	-	-	382	566	-	426	472	-	394	-	-	420.7	566	284
10	D. Solids	mg/1	222	-	-	318	468	-	332	364	-	316	-	-	336.7	468	222
11	S. Solids	mg/1	62	-	-	64	98	-	94	108	-	78	-	-	84.0	108	62
12	Amm. Nitrogen	mg/1	0.06	-	-	0.02	0.12	-	0.12	0.14	-	0.06	-	-	0.1	0.14	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	0.014	-	0.26	0.36	-	0.2	-	-	0.2	0.36	0.01
14	Nitrate Nitrogen	mg/1	0.18	-	-	0.18	0.28	-	0.7	0.9	-	0.45	-	-	0.4	0.9	0.18
15	Phosphate (PO_4)	mg/1	0.325	-	-	0.132	0.375	-	0.775	0.925	-	0.65	-	-	0.5	0.925	0.132
16	Chloride	mg/1	28	-	-	27	11	-	23	24	-	17	-	-	21.7	28	11
17	Sulphate (SO_4)	mg/1	14.81	-	-	5.48	16.64	-	19.96	21.63	-	13.31	-	-	15.3	21.63	5.48
18	T. Alkalinity	mg/1	144	-	-	176	140	-	242	258	-	228	-	-	198.0	258	140
19	T. Hardness	mg/1	140	-	-	170	106	-	254	276	-	236	-	-	197.0	276	106
20	CalciumHardness	mg/1	114	-	-	112	86	-	190	204	-	176	-	-	147.0	204	86
21	Magnesium H.	mg/1	26	-	-	58	20	-	64	72	-	60	-	-	50.0	72	20
22	D. Oxygen	mg/1	7.4	-	-	5.1	8.1	-	7.9	7.7	-	7.4	-	-	7.3	8.1	5.1
23	B.O.D.	mg/1	2.2	-	-	2.7	12	-	5.4	5.8	-	4.2	-	-	5.4	12	2.2
24	C.O.D.	mg/1	28.56	-	-	77.44	106.48	-	69.16	79.04	-	59.28	-	-	70.0	106.48	28.56
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

CAT -C

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2008-2009

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Kamla Park (East side of the lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date		7.4.08	-	-	-	-	-	15.10.08	-	-	19.1.09	-	-	-	-	-
2	Appearance		S.T.	-	-	-	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	30	-	-	-	-	-	-	-	-	-	-	-	30.00	30	30
4	Turbidity	NTU	17	-	-	-	-	-	5	-	-	3	-	-	8.33	17	3
5	Colour	PCS	C.L.	-	-	-	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	-	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.2	-	-	-	-	-	7.5	-	-	7.1	-	-	7.6	8.2	7.1
8	Sp. Conductivity	µMhos/cm.	272	-	-	-	-	-	260	-	-	260	-	-	264	272	260
9	T. Solids	mg/1	232	-	-	-	-	-	329	-	-	241	-	-	267.33	329	232
10	D. Solids	mg/1	190	-	-	-	-	-	171	-	-	188	-	-	183	190	171
11	S. Solids	mg/1	42	-	-	-	-	-	68	-	-	53	-	-	54.33	68	42
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.849	-	-	0.81	-	-	0.83	0.849	0.81
13	Nitrite Nitrogen	mg/1	0.06	-	-	-	-	-	0.035	-	-	0.024	-	-	0.04	0.06	0.024
14	Nitrate Nitrogen	mg/1	0.2	-	-	-	-	-	-	-	-	1.03	-	-	0.62	1.03	0.2
15	Phosphate (PO_4)	mg/1	0.025	-	-	-	-	-	3	-	-	3.4	-	-	2.14	3.4	0.025
16	Chloride	mg/1	20	-	-	-	-	-	27.98	-	-	23.976	-	-	23.99	27.98	20
17	Sulphate (SO_4)	mg/1	10.81	-	-	-	-	-	12.9	-	-	12.4	-	-	12.04	12.9	10.81
18	T. Alkalinity	mg/1	106	-	-	-	-	-	160	-	-	160	-	-	142.00	160	106
19	T. Hardness	mg/1	96	-	-	-	-	-	196	-	-	100	-	-	130.67	196	96
20	Cal.Hardness	mg/1	60	-	-	-	-	-	98	-	-	68	-	-	75.33	98	60
21	Magnesium H.	mg/1	36	-	-	-	-	-	98	-	-	32	-	-	55.33	98	32
22	D. Oxygen	mg/1	7.1	-	-	-	-	-	6.4	-	-	6.6	-	-	6.70	7.1	6.4
23	B.O.D.	mg/1	4.9	-	-	-	-	-	1.5	-	-	-	-	-	3.20	4.9	1.5
24	C.O.D.	mg/1	45	-	-	-	-	-	28.8	-	-	60	-	-	44.60	60	28.8
25	Sodium	mg/1	8	-	-	-	-	-	13	-	-	19	-	-	13.33	19	8
26	Potassium	mg/1	1.125	-	-	-	-	-	9	-	-	3	-	-	4.38	9	1.125
27	T. Coliform	MPN/100ml	160	-	-	-	-	-	170	-	-		-	-	165.00	170	160
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	4	-	-	-	-	-	4.00	4	4
29	T.K.N.	mg/1	8.406	-	-	-	-	-	11.2	-	-	-	-	-	9.803	11.2	8.406

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2008-2009

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.08	-	-	-	-	-	16.10.08	-	-	19.1.09	-	-	-	-	-
2	Appearance		S.T.	-	-	-	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	14	-	-	-	-	-	3	-	-	2	-	-	6.33	14	2
5	Colour	PCS	C.L.	-	-	-	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	O.L.	-	-	-	-	-	O.L.	-	-	O.L.	-	-	-	-	-
7	pH	pH Unit	8.27	-	-	-	-	-	7.5	-	-	7.09	-	-	7.62	8.27	7.09
8	Sp. Conductivity	µMhos/cm.	265	-	-	-	-	-	230	-	-	280	-	-	258.33	280	230
9	T. Solids	mg/1	212	-	-	-	-	-	213	-	-	239	-	-	221.33	239	212
10	D. Solids	mg/1	178	-	-	-	-	-	163	-	-	189	-	-	176.67	189	163
11	S. Solids	mg/1	34	-	-	-	-	-	50	-	-	50	-	-	44.67	50	34
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.158	-	-	0.704	-	-	0.43	0.70	0.158
13	Nitrite Nitrogen	mg/1	0.02	-	-	-	-	-	0.031	-	-	0.031	-	-	0.03	0.03	0.02
14	Nitrate Nitrogen	mg/1	0.15	-	-	-	-	-	-	-	-	1.01	-	-	0.58	1.01	0.15
15	Phosphate (PO_4)	mg/1	0.0125	-	-	-	-	-	3.1	-	-	3	-	-	2.04	3.1	0.0125
16	Chloride	mg/1	20	-	-	-	-	-	30.98	-	-	40.959	-	-	30.65	40.959	20
17	Sulphate (SO_4)	mg/1	9.15	-	-	-	-	-	13.1	-	-	12.9	-	-	11.72	13.1	9.15
18	T. Alkalinity	mg/1	100	-	-	-	-	-	140	-	-	160	-	-	133.33	160	100
19	T. Hardness	mg/1	98	-	-	-	-	-	126	-	-	98	-	-	107.33	126	98
20	CalciumHardness	mg/1	72	-	-	-	-	-	80	-	-	70	-	-	74.00	80	70
21	Magnesium H.	mg/1	26	-	-	-	-	-	46	-	-	28	-	-	33.33	46	26
22	D. Oxygen	mg/1	6.7	-	-	-	-	-	6.2	-	-	6.8	-	-	6.57	6.8	6.2
23	B.O.D.	mg/1	2.8	-	-	-	-	-	1.3	-	-	-	-	-	2.05	2.8	1.3
24	C.O.D.	mg/1	27	-	-	-	-	-	19.2	-	-	50	-	-	32.07	50	19.2
25	Sodium	mg/1	-	-	-	-	-	-	11	-	-	17	-	-	14.00	17	11
26	Potassium	mg/1	-	-	-	-	-	-	7	-	-	5	-	-	6.00	7	5
27	T. Coliform	MPN/100ml	256	-	-	-	-	-	130	-	-	-	-	-	193.00	256	130
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	2	-	-	-	-	-	2.00	2	2
29	T.K.N.	mg/1	-	-	-	-	-	-	8.4	-	-	-	-	-	8.4	8.4	8.4

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2008-2009

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.08	-	-	-	-	-	16.10.08	-	-	19.1.09	-	-	-	-	-
2	Appearance		S.T.	-	-	-	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	30	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	13	-	-	-	-	-	3	-	-	4	-	-	6.67	13	3
5	Colour	PCS	C.L.	-	-	-	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	OL	-	-	-	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.23	-	-	-	-	-	7.5	-	-	7.21	-	-	7.65	8.23	7.21
8	Sp. Conductivity	µMhos/cm.	286	-	-	-	-	-	320	-	-	320	-	-	308.67	320	286
9	T. Solids	mg/1	216	-	-	-	-	-	288	-	-	304	-	-	269.33	304	216
10	D. Solids	mg/1	184	-	-	-	-	-	163	-	-	270	-	-	205.67	270	163
11	S. Solids	mg/1	32	-	-	-	-	-	125	-	-	34	-	-	63.67	125	32
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.731	-	-	0.844	-	-	0.7875	0.844	0.731
13	Nitrite Nitrogen	mg/1	0.04	-	-	-	-	-	0.027	-	-	0.041	-	-	0.036	0.041	0.027
14	Nitrate Nitrogen	mg/1	0.35	-	-	-	-	-	-	-	-	0.81	-	-	0.58	0.81	0.35
15	Phosphate (PO_4)	mg/1	0.05	-	-	-	-	-	2.4	-	-	3.2	-	-	1.883	3.2	0.05
16	Chloride	mg/1	16	-	-	-	-	-	28.98	-	-	31.968	-	-	25.649	31.97	16
17	Sulphate (SO_4)	mg/1	11.64	-	-	-	-	-	12.4	-	-	13.4	-	-	12.48	13.4	11.64
18	T. Alkalinity	mg/1	118	-	-	-	-	-	80	-	-	160	-	-	119.33	160	80
19	T. Hardness	mg/1	98	-	-	-	-	-	106	-	-	84	-	-	96	106	84
20	Cal.Hardness	mg/1	66	-	-	-	-	-	74	-	-	62	-	-	67.33	74	62
21	Magnesium H.	mg/1	32	-	-	-	-	-	32	-	-	22	-	-	28.67	32	22
22	D. Oxygen	mg/1	8.9	-	-	-	-	-	6.9	-	-	6.4	-	-	7.4	8.9	6.4
23	B.O.D.	mg/1	3.4	-	-	-	-	-	2	-	-	-	-	-	2.7	3.4	2
24	C.O.D.	mg/1	36	-	-	-	-	-	38.4	-	-	40	-	-	38.13	40	36
25	Sodium	mg/1	-	-	-	-	-	-	10	-	-	13	-	-	11.5	13	10
26	Potassium	mg/1	-	-	-	-	-	-	6	-	-	2	-	-	4	6	2
27	T. Coliform	MPN/100ml	1600	-	-	-	-	-	200	-	-	-	-	-	900	1600	200
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	4	-	-	-	-	-	4	4	4
29	T.K.N.	mg/1	5.604	-	-	-	-	-	-	-	-	-	-	-	5.604	5.604	5.604
	CAT-B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2008-2009

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	21.7.08	-	-	15.10.08	-	-	19.1.09	-	-	-	-	-
2	Appearance		-	-	-	Clear	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	-	-	-	27	-	-		-	-		-	-	-	-	-
4	Turbidity	NTU	-	-	-	2	-	-	5	-	-	3	-	-	3.33	5	2
5	Colour	PCS	-	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	-	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	-	-	-	7.22	-	-	7	-	-	7.11	-	-	7.11	7.22	7
8	Sp. Conductivity	µMhos/cm.	-	-	-	330	-	-	290	-	-	300	-	-	306.67	330	290
9	T. Solids	mg/1	-	-	-	161	-	-	283	-	-	295	-	-	246.33	295	161
10	D. Solids	mg/1	-	-	-	154	-	-	239	-	-	246	-	-	213	246	154
11	S. Solids	mg/1	-	-	-	7	-	-	44	-	-	49	-	-	33.33	49	7
12	Amm. Nitrogen	mg/1	-	-	-	3.153	-	-	0.841	-	-	0.738	-	-	1.58	3.153	0.738
13	Nitrite Nitrogen	mg/1	-	-	-	1.006	-	-	0.043	-	-	0.04	-	-	0.36	1.01	0.04
14	Nitrate Nitrogen	mg/1	-	-	-	6.895	-	-	-	-	-	0.94	-	-	3.92	6.90	0.94
15	Phosphate (PO_4)	mg/1	-	-	-	1.375	-	-	2.6	-	-	3.3	-	-	2.425	3.3	1.375
16	Chloride	mg/1	-	-	-	22.99	-	-	29.98	-	-	32.967	-	-	28.65	32.967	22.99
17	Sulphate (SO_4)	mg/1	-	-	-	0.175	-	-	12.9	-	-	12.7	-	-	8.59	12.9	0.175
18	T. Alkalinity	mg/1	-	-	-	160	-	-	200	-	-	120	-	-	160	200	120
19	T. Hardness	mg/1	-	-	-	150	-	-	130	-	-	96	-	-	125.33	150	96
20	Cal.Hardness	mg/1	-	-	-	126	-	-	84	-	-	68	-	-	92.67	126	68
21	Magnesium H.	mg/1	-	-	-	24	-	-	46	-	-	28	-	-	32.67	46	24
22	D. Oxygen	mg/1	-	-	-	7.2	-	-	6.5	-	-	6.7	-	-	6.8	7.2	6.5
23	B.O.D.	mg/1	-	-	-	0.6	-	-	1.3	-	-	-	-	-	0.95	1.3	0.6
24	C.O.D.	mg/1	-	-	-	10	-	-	38.4	-	-	30	-	-	26.13	38.4	10
25	Sodium	mg/1	-	-	-	11	-	-	11	-	-	16	-	-	12.67	16	11
26	Potassium	mg/1	-	-	-	8	-	-	9	-	-	3	-	-	6.67	9	3
27	T. Coliform	MPN/100ml	-	-	-	21	-	-	170	-	-	-	-	-	95.5	170	21
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	4	-	-	-	-	-	4	4	4
29	T.K.N.	mg/1	-	-	-	2.802	-	-	-	-	-	-	-	-	2.802	2.802	2.802
	Cat. -B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2008-2009

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.08	-	-	-	-	-	-	-		19.1.09	-	-	-	-	-
2	Appearance		S.T.	-	-	-	-	-	-	-	-	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	13	-	-	-	-	-	-	-	-	5	-	-	9	13	5
5	Colour	PCS	C.L.	-	-	-	-	-	-	-	-	S.blackish	-	-	-	-	-
6	Odour	T. No	OL	-	-	-	-	-	-	-	-	S.Unple.	-	-	-	-	-
7	pH	pH Unit	8.35	-	-	-	-	-	-	-	-	7.03	-	-	7.69	8.35	7.03
8	Sp. Conductivity	µMhos/cm.	289	-	-	-	-	-	-	-	-	280	-	-	284.5	289	280
9	T. Solids	mg/1	244	-	-	-	-	-	-	-	-	152	-	-	198	244	152
10	D. Solids	mg/1	198	-	-	-	-	-	-	-	-	136	-	-	167	198	136
11	S. Solids	mg/1	46	-	-	-	-	-	-	-	-	16	-	-	31	46	16
12	Amm. Nitrogen	mg/1	0.06	-	-	-	-	-	-	-	-	0.011	-	-	0.0355	0.06	0.011
13	Nitrite Nitrogen	mg/1	0.14	-	-	-	-	-	-	-	-	0.011	-	-	0.0755	0.14	0.011
14	Nitrate Nitrogen	mg/1	0.35	-	-	-	-	-	-	-	-	0.08	-	-	0.215	0.35	0.08
15	Phosphate (PO_4)	mg/1	0.087	-	-	-	-	-	-	-	-	2.76	-	-	1.4235	2.76	0.087
16	Chloride	mg/1	23	-	-	-	-	-	-	-	-	11.99	-	-	17.495	23	11.99
17	Sulphate (SO_4)	mg/1	11.64	-	-	-	-	-	-	-	-	12	-	-	11.82	12	11.64
18	T. Alkalinity	mg/1	118	-	-	-	-	-	-	-	-	210	-	-	164	210	118
19	T. Hardness	mg/1	98	-	-	-	-	-	-	-	-	180	-	-	139	180	98
20	CalciumHardness	mg/1	72	-	-	-	-	-	-	-	-	130	-	-	101	130	72
21	Magnesium H.	mg/1	26	-	-	-	-	-	-	-	-	50	-	-	38	50	26
22	D. Oxygen	mg/1	11	-	-	-	-	-	-	-	-	7	-	-	9	11	7
23	B.O.D.	mg/1	6.2	-	-	-	-	-	-	-	-	-	-	-	6.2	6.2	6.2
24	C.O.D.	mg/1	45	-	-	-	-	-	-	-	-	8.84	-	-	26.92	45	8.84
25	Sodium	mg/1	8	-	-	-	-	-	-	-	-	-	-	-	8	8	8
26	Potassium	mg/1	1.125	-	-	-	-	-	-	-	-	-	-	-	1.125	1.125	1.125
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2008-2009

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7..08	-	-	-	-	-	-	-	-	16.01.08	-	-	-	-	-
2	Appearance		S.T.	-	-	-	-	-	-	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	26	-	-	-	-	-	-	-	-	48	-	-	37	48	26
5	Colour	PCS	C.L.	-	-	-	-	-	-	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	OL	-	-	-	-	-	-	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.13	-	-	-	-	-	-	-	-	8.34	-	-	8.235	8.34	8.13
8	Sp. Conductivity	µMhos/cm.	594	-	-	-	-	-	-	-	-	508	-	-	551	594	508
9	T. Solids	mg/1	398	-	-	-	-	-	-	-	-	394	-	-	396	398	394
10	D. Solids	mg/1	360	-	-	-	-	-	-	-	-	316	-	-	338	360	316
11	S. Solids	mg/1	38	-	-	-	-	-	-	-	-	78	-	-	58	78	38
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	0.06	-	-	0.06	0.06	0.06
13	Nitrite Nitrogen	mg/1	0.12	-	-	-	-	-	-	-	-	0.2	-	-	0.16	0.2	0.12
14	Nitrate Nitrogen	mg/1	0.18	-	-	-	-	-	-	-	-	0.45	-	-	0.32	0.45	0.18
15	Phosphate (PO_4)	mg/1	0.375	-	-	-	-	-	-	-	-	0.65	-	-	0.51	0.65	0.38
16	Chloride	mg/1	66	-	-	-	-	-	-	-	-	17	-	-	41.5	66	17
17	Sulphate (SO_4)	mg/1	3.32	-	-	-	-	-	-	-	-	13.31	-	-	8.315	13.31	3.32
18	T. Alkalinity	mg/1	234	-	-	-	-	-	-	-	-	228	-	-	231	234	228
19	T. Hardness	mg/1	248	-	-	-	-	-	-	-	-	236	-	-	242	248	236
20	CalciumHardness	mg/1	104	-	-	-	-	-	-	-	-	176	-	-	140	176	104
21	Magnesium H.	mg/1	144	-	-	-	-	-	-	-	-	60	-	-	102	144	60
22	D. Oxygen	mg/1	4.8	-	-	-	-	-	-	-	-	7.4	-	-	6.1	7.4	4.8
23	B.O.D.	mg/1	2.4	-	-	-	-	-	-	-	-	4.2	-	-	3.3	4.2	2.4
24	C.O.D.	mg/1	18	-	-	-	-	-	-	-	-	59.28	-	-	38.64	59.28	18
25	Sodium	mg/1	34	-	-	-	-	-	-	-	-	-	-	-	34	34	34
26	Potassium	mg/1	8	-	-	-	-	-	-	-	-	-	-	-	8	8	8
27	T. Coliform	MPN/100ml	2400	-	-	-	-	-	-	-	-	-	-	-	2400	2400	2400
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

Water Quality during Year - 2009-2010

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Kamla Park (East side of the lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date		17.4.09	-	-	2.7.09	-	-	30.10.09	-	-	29.1.10	-	-	-	-	-
2	Appearance		Clear	-	-	ST	-	-	Clear	-	-	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	2	-	-	36	-	-	2	-	-	42	-	-	20.50	42	2
5	Colour	PCS	CL	-	-	-	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.28	-	-	7.86	-	-	7.3	-	-	7.87	-	-	7.83	8.28	7.3
8	Sp. Conductivity	µMhos/cm.	280	-	-	366	-	-	290	-	-	230	-	-	291.5	366	230
9	T. Solids	mg/1	156	-	-	382	-	-	310	-	-	280	-	-	282	382	156
10	D. Solids	mg/1	126	-	-	298	-	-	286	-	-	194	-	-	226	298	126
11	S. Solids	mg/1	30	-	-	84	-	-	24	-	-	86	-	-	56	86	24
12	Amm. Nitrogen	mg/1	0.541	-	-	0.64	-	-	0.4	-	-	0.17	-	-	0.438	0.64	0.17
13	Nitrite Nitrogen	mg/1	0.017	-	-	-	-	-	0.14	-	-	-	-	-	0.079	0.14	0.017
14	Nitrate Nitrogen	mg/1	0.1	-	-	0.22	-	-	1.04	-	-	0.44	-	-	0.45	1.04	0.1
15	Phosphate (PO_4)	mg/1	2.11	-	-	0.144	-	-	3.9	-	-	0.22	-	-	1.5935	3.9	0.144
16	Chloride	mg/1	27.39	-	-	28	-	-	29.97	-	-	15	-	-	25.09	29.97	15
17	Sulphate (SO_4)	mg/1	3.13	-	-	10.81	-	-	10.4	-	-	9.5	-	-	8.46	10.81	3.13
18	T. Alkalinity	mg/1	180	-	-	180	-	-	182	-	-	110	-	-	163	182	110
19	T. Hardness	mg/1	120	-	-	188	-	-	190	-	-	114	-	-	153	190	114
20	Cal.Hardness	mg/1	64	-	-	156	-	-	124	-	-	76	-	-	105	156	64
21	Magnesium H.	mg/1	56	-	-	32	-	-	66	-	-	38	-	-	48	66	32
22	D. Oxygen	mg/1	6.2	-	-	7.4	-	-	7.4	-	-	8.8	-	-	7.45	8.8	6.2
23	B.O.D.	mg/1	2.4	-	-	2.5	-	-	4.8	-	-	3.5	-	-	3.3	4.8	2.4
24	C.O.D.	mg/1	28.2	-	-	28.8	-	-	-	-	-	20	-	-	25.67	28.8	20
25	Sodium	mg/1	17	-	-	2	-	-	-	-	-	4	-	-	7.67	17	2
26	Potassium	mg/1	4	-	-	1	-	-	-	-	-	1	-	-	2.00	4	1
27	T. Coliform	MPN/100ml	13	-	-	110	-	-	-	-	-	70	-	-	64.33	110	13
28	F. Coliform	MPN/100ml	4	-	-	-	-	-		-	-	-	-	-	4.00	4	4
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2009-2010

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.4.09	-	-	31.7.09	-	-	5.10.09	-	-	29.1.10	-	-	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	1	-	-	6	-	-	2	-	-	44	-	-	13.25	44	1
5	Colour	PCS	CL	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.23	-	-	7.64	-	-	7.56	-	-	7.9	-	-	7.83	8.23	7.56
8	Sp. Conductivity	µMhos/cm.	270	-	-	350	-	-	230	-	-	210	-	-	265	350	210
9	T. Solids	mg/1	133	-	-	280	-	-	261	-	-	276	-	-	237.5	280	133
10	D. Solids	mg/1	112	-	-	245	-	-	208	-	-	186	-	-	187.75	245	112
11	S. Solids	mg/1	21	-	-	35	-	-	53	-	-	90	-	-	49.75	90	21
12	Amm. Nitrogen	mg/1	0.752	-	-	1.12	-	-	0.48	-	-	0.13	-	-	0.6205	1.12	0.13
13	Nitrite Nitrogen	mg/1	0.007	-	-	0.018	-	-	0.56	-	-	0.47	-	-	0.26375	0.56	0.007
14	Nitrate Nitrogen	mg/1	0.13	-	-	0.25	-	-	0.04	-	-	-	-	-	0.14	0.25	0.04
15	Phosphate (PO_4)	mg/1	2.71	-	-		-	-	1.4	-	-	0.263	-	-	1.46	2.71	0.263
16	Chloride	mg/1	20.99	-	-	56.98	-	-	29.97	-	-	19	-	-	31.74	56.98	19
17	Sulphate (SO_4)	mg/1	12.8	-	-	-	-	-	5	-	-	10.5	-	-	9.433	12.8	5
18	T. Alkalinity	mg/1	140	-	-	230	-	-	211	-	-	100	-	-	170.25	230	100
19	T. Hardness	mg/1	116	-	-	246	-	-	90	-	-	116	-	-	142	246	90
20	CalciumHardness	mg/1	90	-	-	194	-	-	76	-	-	74	-	-	108.5	194	74
21	Magnesium H.	mg/1	26	-	-	52	-	-	14	-	-	42	-	-	33.5	52	14
22	D. Oxygen	mg/1	6.9	-	-	6.5	-	-	5.4	-	-	9	-	-	6.95	9	5.4
23	B.O.D.	mg/1	3.6	-	-	2.2	-	-	3.8	-	-	2.5	-	-	3.025	3.8	2.2
24	C.O.D.	mg/1	18.8	-	-	28.2	-	-	46	-	-	20	-	-	28.25	46	18.8
25	Sodium	mg/1	13	-	-	14	-	-	-	-	-	4	-	-	10.33	14	4
26	Potassium	mg/1	2	-	-	9	-	-	-	-	-	2	-	-	4.33	9	2
27	T. Coliform	MPN/100ml	8	-	-	23	-	-	300	-	-	140	-	-	117.75	300	8
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	4	-	-	-	-	-	4	4	4
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2009-2010

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.4.09	-	-	2.7.10	-	-	6.10.09	-	-	29.1.10	-	-	-	-	-
2	Appearance		Clear	-	-	S.T.	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	2	-	-	34	-	-	2.5	-	-	30	-	-	17.13	34	2
5	Colour	PCS	CL	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.3	-	-	7.88	-	-	7.39	-	-	7.68	-	-	7.81	8.3	7.39
8	Sp. Conductivity	µMhos/cm.	200	-	-	364	-	-	240	-	-	220	-	-	256	364	200
9	T. Solids	mg/1	114	-	-	380	-	-	278	-	-	262	-	-	258.5	380	114
10	D. Solids	mg/1	102	-	-	294	-	-	214	-	-	198	-	-	202	294	102
11	S. Solids	mg/1	12	-	-	86	-	-	64	-	-	64	-	-	56.5	86	12
12	Amm. Nitrogen	mg/1	1.186	-	-	0.43	-	-	0.59	-	-	0.16	-	-	0.5915	1.186	0.16
13	Nitrite Nitrogen	mg/1	0.011	-	-	-	-	-	0.47	-	-	-	-	-	0.2405	0.47	0.011
14	Nitrate Nitrogen	mg/1	0.15	-	-	0.2	-	-	0.1	-	-	0.5	-	-	0.2375	0.5	0.1
15	Phosphate (PO_4)	mg/1	2.76	-	-	0.132	-	-	1.6	-	-	0.209	-	-	1.18	2.76	0.132
16	Chloride	mg/1	24.99	-	-	27	-	-	18.2	-	-	20	-	-	22.55	27	18.2
17	Sulphate (SO_4)	mg/1	12	-	-	9.48	-	-	6	-	-	10	-	-	9.37	12	6
18	T. Alkalinity	mg/1	160	-	-	180	-	-	345.6	-	-	120	-	-	201.4	345.6	120
19	T. Hardness	mg/1	124	-	-	184	-	-	108	-	-	110	-	-	131.5	184	108
20	Cal.Hardness	mg/1	102	-	-	150	-	-	84	-	-	68	-	-	101	150	68
21	Magnesium H.	mg/1	22	-	-	34	-	-	24	-	-	42	-	-	30.5	42	22
22	D. Oxygen	mg/1	6.5	-	-	7.5	-	-	5.5	-	-	8.5	-	-	7	8.5	5.5
23	B.O.D.	mg/1	2.3	-	-	2.7	-	-	4	-	-	2.3	-	-	2.825	4	2.3
24	C.O.D.	mg/1	28.2	-	-	28.8	-	-	55.2	-	-	20	-	-	33.05	55.2	20
25	Sodium	mg/1	16	-	-	2	-	-	-	-	-	5	-	-	7.67	16	2
26	Potassium	mg/1	8	-	-	1	-	-	-	-	-	2	-	-	3.67	8	1
27	T. Coliform	MPN/100ml	13	-	-	90	-	-	500	-	-	80	-	-	170.75	500	13
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	6	-	-	-	-	-	6	6	6
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2009-2010

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		30.4.09	-	-	-	-	-	6.10.09	-	-	29.1.10	-	-	-	-	-
2	Appearance		ST	-	-	-	-	-	Clear	-	-	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	28.5	-	-	-	-	-	-	-	-
4	Turbidity	NTU	36	-	-	-	-	-	4	-	-	42	-	-	27.33	42	4
5	Colour	PCS	Smuddy	-	-	-	-	-	CL	-	-	St. Yellow	-	-	-	-	-
6	Odour	T. No	OL	-	-	-	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.26	-	-	-	-	-	8.42	-	-	7.8	-	-	8.16	8.42	7.8
8	Sp. Conductivity	µMhos/cm.	410	-	-	-	-	-	410	-	-	350	-	-	390	410	350
9	T. Solids	mg/1	296	-	-	-	-	-	314	-	-	392	-	-	334	392	296
10	D. Solids	mg/1	216	-	-	-	-	-	294	-	-	304	-	-	271.33	304	216
11	S. Solids	mg/1	80	-	-	-	-	-	20	-	-	88	-	-	62.67	88	20
12	Amm. Nitrogen	mg/1	0.96	-	-	-	-	-	0.055	-	-	0.18	-	-	0.40	0.96	0.055
13	Nitrite Nitrogen	mg/1	0.9	-	-	-	-	-	0.005	-	-	-	-	-	0.45	0.9	0.005
14	Nitrate Nitrogen	mg/1	1	-	-	-	-	-	0.04	-	-	0.39	-	-	0.48	1	0.04
15	Phosphate (PO_4)	mg/1	0.96	-	-	-	-	-	0.201	-	-	0.28	-	-	0.48	0.96	0.201
16	Chloride	mg/1	31.99	-	-	-	-	-	40.98	-	-	37	-	-	36.66	40.98	31.99
17	Sulphate (SO_4)	mg/1	10.5	-	-	-	-	-	0.762	-	-	11.5	-	-	7.59	11.5	0.762
18	T. Alkalinity	mg/1	270	-	-	-	-	-	400	-	-	190	-	-	286.67	400	190
19	T. Hardness	mg/1	166	-	-	-	-	-	132	-	-	179	-	-	159	179	132
20	Cal.Hardness	mg/1	132	-	-	-	-	-	106	-	-	100	-	-	112.67	132	100
21	Magnesium H.	mg/1	34	-	-	-	-	-	26	-	-	79	-	-	46.33	79	26
22	D. Oxygen	mg/1	6.3	-	-	-	-	-	6.5	-	-	10	-	-	7.6	10	6.3
23	B.O.D.	mg/1	2.7	-	-	-	-	-	1.5	-	-	5.2	-	-	3.13	5.2	1.5
24	C.O.D.	mg/1	28.56	-	-	-	-	-	28	-	-	30	-	-	28.85	30	28
25	Sodium	mg/1	13	-	-	-	-	-	-	-	-	8	-	-	10.5	13	8
26	Potassium	mg/1	1	-	-	-	-	-	-	-	-	2	-	-	1.50	2	1
27	T. Coliform	MPN/100ml	220	-	-	-	-	-	-	-	-	110	-	-	165	220	110
28	F. Coliform	MPN/100ml	6	-	-	-	-	-	-	-	-	-	-	-	6	6	6
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2009-2010

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		27.4.09	-	-	14.7.09	-	-	30.10.09	-	-	29.1.10	-	-	-	-	-
2	Appearance		Turbid	-	-	S.T.	-	-	S.T.	-	-	ST	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	46	-	-	32	-	-	38	-	-	2	-	-	29.50	46	2
5	Colour	PCS	Muddy	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	7.86	-	-	7.87	-	-	7.78	-	-	8.1	-	-	7.9025	8.1	7.78
8	Sp. Conductivity	µMhos/cm.	410	-	-	422	-	-	345	-	-	210	-	-	346.75	422	210
9	T. Solids	mg/1	328	-	-	471	-	-	322	-	-	181	-	-	325.5	471	181
10	D. Solids	mg/1	230	-	-	408	-	-	254	-	-	140	-	-	258	408	140
11	S. Solids	mg/1	98	-	-	63	-	-	68	-	-	41	-	-	67.5	98	41
12	Amm. Nitrogen	mg/1	-	-	-	0.22	-	-	0.13	-	-	0.328	-	-	0.226	0.328	0.13
13	Nitrite Nitrogen	mg/1	0.86	-	-	0.079	-	-	0.04	-	-	0.06	-	-	0.260	0.86	0.04
14	Nitrate Nitrogen	mg/1	-	-	-	0.14	-	-	0.46	-	-	0.14	-	-	0.247	0.46	0.14
15	Phosphate (PO_4)	mg/1	0.98	-	-	0.237	-	-	0.219	-	-	1.1	-	-	0.634	1.1	0.219
16	Chloride	mg/1	55.98	-	-	21	-	-	13	-	-	14	-	-	25.995	55.98	13
17	Sulphate (SO_4)	mg/1	12	-	-	14.36	-	-	8.92	-	-	27	-	-	15.57	27	8.92
18	T. Alkalinity	mg/1	190	-	-	110	-	-	160	-	-	24	-	-	121	190	24
19	T. Hardness	mg/1	152	-	-	120	-	-	104	-	-	140	-	-	129	152	104
20	CalciumHardness	mg/1	108	-	-	78	-	-	56	-	-	120	-	-	90.5	120	56
21	Magnesium H.	mg/1	44	-	-	42	-	-	48	-	-	20	-	-	38.5	48	20
22	D. Oxygen	mg/1	5.3	-	-	7	-	-	7.3	-	-	9.5	-	-	7.275	9.5	5.3
23	B.O.D.	mg/1	3.8	-	-	1.6	-	-	3.9	-	-	1.2	-	-	2.625	3.9	1.2
24	C.O.D.	mg/1	38.4	-	-	18.4	-	-	40	-	-	19.2	-	-	29	40	18.4
25	Sodium	mg/1	-	-	-	12	-	-	-	-	-	18	-	-	15	18	12
26	Potassium	mg/1	-	-	-	8	-	-	-	-	-	5	-	-	6.50	8	5
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	110	-	-	110	110	110
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	4	-	-	4	4	4
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2009-2010

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	-	-	-	29.10.09	-	-	16.01.10	-	-	-	-	-
2	Appearance		-	-	-	-	-	-	Turbid	-	-	Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	-	-	-	-	52	-	-	48	-	-	50.00	52	48
5	Colour	PCS	-	-	-	-	-	-	C.L.	-	-	C.L.	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	-	-	-	-	-	-	7.2	-	-	8.34	-	-	7.77	8.34	7.2
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	-	-	455	-	-	508	-	-	481.5	508	455
9	T. Solids	mg/1	-	-	-	-	-	-	424	-	-	394	-	-	409	424	394
10	D. Solids	mg/1	-	-	-	-	-	-	344	-	-	316	-	-	330	344	316
11	S. Solids	mg/1	-	-	-	-	-	-	80	-	-	78	-	-	79	80	78
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.24	-	-	0.06	-	-	0.15	0.24	0.06
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.38	-	-	0.2	-	-	0.29	0.38	0.2
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	0.67	-	-	0.45	-	-	0.56	0.67	0.45
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.36	-	-	0.65	-	-	0.505	0.65	0.36
16	Chloride	mg/1	-	-	-	-	-	-	26	-	-	17	-	-	21.5	26	17
17	Sulphate (SO_4)	mg/1	-	-	-	-	-	-	12.3	-	-	13.31	-	-	12.805	13.31	12.3
18	T. Alkalinity	mg/1	-	-	-	-	-	-	270	-	-	228	-	-	249	270	228
19	T. Hardness	mg/1	-	-	-	-	-	-	182	-	-	236	-	-	209	236	182
20	CalciumHardness	mg/1	-	-	-	-	-	-	126	-	-	176	-	-	151	176	126
21	Magnesium H.	mg/1	-	-	-	-	-	-	56	-	-	60	-	-	58	60	56
22	D. Oxygen	mg/1	-	-	-	-	-	-	3.9	-	-	7.4	-	-	5.65	7.4	3.9
23	B.O.D.	mg/1	-	-	-	-	-	-	4.5	-	-	4.2	-	-	4.35	4.5	4.2
24	C.O.D.	mg/1	-	-	-	-	-	-	60	-	-	59.28	-	-	59.64	60	59.28
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2010-2011

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.10	-	-	14.7.10	-	-	27.10.10	-	-	1.1.11	-	-	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	-	-	-	28	-	-	28	-	-		-	-	28.00	28	28
4	Turbidity	NTU	-	-	-	-	-	-	-	-	-	-	-	-			
5	Colour	PCS	CL	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.18	-	-	8.23	-	-	7.41	-	-	8.01	-	-	7.96	8.23	7.41
8	Sp. Conductivity	µMhos/cm.	240	-	-	240	-	-	240	-	-	264	-	-	246	264	240
9	T. Solids	mg/1	154	-	-	187	-	-	226	-	-	180	-	-	186.75	226	154
10	D. Solids	mg/1	150	-	-	150	-	-	168	-	-	160	-	-	157	168	150
11	S. Solids	mg/1	4	-	-	37	-	-	58	-	-	20	-	-	29.75	58	4
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.518	-	-	-	-	-	0.518	0.518	0.518
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	0.28	-	-			
14	Nitrate Nitrogen	mg/1	0.16	-	-	4.3	-	-	-	-	-	-	-	-	2.23	4.30	0.16
15	Phosphate (PO_4)	mg/1	0.1	-	-	0.5	-	-	0.95	-	-	0.48	-	-	0.51	0.95	0.1
16	Chloride	mg/1	43.28	-	-	19.14	-	-	34.98	-	-	19.98	-	-	29.345	43.28	19.14
17	Sulphate (SO_4)	mg/1	10	-	-	2.5	-	-	8.16	-	-	4.3	-	-	6.24	10.00	2.5
18	T. Alkalinity	mg/1	310	-	-	90	-	-	150	-	-	110	-	-	165	310	90
19	T. Hardness	mg/1	94	-	-	180	-	-	134	-	-	146	-	-	138.5	180	94
20	CalciumHardness	mg/1	46	-	-	44	-	-	96	-	-	48	-	-	58.5	96	44
21	Magnesium H.	mg/1	48	-	-	136	-	-	36	-	-	98	-	-	79.5	136	36
22	D. Oxygen	mg/1	8.7	-	-	7.4	-	-	6.5	-	-	9	-	-	7.9	9	6.5
23	B.O.D.	mg/1	3.4	-	-	3.6	-	-	1.2	-	-	4.2	-	-	3.1	4.2	1.2
24	C.O.D.	mg/1	35.04	-	-	35.2	-	-	20	-	-	18.4	-	-	27.16	35.2	18.4
25	Sodium	mg/1	5	-	-	-	-	-	13	-	-	-	-	-	9	13	5
26	Potassium	mg/1	3	-	-	-	-	-	8	-	-	-	-	-	5.5	8	3
27	T. Coliform	MPN/100ml	1600	-	-	1600	-	-	1600	-	-	13	-	-	1203.25	1600	13
28	F. Coliform	MPN/100ml	2	-	-	17	-	-	2	-	-	2	-	-	5.75	17	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2010-2011

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.10	-	-	15.7.10	-	-	27.10.10	-	-	1.1.11	-	-	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	30	-	-	29	-	-	28	-	-		-	-	29.00	30	28
4	Turbidity	NTU	-	-	-	-	-	-	-	-	-	-	-	-			
5	Colour	PCS	CL	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	8.04	-	-	8.31	-	-	7.41	-	-	8.38	-	-	8.04	8.38	7.41
8	Sp. Conductivity	µMhos/cm.	190	-	-	230	-	-	280	-	-	270	-	-	242.5	280	190
9	T. Solids	mg/1	136	-	-	174	-	-	299	-	-	194	-	-	200.75	299	136
10	D. Solids	mg/1	130	-	-	150	-	-	187	-	-	160	-	-	156.75	187	130
11	S. Solids	mg/1	6	-	-	24	-	-	112	-	-	34	-	-	44	112	6
12	Amm. Nitrogen	mg/1		-	-	-	-	-	0.071	-	-	0.21	-	-	0.1405	0.21	0.071
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.032	-	-	-	-	-	0.03	0.03	0.032
14	Nitrate Nitrogen	mg/1	0.17	-	-	4.8	-	-	-	-	-	-	-	-	2.49	4.80	0.17
15	Phosphate (PO_4)	mg/1	-	-	-	0.1	-	-	2.3	-	-	0.39	-	-	0.93	2.3	0.1
16	Chloride	mg/1	18.71	-	-	24.44	-	-	33.98	-	-	22.97	-	-	25.025	33.98	18.71
17	Sulphate (SO_4)	mg/1	12.5	-	-	3.2	-	-	10.1	-	-	4.8	-	-	7.65	12.50	3.2
18	T. Alkalinity	mg/1	380	-	-	60	-	-	110	-	-	120	-	-	167.5	380	60
19	T. Hardness	mg/1	124	-	-	94	-	-	112	-	-	126	-	-	114	126	94
20	Cal.Hardness	mg/1	40	-	-	48	-	-	84	-	-	58	-	-	57.5	84	40
21	Magnesium H.	mg/1	84	-	-	46	-	-	28	-	-	68	-	-	56.5	84	28
22	D. Oxygen	mg/1	8.5	-	-	9.3	-	-	7.1	-	-	10.1	-	-	8.75	10.1	7.1
23	B.O.D.	mg/1	4.7	-	-	2.9	-	-	2.1	-	-	5.5	-	-	3.8	5.5	2.1
24	C.O.D.	mg/1	43.8	-	-	17.6	-	-	40	-	-	55.2	-	-	39.15	55.2	17.6
25	Sodium	mg/1	9	-	-	-	-	-	12	-	-	-	-	-	10.5	12	9
26	Potassium	mg/1	3	-	-	-	-	-	7	-	-	-	-	-	5	7	3
27	T. Coliform	MPN/100ml	900	-	-	900	-	-	800	-	-	14	-	-	653.5	900	14
28	F. Coliform	MPN/100ml	2	-	-	9	-	-	2	-	-	4	-	-	4.25	9	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2010-2011

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.10	-	-	19.7.10	-	-	27.10.10	-	-	1.1.11	-	-	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	-	-	-	28	-	-	29	-	-		-	-	28.50	29	28
4	Turbidity	NTU	-	-	-	-	-	-	-	-	-	-	-	-			
5	Colour	PCS	CL	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	7.75	-	-	7.7	-	-	7.43	-	-	8.35	-	-	7.81	8.35	7.43
8	Sp. Conductivity	µMhos/cm.	260	-	-	110	-	-	120	-	-	270	-	-	190	270	110
9	T. Solids	mg/1	171	-	-	278	-	-	287	-	-	194	-	-	232.5	287	171
10	D. Solids	mg/1	170	-	-	256	-	-	263	-	-	170	-	-	214.75	263	170
11	S. Solids	mg/1	1	-	-	22	-	-	24	-	-	24	-	-	17.75	24	1
12	Amm. Nitrogen	mg/1	-	-	-	0.11	-	-	0.613	-	-	0.21	-	-	0.311	0.613	0.11
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.038	-	-	-	-	-	0.04	0.04	0.038
14	Nitrate Nitrogen	mg/1	0.24	-	-	-	-	-	-	-	-	-	-	-	0.24	0.24	0.24
15	Phosphate (PO_4)	mg/1	0.03	-	-	BDL	-	-	2.8	-	-	0.38	-	-	1.07	2.8	0.03
16	Chloride	mg/1	23	-	-	22.99	-	-	25.99	-	-	21.97	-	-	23.49	25.99	21.97
17	Sulphate (SO_4)	mg/1	13.5	-	-	5	-	-	7.2	-	-	3.9	-	-	7.4	13.50	3.9
18	T. Alkalinity	mg/1	340	-	-	140	-	-	160	-	-	140	-	-	195	340	140
19	T. Hardness	mg/1	116	-	-	270	-	-	276	-	-	140	-	-	200.5	276	116
20	Cal.Hardness	mg/1	58	-	-	230	-	-	232	-	-	42	-	-	140.5	232	42
21	Magnesium H.	mg/1	58	-	-	40	-	-	44	-	-	98	-	-	60	98	40
22	D. Oxygen	mg/1	1.6	-	-	4.5	-	-	-	-	-	9.8	-	-	5.3	9.8	1.6
23	B.O.D.	mg/1	0.2	-	-	2.2	-	-	2.1	-	-	4.5	-	-	2.25	4.5	0.2
24	C.O.D.	mg/1	17.52	-	-	48.4	-	-	40	-	-	36.8	-	-	35.68	48.4	17.52
25	Sodium	mg/1	5	-	-	25	-	-	22	-	-	-	-	-	17.33	25	5
26	Potassium	mg/1	2	-	-	12	-	-	10	-	-	-	-	-	8	12	2
27	T. Coliform	MPN/100ml	900	-	-	1600	-	-	1600	-	-	13	-	-	1028.25	1600	13
28	F. Coliform	MPN/100ml	3	-	-	12	-	-	10	-	-	4	-	-	7.25	12	3
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2010-2011

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		13.4.10	-	-	-	10.8.10	-	-	29.11.10	-	-	3.2.11	-	-	-	-
2	Appearance		S.Turbid	-	-	-	S.Turbid	-	-	S.Turbid	-	-	S.Turbid	-	-	-	-
3	Temperature	oC	-	-	-	-	28	-	-	28	-	-	-	-	28.00	28	28
4	Turbidity	NTU	-	-	-	-	-	-	-	-	-	-	12	-	12.00	12	12
5	Colour	PCS	CL	-	-	-	CL	-	-	CL	-	-	L.Yellow	-	-	-	-
6	Odour	T. No	OL	-	-	-	OL	-	-	OL	-	-	Sl.Unple.	-	-	-	-
7	pH	pH Unit	8.5	-	-	-	8.86	-	-	7.54	-	-	7.74	-	8.16	8.86	7.54
8	Sp. Conductivity	µMhos/cm.	220	-	-	-	259	-	-	522	-	-	240	-	310.25	522	220
9	T. Solids	mg/1	190	-	-	-	184	-	-	345	-	-	270	-	247.25	345	184
10	D. Solids	mg/1	150	-	-	-	160	-	-	290	-	-	192	-	198	290	150
11	S. Solids	mg/1	40	-	-	-	24	-	-	55	-	-	78	-	49.25	78	24
12	Amm. Nitrogen	mg/1	-	-	-	-	1.8	-	-	0.15	-	-	0.37	-	0.77333	1.8	0.15
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	0.16	-	0.16	0.16	0.16
14	Nitrate Nitrogen	mg/1	0.47	-	-	-	0.3	-	-	-	-	-	0.9	-	0.56	0.90	0.3
15	Phosphate (PO_4)	mg/1	0.06	-	-	-	1	-	-	0.36	-	-	1	-	0.61	1	0.06
16	Chloride	mg/1	14	-	-	-	23.39	-	-	20	-	-	20.99	-	19.595	23.39	14
17	Sulphate (SO_4)	mg/1	13	-	-	-	6	-	-	10	-	-	5.7	-	8.675	13.00	5.7
18	T. Alkalinity	mg/1	130	-	-	-	180	-	-	150	-	-	56	-	129	180	56
19	T. Hardness	mg/1	90	-	-	-	136	-	-	98	-	-	170	-	123.5	170	90
20	CalciumHardness	mg/1	50	-	-	-	40	-	-	54	-	-	106	-	62.5	106	40
21	Magnesium H.	mg/1	40	-	-	-	96	-	-	44	-	-	64	-	61	96	40
22	D. Oxygen	mg/1	9.5	-	-	-	6	-	-	7.2	-	-	13.6	-	9.075	13.6	6
23	B.O.D.	mg/1	2.3	-	-	-	3.1	-	-	2.5	-	-	4	-	2.975	4	2.3
24	C.O.D.	mg/1	27.6	-	-	-	20	-	-	30	-	-	9.6	-	21.8	30	9.6
25	Sodium	mg/1	23	-	-	-	1	-	-	-	-	-	12.75	-	12.25	23	1
26	Potassium	mg/1	6	-	-	-	3	-	-	-	-	-	4.32	-	4.44	6	3
27	T. Coliform	MPN/100ml	-	-	-	-	1600	-	-	-	-	-	110	-	855	1600	110
28	F. Coliform	MPN/100ml	-	-	-	-	23	-	-	-	-	-	6	-	14.5	23	6
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2010-2011

Natural Water Resource : Upper Lake, at Kamla Park Near w/s.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23.4.10	-	-	15.7.10	-	-	27.10.10	-	-	1.1.11	-	-	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	Clear	-	-	Clear	-	-	-	-	-
3	Temperature	oC	36	-	-	29	-	-	-	-	-	-	-	-	32.5	36	29
4	Turbidity	NTU	-	-	-	-	-	-	-	-	-	-	-	-			
5	Colour	PCS	CL	-	-	CL	-	-	CL	-	-	CL	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	OL	-	-	OL	-	-	-	-	-
7	pH	pH Unit	7.96	-	-	8.14	-	-	7.61	-	-	7.59	-	-	7.83	8.14	7.59
8	Sp. Conductivity	µMhos/cm.	240	-	-	260	-	-	280	-	-	308	-	-	272	308	240
9	T. Solids	mg/1	252	-	-	214	-	-	249	-	-	245	-	-	240	252	214
10	D. Solids	mg/1	190	-	-	170	-	-	174	-	-	200	-	-	183.5	200	170
11	S. Solids	mg/1	62	-	-	44	-	-	75	-	-	45	-	-	56.5	75	44
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.41	-	-	0.23	-	-	0.32	0.41	0.23
13	Nitrite Nitrogen	mg/1		-	-	-	-	-	-	-	-	-	-	-			
14	Nitrate Nitrogen	mg/1	0.57	-	-	4.6	-	-	-	-	-	-	-	-	2.585	4.6	0.57
15	Phosphate (PO_4)	mg/1	0.02	-	-	0.4	-	-	2.3	-	-	0.43	-	-	0.79	2.3	0.02
16	Chloride	mg/1	17	-	-	22.32	-	-	30.99	-	-	20.97	-	-	22.82	30.99	17
17	Sulphate (SO_4)	mg/1	14	-	-	3	-	-	1.1	-	-	4.4	-	-	5.625	14.00	1.1
18	T. Alkalinity	mg/1	140	-	-	70	-	-	180	-	-	140	-	-	132.5	180	70
19	T. Hardness	mg/1	92	-	-	100	-	-	202	-	-	160	-	-	138.5	202	92
20	CalciumHardness	mg/1	58	-	-	54	-	-	98	-	-	62	-	-	68	98	54
21	Magnesium H.	mg/1	34	-	-	46	-	-	104	-	-	98	-	-	70.5	104	34
22	D. Oxygen	mg/1	8.2	-	-	6.9	-	-	6.1	-	-	8.1	-	-	7.325	8.2	6.1
23	B.O.D.	mg/1	3.2	-	-	2.7	-	-	2.3	-	-	4.1	-	-	3.075	4.1	2.3
24	C.O.D.	mg/1	27.6	-	-	26.4	-	-	30	-	-	18.4	-	-	25.6	30	18.4
25	Sodium	mg/1	24	-	-	-	-	-	15	-	-	-	-	-	19.5	24	15
26	Potassium	mg/1	4	-	-	-	-	-	10	-	-	-	-	-	7	10	4
27	T. Coliform	MPN/100ml	110	-	-	900	-	-	900	-	-	11	-	-	480.25	900	11
28	F. Coliform	MPN/100ml	-	-	-	14	-	-	16	-	-	2	-	-	10.6667	16	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2010-2011

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Dry	-	-	18.7.10	N. R.	-	-	Dry	-	-	4.2.11	-	-	-	-
2	Appearance		-	-	-	Clear	-	-	-	-	-	-	Turbid	-	-	-	-
3	Temperature	oC	-	-	-	27	-	-	-	-	-	-	-	-	27	27	27
4	Turbidity	NTU	-	-	-	-	-	-	-	-	-	-	22	-	22	22	22
5	Colour	PCS	-	-	-	CL	-	-	-	-	-	-	Green	-	-	-	-
6	Odour	T. No	-	-	-	OL	-	-	-	-	-	-	Unple	-	-	-	-
7	pH	pH Unit	-	-	-	8.5	-	-	-	-	-	-	7.2	-	7.85	8.5	7.2
8	Sp. Conductivity	µMhos/cm.	-	-	-	210	-	-	-	-	-	-	650	-	430	650	210
9	T. Solids	mg/1	-	-	-	266	-	-	-	-	-	-	564	-	415	564	266
10	D. Solids	mg/1	-	-	-	182	-	-	-	-	-	-	512	-	347	512	182
11	S. Solids	mg/1	-	-	-	84	-	-	-	-	-	-	52	-	68	84	52
12	Amm. Nitrogen	mg/1	-	-	-	0.512	-	-	-	-	-	-	0.8	-	0.656	0.8	0.512
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	0.45	-	0.45	0.45	0.45
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	0.99	-	0.99	0.99	0.99
15	Phosphate (PO_4)	mg/1	-	-	-	0.99	-	-	-	-	-	-	1.21	-	1.10	1.21	0.99
16	Chloride	mg/1	-	-	-	26.99	-	-	-	-	-	-	48.99	-	37.99	48.99	26.99
17	Sulphate (SO_4)	mg/1	-	-	-	11	-	-	-	-	-	-	39.3	-	25.15	39.30	11
18	T. Alkalinity	mg/1	-	-	-	140	-	-	-	-	-	-	62	-	101	140	62
19	T. Hardness	mg/1	-	-	-	130	-	-	-	-	-	-	172	-	151	172	130
20	CalciumHardness	mg/1	-	-	-	120	-	-	-	-	-	-	122	-	121	122	120
21	Magnesium H.	mg/1	-	-	-	30	-	-	-	-	-	-	50	-	40	50	30
22	D. Oxygen	mg/1	-	-	-	6.4	-	-	-	-	-	-	12.2	-	9.3	12.2	6.4
23	B.O.D.	mg/1	-	-	-	2.2	-	-	-	-	-	-	0.8	-	1.5	2.2	0.8
24	C.O.D.	mg/1	-	-	-	48.4	-	-	-	-	-	-	76.8	-	62.6	76.8	48.4
25	Sodium	mg/1	-	-	-	33	-	-	-	-	-	-	44	-	38.5	44	33
26	Potassium	mg/1	-	-	-	22	-	-	-	-	-	-	23	-	22.5	23	22
27	T. Coliform	MPN/100ml	-	-	-	1600	-	-	-	-	-	-	1600	-	1600	1600	1600
28	F. Coliform	MPN/100ml	-	-	-	70	-	-	-	-	-	-	11	-	40.5	70	11
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2011-2012

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		2.4.11	-	-	1.7.11	-	-	1.10.11	-	-	10.1.12	3.2.12	-	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	27	-	-	2	-	-	14	-	-	53	7	-	20.60	53	2
5	Colour	PCS	CL	-	-	CL	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	8.38	-	-	8.39	-	-	8.15	-	-	7.87	7.59	-	8.08	8.39	7.59
8	Sp. Conductivity	µMhos/cm.	260	-	-	290	-	-	251	-	-	295	476	-	314.40	476	251
9	T. Solids	mg/1	200	-	-	214	-	-	192	-	-	194	397	-	239.40	397	192
10	D. Solids	mg/1	180	-	-	200	-	-	136	-	-	146	352	-	202.80	352	136
11	S. Solids	mg/1	20	-	-	14	-	-	56	-	-	48	45	-	36.60	56	14
12	Amm. Nitrogen	mg/1	0.022	-	-	0.15	-	-	0.18	-	-	0.11	0.0396	-	0.10	0.18	0.022
13	Nitrite Nitrogen	mg/1	0.051	-	-	0.334	-	-	0.3	-	-	0.36	1.56	-	0.52	1.56	0.051
14	Nitrate Nitrogen	mg/1	3.025	-	-	7.24	-	-	1.3	-	-	BDL	14.3	-	6.47	14.30	1.3
15	Phosphate (PO_4)	mg/1	0.01	-	-	BDL	-	-	0.95	-	-	1.18	BDL	-	0.71	1.18	0.01
16	Chloride	mg/1	23.99	-	-	19.98	-	-	19.99	-	-	25.6	36.37	-	25.19	36.37	19.98
17	Sulphate (SO_4)	mg/1	3.75	-	-	27.75	-	-	13.5	-	-	12	21.5	-	15.70	27.75	3.75
18	T. Alkalinity	mg/1	108	-	-	102	-	-	118	-	-	146	220	-	138.80	220	102
19	T. Hardness	mg/1	92	-	-	120	-	-	112	-	-	106	190	-	124.00	190	92
20	CalciumHardness	mg/1	46	-	-	90	-	-	80	-	-	64	136	-	83.20	136	46
21	Magnesium H.	mg/1	46	-	-	30	-	-	32	-	-	42	54	-	40.80	54	30
22	D. Oxygen	mg/1	6.7	-	-	7.1	-	-	6.3	-	-	11.6	13.1	-	8.96	13.1	6.3
23	B.O.D.	mg/1	4.5	-	-	4.4	-	-	1.7	-	-	3.4	4.9	-	3.78	4.9	1.7
24	C.O.D.	mg/1	35.2	-	-	36.8	-	-	19.76	-	-	19.2	39.2	-	30.03	39.2	19.2
25	Sodium	mg/1	-	-	-	8.5	-	-	15	-	-	-	8.75	-	10.75	15	8.5
26	Potassium	mg/1	-	-	-	2.5	-	-	0.625	-	-	-	6.75	-	3.29	6.75	0.625
27	T. Coliform	MPN/100ml	1600	-	-	1600	-	-	60	-	-	40	86	-	677.2	1600	40
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	7	-	-	7	7	7
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2011-2012

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		2.4.11	-	-	1.7.11	-	-	1.10.11	-	-	10.1.12	3.2.12	5.3.12	-	-	-
2	Appearance		Clear	-	-	Clear	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	20	-	-	20	-	-	16	-	-	16	8	25	17.5	25	8
5	Colour	PCS	CL	-	-	CL	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	8.48	-	-	7.93	-	-	7.89	-	-	7.86	7.77	7.93	8.0	8.48	7.77
8	Sp. Conductivity	µMhos/cm.	268	-	-	510	-	-	258	-	-	246	472	231	330.8	510	231
9	T. Solids	mg/1	218	-	-	470	-	-	200	-	-	232	402	262	297.3	470	200
10	D. Solids	mg/1	190	-	-	350	-	-	142	-	-	188	360	202	238.7	360	142
11	S. Solids	mg/1	28	-	-	120	-	-	58	-	-	44	42	60	58.7	120	28
12	Amm. Nitrogen	mg/1	0.008	-	-	0.154	-	-	0.16	-	-	-	1.254	0.0594	0.3	1.254	0.008
13	Nitrite Nitrogen	mg/1	1.05	-	-	0.74	-	-	0.2	-	-	0.1	1.6	0.05	0.6	1.60	0.05
14	Nitrate Nitrogen	mg/1	3.837	-	-	6.15	-	-	1	-	-	0.7	16.1	0.1	4.6	16.10	0.1
15	Phosphate (PO_4)	mg/1	0.075	-	-	0.06	-	-	0.75	-	-	0.792	BDL	0.396	0.4	0.792	0.06
16	Chloride	mg/1	24.99	-	-	47.95	-	-	23.99	-	-	13.49	34.45	23.92	28.1	47.95	13.49
17	Sulphate (SO_4)	mg/1	1.75	-	-	58	-	-	12.5	-	-	8.75	23.5	4.37	18.1	58.00	1.75
18	T. Alkalinity	mg/1	110	-	-	138	-	-	106	-	-	120	212	114	133.3	212	106
19	T. Hardness	mg/1	88	-	-	180	-	-	110	-	-	140	208	124	141.7	208	88
20	Cal.Hardness	mg/1	46	-	-	92	-	-	74	-	-	112	138	64	87.7	138	46
21	Magnesium H.	mg/1	42	-	-	88	-	-	36	-	-	28	70	60	54.0	88	28
22	D. Oxygen	mg/1	10.5	-	-	9	-	-	6.8	-	-	7.3	-	9.8	8.7	10.5	6.8
23	B.O.D.	mg/1	4.2	-	-	4.6	-	-	1.9	-	-	1.5	5.5	2	3.3	5.5	1.5
24	C.O.D.	mg/1	35.2	-	-	55.2	-	-	19.76	-	-	19.2	-	18.4	29.6	55.2	18.4
25	Sodium	mg/1	-	-	-	16	-	-	10	-	-	5.37	9.125	8.12	9.7	16	5.37
26	Potassium	mg/1	-	-	-	6.25	-	-	0.625	-	-	2.5	6.75	2.75	3.8	6.75	0.625
27	T. Coliform	MPN/100ml	170	-	-	1600	-	-	90	-	-	26	21	-	381.4	1600	21
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	2	-	-	2.0	2	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2011-2012

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.11	-	-	1.7.11	-	-	-	-	-	10.1.12	3.2.12	5.3.12	-	-	-
2	Appearance		Clear	-	-	S.Turbid	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	12	-	-	22	-	-	-	-	-	10	4	30	15.60	30	4
5	Colour	PCS	CL	-	-	CL	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	7.72	-	-	7.53	-	-	-	-	-	7.86	7.7	7.86	7.73	7.86	7.53
8	Sp. Conductivity	µMhos/cm.	290	-	-	410	-	-	-	-	-	246	259	233	287.6	410	233
9	T. Solids	mg/1	204	-	-	276	-	-	-	-	-	236	262	264	248.4	276	204
10	D. Solids	mg/1	182	-	-	260	-	-	-	-	-	190	246	200	215.6	260	182
11	S. Solids	mg/1	21	-	-	26	-	-	-	-	-	46	16	64	34.6	64	16
12	Amm. Nitrogen	mg/1	Nil	-	-	0.083	-	-	-	-	-	-	-	0.112	0.10	0.112	0.083
13	Nitrite Nitrogen	mg/1	-	-	-	0.668	-	-	-	-	-	0.06	105	0.11	26.46	105	0.06
14	Nitrate Nitrogen	mg/1	0.054	-	-	8.4	-	-	-	-	-	1.84	0.3	0.425	2.20	8.40	0.054
15	Phosphate (PO_4)	mg/1	0.28	-	-	0.39	-	-	-	-	-	363	0.396	0.17	72.85	363	0.17
16	Chloride	mg/1	22.63	-	-	41.95	-	-	-	-	-	15.49	11.31	23.92	23.06	41.95	11.31
17	Sulphate (SO_4)	mg/1	5.6	-	-	10.5	-	-	-	-	-	8.75	9.5	7.5	8.37	10.50	5.6
18	T. Alkalinity	mg/1	180	-	-	206	-	-	-	-	-	122	142	114	152.8	206	114
19	T. Hardness	mg/1	150	-	-	176	-	-	-	-	-	148	140	130	148.8	176	130
20	Cal.Hardness	mg/1	44	-	-	112	-	-	-	-	-	118	90	80	88.8	118	44
21	Magnesium H.	mg/1	106	-	-	64	-	-	-	-	-	30	50	50	60	106	30
22	D. Oxygen	mg/1	4.8	-	-	3.4	-	-	-	-	-	10.5	8	10.5	7.44	10.5	3.4
23	B.O.D.	mg/1	9	-	-	1.6	-	-	-	-	-	1.6	1.7	2	3.18	9	1.6
24	C.O.D.	mg/1	36	-	-	27.6	-	-	-	-	-	19.2	19.26	27.6	25.932	36	19.2
25	Sodium	mg/1	-	-	-	10.5	-	-	-	-	-	10	8.12	8.12	9.185	10.5	8.12
26	Potassium	mg/1	-	-	-	5.25	-	-	-	-	-	2	1.38	2.12	2.6875	5.25	1.38
27	T. Coliform	MPN/100ml	17	-	-	1600	-	-	-	-	-	33	50	26	345.2	1600	17
28	F. Coliform	MPN/100ml	2	-	-	-	-	-	-	-	-	2	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2011-2012

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	-	6.8.11	-	-	8.11.11	-	-	3.2.12	-	-	-	-
2	Appearance		-	-	-	-	Sl.Turbid	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	-	-	28	-	-	3	-	-	8.14	-	13.05	28	3
5	Colour	PCS	-	-	-	-	Sl.muddy	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	OL	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	-	7.94	-	-	7.99	-	-	7.61	-	7.85	7.99	7.61
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	510	-	-	306	-	-	-	-	408	510	306
9	T. Solids	mg/1	-	-	-	-	360	-	-	192	-	-	198	-	250	360	192
10	D. Solids	mg/1	-	-	-	-	284	-	-	174	-	-	181	-	213	284	174
11	S. Solids	mg/1	-	-	-	-	76	-	-	18	-	-	17	-	37	76	17
12	Amm. Nitrogen	mg/1	-	-	-	-	0.14	-	-	BDL	-	-	0.2046	-	0.17	0.2046	0.14
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	-	BDL	-	-	0.02	-	0.02	0.02	0.02
14	Nitrate Nitrogen	mg/1	-	-	-	-	1.5	-	-	0.6	-	-	1.6	-	1.23	1.60	0.6
15	Phosphate (PO_4)	mg/1	-	-	-	-	0.95	-	-	0.33	-	-	BDL	-	0.64	0.95	0.33
16	Chloride	mg/1	-	-	-	-	22.99	-	-	9.4	-	-	13.4	-	15.3	22.99	9.4
17	Sulphate (SO_4)	mg/1	-	-	-	-	14.5	-	-	11.75	-	-	11.75	-	12.7	14.50	11.75
18	T. Alkalinity	mg/1	-	-	-	-	214	-	-	114	-	-	122	-	150.0	214	114
19	T. Hardness	mg/1	-	-	-	-	176	-	-	62	-	-	120	-	119.3	176	62
20	CalciumHardness	mg/1	-	-	-	-	132	-	-	38	-	-	82	-	84.0	132	38
21	Magnesium H.	mg/1	-	-	-	-	44	-	-	24	-	-	38	-	35.3	44	24
22	D. Oxygen	mg/1	-	-	-	-	5.7	-	-	6.1	-	-	9.3	-	7.0	9.3	5.7
23	B.O.D.	mg/1	-	-	-	-	2.7	-	-	1.2	-	-	1.7	-	1.9	2.7	1.2
24	C.O.D.	mg/1	-	-	-	-	27.24	-	-	21.36	-	-	9.8	-	19.5	27.24	9.8
25	Sodium	mg/1	-	-	-	-	-	-	-	-	-	-	4	-	4.0	4	4
26	Potassium	mg/1	-	-	-	-	-	-	-	-	-	-	2.25	-	2.25	2.25	2.25
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	28	-	28	28	28
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. A																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2011-2012

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		2.4.11	-	-	1.7.11	-	-	1.10.11	-	-	10.1.12	3.2.12	5.3.12	-	-	-
2	Appearance		Clear	-	-	Sl.Turbid	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	33	-	-	12	-	-	13	-	-	46	9	14	21.17	46	9
5	Colour	PCS	CL	-	-	CL	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	OL	-	-	OL	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	8.23	-	-	8.49	-	-	7.75	-	-	7.92	8	8.02	8.07	8.49	7.75
8	Sp. Conductivity	µMhos/cm.	264	-	-	280	-	-	276	-	-	261	470	228	296.5	470	228
9	T. Solids	mg/1	205	-	-	311	-	-	206	-	-	150	396	236	250.67	396	150
10	D. Solids	mg/1	180	-	-	121	-	-	154	-	-	102	358	200	185.83	358	102
11	S. Solids	mg/1	25	-	-	190	-	-	52	-	-	48	38	36	64.83	190	25
12	Amm. Nitrogen	mg/1	0.023	-	-	0.218	-	-	0.2	-	-	0.19	BDL	0.0594	0.138	0.218	0.023
13	Nitrite Nitrogen	mg/1	0.33	-	-	0.085	-	-	0.25	-	-	BDL	0.4	0.055	0.224	0.4	0.055
14	Nitrate Nitrogen	mg/1	3.623	-	-	4.92	-	-	1.2	-	-	BDL	14.5	0.25	4.899	14.5	0.25
15	Phosphate (PO_4)	mg/1	0.25	-	-	BDL	-	-	0.85	-	-	1.22	BDL	-	0.773	1.22	0.25
16	Chloride	mg/1	23.99	-	-	5.99	-	-	16.99	-	-	12.9	37.32	20.1	19.55	37.32	5.99
17	Sulphate (SO_4)	mg/1	4	-	-	20	-	-	10	-	-	10.5	24	4.27	12.13	24.00	4
18	T. Alkalinity	mg/1	104	-	-	104	-	-	122	-	-	162	208	128	138.00	208	104
19	T. Hardness	mg/1	96	-	-	110	-	-	124	-	-	62	190	124	117.67	190	62
20	CalciumHardness	mg/1	52	-	-	58	-	-	86	-	-	32	140	76	74	140	32
21	Magnesium H.	mg/1	44	-	-	52	-	-	38	-	-	30	50	48	43.67	52	30
22	D. Oxygen	mg/1	6.5	-	-	9.2	-	-	6.5	-	-	8.5	12.5	9.1	8.72	12.5	6.5
23	B.O.D.	mg/1	4.1	-	-	4.3	-	-	2.4	-	-	0.9	4.8	2	3.08	4.8	0.9
24	C.O.D.	mg/1	35.2	-	-	46	-	-	29.64	-	-	28.8	38.2	27.5	34.22	46	27.5
25	Sodium	mg/1	-	-	-	6	-	-	13	-	-	-	8.75	5	8.19	13	5
26	Potassium	mg/1	-	-	-	3.5	-	-	0.625	-	-	-	8.75	0.01	3.22	8.75	0.01
27	T. Coliform	MPN/100ml	70	-	-	900	-	-	110	-	-	50	120	26	212.67	900	26
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	6	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2011-2012

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	-	-	-	1.10.11	8.11.11	-	-	3.2.12	-	-	-	-
2	Appearance		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	-	-	-	-	13	2	-	-	4	-	6.3	13	2
5	Colour	PCS	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	-	-	-	7.75	7.95	-	-	7.96	-	7.9	7.96	7.75
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	-	-	329	314	-	-	253	-	298.7	329	253
9	T. Solids	mg/1	-	-	-	-	-	-	226	182	-	-	262	-	223.3	262	182
10	D. Solids	mg/1	-	-	-	-	-	-	174	136	-	-	244	-	184.7	244	136
11	S. Solids	mg/1	-	-	-	-	-	-	52	46	-	-	18	-	38.7	52	18
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.2	BDL	-	-	BDL	-	0.2	0.2	0.2
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.25	-	-	-	0.05	-	0.2	0.25	0.05
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	1.2	0.61	-	-	0.38	-	0.7	1.20	0.38
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.75	0.72	-	-	0.594	-	0.7	0.75	0.594
16	Chloride	mg/1	-	-	-	-	-	-	20.99	7.5	-	-	16.31	-	14.9	20.99	7.5
17	Sulphate (SO_4)	mg/1	-	-	-	-	-	-	14	14	-	-	6.25	-	11.4	14.00	6.25
18	T. Alkalinity	mg/1	-	-	-	-	-	-	138	116	-	-	140	-	131.3	140	116
19	T. Hardness	mg/1	-	-	-	-	-	-	140	60	-	-	170	-	123.3	170	60
20	CalciumHardness	mg/1	-	-	-	-	-	-	102	40	-	-	100	-	80.7	102	40
21	Magnesium H.	mg/1	-	-	-	-	-	-	38	20	-	-	70	-	42.7	70	20
22	D. Oxygen	mg/1	-	-	-	-	-	-	6.1	6.5	-	-	8.5	-	7.0	8.5	6.1
23	B.O.D.	mg/1	-	-	-	-	-	-	1.8	1.4	-	-	1.8	-	1.7	1.8	1.4
24	C.O.D.	mg/1	-	-	-	-	-	-	19.76	14.26	-	-	19.26	-	17.8	19.76	14.26
25	Sodium	mg/1	-	-	-	-	-	-	14	-	-	-	10.25	-	12.1	14	10.25
26	Potassium	mg/1	-	-	-	-	-	-	0.625	-	-	-	1.37	-	1.0	1.37	0.625
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	60	-	-	-	40	-	50	60	40
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-		-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. A																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2012-2013

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		27.4.12	-	-	18.7.12	13.8.12	3.9.12	-	7.11.12	-	15.1.13	-	5.3.13	-	-	-
2	Appearance		Sl.Turbid	-	-	-	Sl.Turbid	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	5	-	-	28	9	8	-	8	-	16	-	19	13.29	28	5
5	Colour	PCS	CL	-	-	-	Sl.Muddy	CL	-	Sl.Greenish	-	Sl.Greenish	-	Sl.Greenish	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	7.66	-	-	7.88	7.5	8.49	-	7.91	-	8.49	-	8.11	8.01	8.49	7.5
8	Sp. Conductivity	µMhos/cm.	282	-	-	309	310	236	-	-	-	263	-	284	280.67	310	236
9	T. Solids	mg/1	263	-	-	264	210	174	-	141	-	216	-	235	214.71	264	141
10	D. Solids	mg/1	250	-	-	178	180	148	-	130	-	178	-	213	182.43	250	130
11	S. Solids	mg/1	13	-	-	86	30	26	-	14	-	38	-	22	32.71	86	13
12	Amm. Nitrogen	mg/1	0.03	-	-	0.4	0.475	0.45	-	ND	-	0.055	-	0.055	0.24	0.475	0.03
13	Nitrite Nitrogen	mg/1	0.009	-	-	0.3	0.5	0.4	-	0.63	-	0.144	-	0.054	0.29	0.63	0.009
14	Nitrate Nitrogen	mg/1	0.111	-	-	1.5	1.8	1.6	-	ND	-	0.45	-	2.1	1.26	2.1	0.111
15	Phosphate (PO_4)	mg/1	1.51	-	-	1.15	1.25	1.2	-	ND	-	0.24	-	BDL	1.07	1.51	0.24
16	Chloride	mg/1	21.05	-	-	34.98	20.7	17.99	-	20.9	-	14.994	-	33.74	23.48	34.98	14.994
17	Sulphate (SO_4)	mg/1	4.3	-	-	14	18.4	17	-	6.6	-	4.5	-	7.5	10.33	18.4	4.3
18	T. Alkalinity	mg/1	104	-	-	154	154	120	-	124	-	152	-	122	132.86	154	104
19	T. Hardness	mg/1	100	-	-	120	112	100	-	118	-	128	-	146	117.71	146	100
20	CalciumHardness	mg/1	68	-	-	44	82	66	-	78	-	84	-	102	74.86	102	44
21	Magnesium H.	mg/1	32	-	-	76	30	34	-	40	-	44	-	44	42.86	76	30
22	D. Oxygen	mg/1	6	-	-	5.1	5.3	6.7	-	6.6	-	11.8	-	7.6	7.01	11.8	5.1
23	B.O.D.	mg/1	2.4	-	-	4	2.5	2.2	-	4.4	-	7.2	-	1.7	3.49	7.2	1.7
24	C.O.D.	mg/1	9.68	-	-	36.64	28.56	29.4	-	38.4	-	40	-	10	27.53	40	9.68
25	Sodium	mg/1	na	-	-	17	17.5	17.5	-	4.6	-	-	-	-	14.15	17.5	4.6
26	Potassium	mg/1	na	-	-	20	6.5	6.5	-	2.2	-	-	-	-	8.80	20	2.2
27	T. Coliform	MPN/100ml	13	-	-	140	170	140	-	130	-	170	-	80	120.43	170	13
28	F. Coliform	MPN/100ml	0	-	-	-	6	6	-	Nil	-	12	-	2	5.20	12	0
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2012-2013

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		27.4.12	-	-	18.7.12	13.8.12	3.9.12	-	8.11.12	-	15.1.13	-	5.3.13	-	-	-
2	Appearance		Sl.Turbid	-	-	-	Sl.Turbid	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	7	-	-	26	8	14	-	4	-	14	-	6	11.29	26	4
5	Colour	PCS	CL	-	-	-	Sl.muddy	Sl.Greenis	-	CL	-	CL	-	Sl.Greenis	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	7.75	-	-	8.09	7.18	8.38	-	7.66	-	8.49	-	8.13	7.95	8.49	7.18
8	Sp. Conductivity	µMhos/cm.	286	-	-	301	323	232	-	275	-	259	-	295	281.57	323	232
9	T. Solids	mg/1	348	-	-	262	224	182	-	122	-	203	-	252	227.57	348	122
10	D. Solids	mg/1	325	-	-	180	196	140	-	107	-	182	-	226	193.71	325	107
11	S. Solids	mg/1	23	-	-	82	28	42	-	15	-	21	-	26	33.86	82	15
12	Amm. Nitrogen	mg/1	0.07	-	-	0.33	0.65	0.4	-	0.1	-	0.11	-	0.05	0.24	0.65	0.05
13	Nitrite Nitrogen	mg/1	0.025	-	-	0.45	0.55	0.45	-	0.11	-	0.1	-	ND	0.28	0.55	0.025
14	Nitrate Nitrogen	mg/1	0.163	-	-	1.25	1.9	1.7	-	0.38	-	0.4	-	1.5	1.04	1.90	0.163
15	Phosphate (PO_4)	mg/1	1.53	-	-	0.924	1.4	1	-	ND	-	0.32	-	BDL	1.03	1.53	0.32
16	Chloride	mg/1	25.84	-	-	33.98	25.21	21.99	-	21.99	-	15.996	-	38.56	26.224	38.56	15.996
17	Sulphate (SO_4)	mg/1	6.2	-	-	13.75	20	16	-	6.75	-	2.5	-	8.75	10.564	20.00	2.5
18	T. Alkalinity	mg/1	120	-	-	148	174	126	-	126	-	150	-	114	136.857	174	114
19	T. Hardness	mg/1	136	-	-	102	120	102	-	92	-	150	-	112	116.286	150	92
20	Cal.Hardness	mg/1	80	-	-	40	90	58	-	64	-	90	-	82	72	90	40
21	Magnesium H.	mg/1	56	-	-	62	30	44	-	28	-	60	-	30	44.286	62	28
22	D. Oxygen	mg/1	8.9	-	-	5.2	5	6.8	-	7	-	11.8	-	8.1	7.543	11.8	5
23	B.O.D.	mg/1	7.3	-	-	2.8	4.1	2.5	-	1.8	-	7.3	-	1.4	3.886	7.3	1.4
24	C.O.D.	mg/1	29.04	-	-	18.32	47.6	29.4	-	9.6	-	30	-	10	24.851	47.6	9.6
25	Sodium	mg/1	na	-	-	18.37	25	16.25	-	16.5	-	-	-		19.03	25	16.25
26	Potassium	mg/1	na	-	-	20	10	6.5	-	2.2	-	-	-	-	9.675	20	2.2
27	T. Coliform	MPN/100ml	33	-	-	80	220	130	-	80	-	110	-	110	109	220	33
28	F. Coliform	MPN/100ml	6	-	-	-	8	6	-	Nil	-	12	-	2	6.8	12	2
29	T.K.N.	mg/1	na	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. D																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2012-2013

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	18.7.12	13.8.12	3.9.12	-	8.11.12	-	15.1.13	-	5.3.13	-	-	-
2	Appearance		-	-	-	-	Sl.Turbid	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	-	30	7	7	-	5	-	18	-	19	14.33	30	5
5	Colour	PCS	-	-	-	-	Sl. Muddy	Sl.Greenis	-	CL	-	Sl.Greenish	-	Sl.Greenis	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	7.92	7.35	8.42	-	7.74	-	8.5	-	8.18	8.02	8.5	7.35
8	Sp. Conductivity	µMhos/cm.	-	-	-	291	282	229	-	285	-	271	-	284	273.67	291	229
9	T. Solids	mg/1	-	-	-	260	192	156	-	131	-	180	-	234	192.17	260	131
10	D. Solids	mg/1	-	-	-	170	168	136	-	110	-	175	-	216	162.5	216	110
11	S. Solids	mg/1	-	-	-	90	24	20	-	21	-	5	-	18	29.67	90	5
12	Amm. Nitrogen	mg/1	-	-	-	0.528	0.4	0.35	-	0.11	-	0.11	-	0.095	0.2655	0.528	0.095
13	Nitrite Nitrogen	mg/1	-	-	-	0.55	0.45	0.3	-	0.18	-	0.0396	-	2.6	0.69	2.6	0.0396
14	Nitrate Nitrogen	mg/1	-	-	-	1.7	1.5	1.5	-	0.94	-	0.3	-	ND	1.19	1.70	0.3
15	Phosphate (PO_4)	mg/1	-	-	-	1	1	0.8	-	ND	-	0.23	-	BDL	0.76	1	0.23
16	Chloride	mg/1	-	-	-	25.99	22.51	16.99	-	32.98	-	9.996	-	53.03	26.916	53.03	9.996
17	Sulphate (SO_4)	mg/1	-	-	-	17	15	14.4	-	7.75	-	2	-	10	11.025	17.00	2
18	T. Alkalinity	mg/1	-	-	-	168	148	132	-	96	-	152	-	112	134.67	168	96
19	T. Hardness	mg/1	-	-	-	122	110	100	-	84	-	150	-	128	115.67	150	84
20	Cal.Hardness	mg/1	-	-	-	38	74	62	-	76	-	94	-	98	73.67	98	38
21	Magnesium H.	mg/1	-	-	-	84	36	38	-	8	-	56	-	30	42	84	8
22	D. Oxygen	mg/1	-	-	-	5.5	6.5	7.1	-	6.5	-	11.4	-	7.2	7.367	11.4	5.5
23	B.O.D.	mg/1	-	-	-	4.1	2.7	1.9	-	7.2	-	6.9	-	1.3	4.017	7.2	1.3
24	C.O.D.	mg/1	-	-	-	45.8	28.56	19.6	-	48	-	60	-	10	35.327	60	10
25	Sodium	mg/1	-	-	-	15	15	15	-	9.75	-	-	-		13.688	15	9.75
26	Potassium	mg/1	-	-	-	16.75	6.5	3.125	-	2	-	-	-	-	7.094	16.75	2
27	T. Coliform	MPN/100ml	-	-	-	130	110	110	-	110	-	110	-	80	108.333	130	80
28	F. Coliform	MPN/100ml	-	-	-	-	6	6	-	2	-	6	-	2	4.4	6	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2012-2013

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	-	16.8.12	-	-	8.11.12	-	-	-	-	-	-	-
2	Appearance		-	-	-	-	Sl.Turbid	-	-	Sl.Turbid	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	-	-	12	-	-	5	-	-	-	-	8.50	12	5
5	Colour	PCS	-	-	-	-	CL	-	-	CL	-	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	-	7.32	-	-	7.68	-	-	-	-	7.50	7.68	7.32
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	349	-	-	-	-	-	-	-	349	349	349
9	T. Solids	mg/1	-	-	-	-	250	-	-	195	-	-	-	-	222.5	250	195
10	D. Solids	mg/1	-	-	-	-	168	-	-	170	-	-	-	-	169	170	168
11	S. Solids	mg/1	-	-	-	-	82	-	-	25	-	-	-	-	53.5	82	25
12	Amm. Nitrogen	mg/1	-	-	-	-	1.2	-	-	0.4	-	-	-	-	0.8	1.2	0.4
13	Nitrite Nitrogen	mg/1	-	-	-	-	0.04	-	-	0.58	-	-	-	-	0.31	0.58	0.04
14	Nitrate Nitrogen	mg/1	-	-	-	-	0.42	-	-	ND	-	-	-	-	0.42	0.42	0.42
15	Phosphate (PO_4)	mg/1	-	-	-	-	0.02	-	-	2.1	-	-	-	-	1.06	2.1	0.02
16	Chloride	mg/1	-	-	-	-	36.98	-	-	18.99	-	-	-	-	27.985	36.98	18.99
17	Sulphate (SO_4)	mg/1	-	-	-	-	18.6	-	-	0.21	-	-	-	-	9.405	18.60	0.21
18	T. Alkalinity	mg/1	-	-	-	-	194	-	-	48	-	-	-	-	121	194	48
19	T. Hardness	mg/1	-	-	-	-	144	-	-	102	-	-	-	-	123	144	102
20	CalciumHardness	mg/1	-	-	-	-	128	-	-	82	-	-	-	-	105	128	82
21	Magnesium H.	mg/1	-	-	-	-	16	-	-	20	-	-	-	-	18	20	16
22	D. Oxygen	mg/1	-	-	-	-	4.7	-	-	5.9	-	-	-	-	5.3	5.9	4.7
23	B.O.D.	mg/1	-	-	-	-	15	-	-	3.6	-	-	-	-	9.3	15	3.6
24	C.O.D.	mg/1	-	-	-	-	107.8	-	-	19.6	-	-	-	-	63.7	107.8	19.6
25	Sodium	mg/1	-	-	-	-	22.5	-	-	6.6	-	-	-	-	14.55	22.5	6.6
26	Potassium	mg/1	-	-	-	-	10	-	-	2	-	-	-	-	6	10	2
27	T. Coliform	MPN/100ml	-	-	-	-	140	-	-	130	-	-	-	-	135	140	130
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	2	-	-	-	-	2	2	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. D																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2012-2013

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		27.4.12	-	-	18.7.12	13.8.12	3.9.12	-	8.11.12	-	15.1.13	-	5.3.13	-	-	-
2	Appearance		Sl.Turbid	-	-	-	Sl.Turbid	Sl.Turbid	-	Sl.Turbid	-	Sl.Turbid	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	8	-	-	13	6	8	-	6	-	13	-	5	8.429	13	5
5	Colour	PCS	CL	-	-	-	Sl.Muddy	Sl.Muddy	-	-	-	CL	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	7.81	-	-	7.73	7.43	8.47	-	7.96	-	8.5	-	8.16	8.0	8.5	7.43
8	Sp. Conductivity	µMhos/cm.	287	-	-	297	306	229	-	267	-	260	-	296	277.4	306	229
9	T. Solids	mg/1	368	-	-	218	196	158	-	172	-	182	-	250	220.6	368	158
10	D. Solids	mg/1	339	-	-	174	176	134	-	160	-	159	-	222	194.9	339	134
11	S. Solids	mg/1	29	-	-	44	20	24	-	12	-	23	-	28	25.7	44	12
12	Amm. Nitrogen	mg/1	0.05	-	-	0.36	0.35	0.3	-	ND	-	0.055	-	0.075	0.2	0.36	0.05
13	Nitrite Nitrogen	mg/1	0.02	-	-	0.4	0.4	0.35	-	0.34	-	0.136	-	-	0.3	0.4	0.02
14	Nitrate Nitrogen	mg/1	0.155	-	-	1.6	1.7	1.3	-	ND	-	0.0029	-	2.2	1.2	2.2	0.0029
15	Phosphate (PO_4)	mg/1	1.54	-	-	1.2	1.2	0.95	-	7.96	-	0.27	-	BDL	2.2	7.96	0.27
16	Chloride	mg/1	26.8	-	-	23.99	18	15.99	-	14.6	-	15.995	-	43.39	22.7	43.39	14.6
17	Sulphate (SO_4)	mg/1	6.5	-	-	15	17	14	-	9.8	-	3.75	-	4.5	10.1	17	3.75
18	T. Alkalinity	mg/1	114	-	-	158	146	118	-	108	-	144	-	116	129.1	158	108
19	T. Hardness	mg/1	124	-	-	100	106	96	-	124	-	130	-	184	123.4	184	96
20	CalciumHardness	mg/1	70	-	-	64	80	60	-	82	-	90	-	152	85.4	152	60
21	Magnesium H.	mg/1	54	-	-	36	26	36	-	42	-	40	-	32	38.000	54	26
22	D. Oxygen	mg/1	7.4	-	-	5.3	5.5	7.3	-	6.8	-	11.6	-	8.1	7.429	11.6	5.3
23	B.O.D.	mg/1	4.8	-	-	2.7	2.3	1.7	-	2.4	-	6.9	-	1.2	3.143	6.9	1.2
24	C.O.D.	mg/1	19.36	-	-	27.48	28.56	19.6	-	19.2	-	30	-	10	22.029	30	10
25	Sodium	mg/1	na	-	-	12.37	16.25	11.25	-	7.2	-	-	-	-	11.768	16.25	7.2
26	Potassium	mg/1	na	-	-	16.75	6.5	3.125	-	1	-	-	-	-	6.844	16.75	1
27	T. Coliform	MPN/100ml	27	-	-	110	140	90	-	70	-	130	-	70	91	140	27
28	F. Coliform	MPN/100ml	4	-	-	-	6	4	-	-	-	9	-	2	5	9	2
29	T.K.N.	mg/1	na	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2012-2013

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		-	-	-	-	16.8.12	14.9.12	-	7.11.12	-	-	-	-	-	-	-
2	Appearance		-	-	-	-	Sl.Turbid	Sl.Turbid	-	Sl.Turbid	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	-	-	-	-	5	10	-	9	-	-	-	-	8	10	5
5	Colour	PCS	-	-	-	-	Sl.Muddy	CL	-	CL	-	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	-	7.5	7.72	-	7.49	-	-	-	-	7.57	7.72	7.49
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	838	738	-	366	-	-	-	-	647.33	838	366
9	T. Solids	mg/1	-	-	-	-	718	464	-	240	-	-	-	-	474	718	240
10	D. Solids	mg/1	-	-	-	-	676	320	-	220	-	-	-	-	405.33	676	220
11	S. Solids	mg/1	-	-	-	-	42	144	-	20	-	-	-	-	68.67	144	20
12	Amm. Nitrogen	mg/1	-	-	-	-	0.44	0.71	-	ND	-	-	-	-	0.575	0.71	0.44
13	Nitrite Nitrogen	mg/1	-	-	-	-	0.15	BDL	-	0.38	-	-	-	-	0.265	0.38	0.15
14	Nitrate Nitrogen	mg/1	-	-	-	-	0.49	0.6	-	0.52	-	-	-	-	0.537	0.6	0.49
15	Phosphate (PO_4)	mg/1	-	-	-	-	ND	0.65	-	ND	-	-	-	-	0.65	0.65	0.65
16	Chloride	mg/1	-	-	-	-	13.99	35.2	-	42.98	-	-	-	-	30.72	42.98	13.99
17	Sulphate (SO_4)	mg/1	-	-	-	-	216	164	-	10.75	-	-	-	-	130.25	216	10.75
18	T. Alkalinity	mg/1	-	-	-	-	176	130	-	158	-	-	-	-	154.67	176	130
19	T. Hardness	mg/1	-	-	-	-	400	258	-	180	-	-	-	-	279.33	400	180
20	CalciumHardness	mg/1	-	-	-	-	300	198	-	152	-	-	-	-	216.67	300	152
21	Magnesium H.	mg/1	-	-	-	-	100	60	-	28	-	-	-	-	62.67	100	28
22	D. Oxygen	mg/1	-	-	-	-	8.8	11	-	7.2	-	-	-	-	9	11	7.2
23	B.O.D.	mg/1	-	-	-	-	11	2.5	-	11	-	-	-	-	8.17	11	2.5
24	C.O.D.	mg/1	-	-	-	-	68.6	19.2	-	57.6	-	-	-	-	48.47	68.6	19.2
25	Sodium	mg/1	-	-	-	-	30	9.3	-	7.75	-	-	-	-	15.68	30	7.75
26	Potassium	mg/1	-	-	-	-	13.12	33.25	-	1.5	-	-	-	-	15.96	33.25	1.5
27	T. Coliform	MPN/100ml	-	-	-	-	300	900	-	80	-	-	-	-	426.67	900	80
28	F. Coliform	MPN/100ml	-	-	-	-	-	2	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. C																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2013-2014

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal) **XGN ID-22114**

Description of Sampling Station : Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Par	Unit	April(25-04-2013)	May(05-05-2013)	June(06-06-2013)	July(02-07-2013)	Aug(05-08-2013)	Sep()	Oct()	Nov()	Dec(09-12-2013)	Jan()	Feb(18-02-2014)	March()	Av.	Maxi.	Mini.
1	Acidity	mg/l	20	20	20	20	ND				30		30		23.33	30	20
2	Alkalinity	mg/l	118	140	140	130	168				128		146		138.57	168	118
3	Ammo..Nitrogen	mg/l	0.02	BDL	BDL	0.085	0.044				0.025		0.035		0.04	0.085	0.02
4	B.O.D	mg/l	0.8	1.4	1.5	1.7	2.8				1.3		2.5		1.71	2.8	0.8
5	Boron	mg/l	BDL	BDL	BDL	BDL	ND				BDL		BDL				
6	Calcium	mg/l	27.23	24.82	20.02	24	62.47				34.44		36.84		32.83	62.47	20.02
7	Calcium Hardness	mg/Lts	68	62	50	60	156				86		92		82.00	156	50
8	Chemical Oxygen	mg/l	9.88	19.04	28.2	20	not				19.76		20		19.48	28.2	9.88
9	Chloride	mg/l	18.99	18.85	21.52	19.99	16.44				11.48		14.46		17.39	21.52	11.48
10	Conductivity	micromho/cm	287	267	257	228	293				263		287		268.86	293	228
11	Dis.Oxygen	mg/l	6.7	8.6	7.2	7	2.8				7.6		7.2		6.729	8.6	2.8
12	Fecal Coliform	MPN/100 ml	4	2	2	2	ND				<2		2		2.4	4	2
13	Iron	mg/l	0.26												0.26	0.26	0.26
14	Magnesium	mg/l	13.11	18.45	13.11	4.878	11.65				8.74		16.51		12.35	18.45	4.878
15	Magnesium Hard.	Mg/Lts	54	76	54	20	48				36		68		50.86	76	20
16	Nitrate	mg/l	1.95	1.75	1.45	3	1.12				2.2		3.65		2.16	3.65	1.12
17	Nitrite	mg/l	0.005	0.01	0.025	0.015	0.045				BDL		0.03		0.0217	0.05	0.005
18	pH	pH Units	8.15	8.05	8.78	8.64	9.03				7.95		7.86		8.35	9.03	7.86
19	Phosphate	mg/l	0.099	0.231	BDL	0.165	0.18				0.033		BDL		0.1416	0.231	0.033
20	Potassium	mg/l	Instrument out of order				ND				3.06		4.13		3.595	4.13	3.06
21	Sodium	mg/l					ND				8.93		10.71		9.82	10.71	8.93
22	Sulphate	mg/l	9.75	2.75	5	7.75	9.5				7.75		6		6.929	9.75	2.75
23	Suspended Solids	mg/l	28	18	6	85	43				18		17		30.714	85	6
24	Total coliform	MPN/100 ml	90	110	90	90	ND				30		30		73.33	110	30
25	T. D. S.	mg/l	152	146	160	136	162				130		200		155.14	200	130
26	Total Hardn.	mg/l	122	138	104	80	204				122		160		132.86	204	80
27	Total Kjeldahl	mg/l	Instrument out of order				ND										
28	Total Solids	mg/l	180	164	166	221	205				148		217		185.86	221	148
29	Turbidity	N.T.U.	9	7	2	28	8				10		8		10.29	28	2
30	Zinc	mg/l	0.03												0.03	0.03	0.03
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2013-2014

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Par	Unit	April(25-04-2013)	May(05-05-2013)	June(06-06-2013)	July(02-07-2013)	Aug(05-08-2013)	Sep()	Oct()	Nov()	Dec(09-12-2013)	Jan()	Feb(18-02-2014)	March()	Av.	Maxi.	Mini.
1	Acidity	mg/l	20	30	20	30	ND				30		20		25.0	30	20
2	Alkalinity	mg/l	124	130	NIL	128	170				130		140		137.0	170	124
3	Ammo. Nitrogen	mg/l	BDL	BDL	BDL	0.09	0.045				0.115		0.035		0.071	0.115	0.035
4	B.O.D	mg/l	0.6	0.9	1.3	1.5	2.8				3.7		2.6		1.914	3.7	0.6
5	Boron	mg/l	BDL	BDL	BDL	BDL	ND				BDL		BDL				
6	Calcium	mg/l	29.63	24.82	23.22	27.2	63.27				35.24		47.256		35.805	63.27	23.22
7	Calcium Hard.	Mg/Lts	74	62	58	68	158				88		118		89.43	158	58
8	C. O. D.	mg/l	9.4	9.32	18.8	20	not working				19.76		29.64		17.82	29.64	9.32
9	Chloride	mg/l	19.99	17.99	21.52	26.99	18.5				11.48		21.21		19.67	26.99	11.48
10	Cobalt	mg/l	ND														
11	Conductivity	micromho/cm	291	297	254	281	294				269		280		280.86	297	254
12	Dissolved Oxygen	mg/l	5.8	7.8	7	4.7	8.8				6.7		7.5		6.900	8.8	4.7
13	Fecal Coliform	MPN/100 ml	Nil	<2	2	6	ND				<2		6		4.67	6	2
14	Magnesium	mg/l	12.14	10.66	9.71	7.804	13.11				6.79		4.371		9.23	13.11	4.371
15	Mg Hard..	Mg/Lts	50	44	40	32	54				28		18		38	54	18
16	Nitrate	mg/l	0.113	2	1.35	5.9	1.13				1.75		2.55		2.11	5.9	0.113
17	Nitrite	mg/l	0.137	BDL	BDL	0.095	0.046				0.055		0.025		0.072	0.137	0.025
18	pH	pH Units	8.3	8.25	8.78	7.72	9.05				7.6		7.93		8.233	9.05	7.6
19	Phosphate	mg/l	0.047	BDL	0.099	0.033	0. 20				0.066		1.22		0.293	1.22	0.033
20	Potassium	mg/l	Instrument out of order				ND				4.11		3.59		3.850	4.11	3.59
21	Sodium	mg/l					ND				15.61		21.9		18.76	21.9	15.61
22	Sulphate	mg/l	BDL	10	5	10.75	9.6				9.5		6.25		8.52	10.75	5
23	Susp. Solids	mg/l	15	11	18	55	49				14		38		28.57	55	11
24	Total coliform	MPN/100 ml	90	70	80	130	ND				33		110		85.50	130	33
25	T. D. S.	mg/l	208	156	146	187	165				138		160		165.71	208	138
26	Total Hard.	mg/l	128	106	98	100	212				116		136		128.00	212	98
27	Total Kjeldahl Nitrogen	mg/l	Instrument out of order				ND				Instrument is out of		Instrument is out of				
28	Total Solids	mg/l	223	167	164	242	214				152		198		194.29	242	152
29	Turbidity	N.T.U.	7	4	4	17	9				14		12		9.571	17	4
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2013-2014

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Par	Unit	April(25-04-2013)	May(05-05-2013)	June(06-06-2013)	July(02-07-2013)	Aug(05-08-2013)	Sep()	Oct()	Nov()	Dec(09-12-2013)	Jan()	Feb(18-02-2014)	March()	Av.	Maxi.	Mini.
1	Acidity as Caco3	mg/l	20	20	20	30	ND				40		20		25.00	40	20
2	Alkalinity	mg/l	134	124	126	130	164				130		146		136.29	164	124
3	Ammo.Nitrogen	mg/l	BDL	BDL	BDL	0.35	0.035				0.15		0.075		0.15	0.35	0.035
4	B.O.D	mg/l	2	1.1	1.4	1.8	2.3				4.1		1		1.96	4.1	1
5	Boron	mg/l	BDL	BDL	BDL	0.4	ND				BDL		BDL		0.40	0.4	0.4
6	Calcium	mg/l	28.83	23.22	20.02	36	57.66				34.44		32.038		33.17	57.66	20.02
7	Calcium Hardness	Mg/Lts	72	58	50	90	144				86		80		82.86	144	50
8	Chemical Oxygen	mg/l	18.8	9.32	28.2	20	not				29.64		9.88		19.31	29.64	9.32
9	Chloride	mg/l	19.99	18.99	20.54	22.99	15.42				11.48		16.39		17.97	22.99	11.48
10	Conductivity	micromho/cm	296	300	261	277	280				266		274		279.14	300	261
11	D . Oxygen	mg/l	3.8`	8.6	6.8	5.3	7.7				6.5		7.8		7.12	8.6	5.3
12	Fecal Coliform	MPN/100 ml	nil	<2	4	6	ND				<2		4		4.67	6	4
13	Magnesium	mg/l	14.57	11.17	11.65	8.78	16.02				9.71		13.598		12.21	16.02	8.78
14	Magnesium Hard.	Mg/Lts	60	46	48	36	66				40		56		50.29	66	36
15	Mangenese	mg/l	0.01												0.01	0.01	0.01
16	Nitrate	mg/l	1.35	2.25	1.4	4.2	0.83				2.4		2.2		2.09	4.2	0.83
17	Nitrite	mg/l	0.135	BDL	0.01	0.04	0.027				BDL		0.01		0.04	0.135	0.01
18	pH	pH Units	8.3	8.28	8.78	7.32	8.31				7.56		7.87		8.06	8.78	7.32
19	Phosphate	mg/l	0.297	BDL	0.825	0.33	0.13				0.066		0.462		0.35	0.825	0.066
20	Potassium	mg/l	Instrument out of order				ND				3.35		2.97		3.16	3.35	2.97
21	Sodium	mg/l	Instrument out of order				ND				10.46		17.53		14.00	17.53	10.46
22	Sulphate	mg/l	BDL	7.75	4	11.75	6.6				8.5		4.75		7.23	11.75	4
23	Suspended Solids	mg/l	14	9	18	57	35				22		25		25.71	57	9
24	Total coliform	MPN/100 ml	110	30	140	140	nd				50		90		93.33	140	30
25	T. D. S.	mg/l	154	157	147	172	148				134		147		151.29	172	134
26	Total Hard.	mg/l	132	104	98	126	210				126		136		133.14	210	98
27	Total Kjeldahl	mg/l	Instrument out of order				ND				Instrument		Instrument				
28	Total Solids	mg/l	168	166	165	229	183				156		172		177.00	229	156
29	Turbidity	N.T.U.	6	3	4	18	7				14		8		8.57	18	3
30	Zinc	mg/l	ND														
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2013-2014

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Par	Unit	April()	May()	June()	July()	Aug(05-08-2013)	Sep()	Oct()	Nov()	Dec()	Jan()	Feb(18-02-2014)	March()	Av.	Maxi.	Mini.
1	Acidity as Caco3	mg/l					ND						20		20.00	20	20
2	Alkalinity as Caco3	mg/l					261						160		210.50	261	160
3	Ammonical Nitrogen	mg/l					0.034						BDL		0.03	0.034	0.034
4	B.O.D (3 Days 27oC)	mg/l					2.2						1.8		2.00	2.2	1.8
5	Boron	mg/l					ND						BDL				
6	Calcium	mg/l					54.46						37.64		46.05	54.46	37.64
7	Calcium Hardness as CACO3	Mg/Lts					136						94		115	136	94
8	Chemical Oxygen Demand	mg/l					not working						10		10.00	10	10
9	Chloride	mg/l					13.36						12.53		12.95	13.36	12.53
10	Conductivity	micromho/cm					261						293		277.00	293	261
11	Dissolved Oxygen	mg/l					7.5						7.7		7.6	7.7	7.5
12	Magnesium	mg/l					13.11						16.99		15.05	16.99	13.11
13	Magnesium Hardness as	Mg/Lts					54						70		62.00	70	54
14	Nitrate	mg/l					0.74						3.3		2.02	3.3	0.74
15	Nitrite	mg/l					0.025						BDL		0.03	0.025	0.025
16	P-Alkanity	--					nd						NIL				
17	pH	pH Units					8.03						8.02		8.025	8.03	8.02
18	Phosphate	mg/l					0.12						BDL		0.12	0.12	0.12
19	Potassium	mg/l					ND						3.74		3.74	3.74	3.74
20	Sodium	mg/l					ND						10.51		10.51	10.51	10.51
21	Sulphate	mg/l					6.4						8		7.20	8	6.4
22	Suspended Solids	mg/l					23						18		20.50	23	18
23	Total Dissolved Solids	mg/l					140						177		158.50	177	140
24	Total Hardness as CaCo3	mg/l					190						164		177.00	190	164
25	Total Kjeldahl Nitrogen	mg/l					ND						INSTRUMENT				
26	Total Solids	mg/l					163						195		179.00	195	163
27	Turbidity	N.T.U.					5						14		9.50	14	5
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2013-2014

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal. **XGN ID-22025**

Description of Sampling Station :

S.No.	Par	Unit	April(25-04-2013)	May(05-05-2013)	June(06-06-2013)	July(02-07-2013)	Aug(05-08-2013)	Sep()	Oct()	Nov()	Dec(09-12-2013)	Jan()	Feb(18-02-2014)	March()	Av.	Maxi.	Mini.
1	Acidity	mg/l	10	30	20	20	ND				30		20		21.67	30	10
2	Alkalinity	mg/l	140	128	136	116	172				118		150		137.14	172	116
3	Ammo. Nitrogen	mg/l	BDL	BDL	BDL	0.1	0.046				BDL		0.025		0.06	0.1	0.025
4	B.O.D	mg/l	1.5	1.2	1.6	2.3	3				2.3		2		1.99	3	1.2
5	Boron	mg/l	BDL	BDL	BDL	BDL	ND				BDL		BDL				
6	Calcium	mg/l	25.63	25.63	20.02	21.6	64.07				32.03		36.04		32.15	64.07	20.02
7	Calcium Hard.	Mg/Lts	64	64	50	54	160				80		90		80.29	160	50
8	Chemical Oxygen Demand	mg/l	18.8	9.33	28.2	30	not working				9.88		20		19.37	30	9.33
9	Chloride	mg/l	17.99	18.99	21.52	24.99	19.53				11.48		15.42		18.56	24.99	11.48
10	Conductivity	micromho/c	293	291	265	231	299				246		280		272.14	299	231
11	D. Oxygen	mg/l	7.5	8.1	7.4	6.7	8.9				7		7.7		7.61	8.9	6.7
12	Fecal Coliform	MPN/100 ml	Nil	<2	4	4	ND				<2		4		4.00	4	4
13	Magnesium	mg/l	14.08	12.62	10.68	7.317	16.99				10.19		17.48		12.77	17.48	7.317
14	Mg Hard.	Mg/Lts	58	52	44	30	70				42		72		52.57	72	30
15	Mangenese	mg/l	0.02												0.02	0.02	0.02
16	Nitrate	mg/l	1.55	2.3	1.4	3.2	1.31				1.85		2.7		2.04	3.2	1.31
17	Nitrite	mg/l	0.217	BDL	0.03	0.025	0.047				0.15		BDL		0.09	0.217	0.025
18	P-Alkanity	--	NIl	NIL	NIL	NIL	nd				NIL		NIL				
19	pH	pH Units	8.32	8.28	8.61	8.49	9.09				7.94		7.87		8.37	9.09	7.87
20	Phosphate	mg/l	0.231	BDL	0.033	0.165	0.21				0.66		BDL		0.2598	0.66	0.033
21	Potassium	mg/l	Instrument out of order				ND				2.07		3.72		2.895	3.72	2.07
22	Sodium	mg/l	Instrument out of order				ND				7.72		9.89		8.805	9.89	7.72
23	Sulphate	mg/l	BDL	7.75	5.5	10	9.7				8.25		5.25		7.742	10	5.25
24	Suspended Solids	mg/l	10	9	6	76	52				14		18		26.429	76	6
25	Total coliform	MPN/100 ml	110	50	110	110	nd				50		50		80.000	110	50
26	T. D. S.	mg/l	168	160	168	140	168				158		181		163.29	181	140
27	Total Hardn.	mg/l	122	116	94	84	230				122		162		132.86	230	84
28	Total Solids	mg/l	178	169	174	216	220				172		199		189.71	220	169
29	Turbidity	N.T.U.	4	3	2	25	11				10		11		9.429	25	2
30	Zinc	mg/l	0.08												0.080	0.08	0.08
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2013-2014

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka,Bairagarh.

S.No.	Par	Unit	April(25-04-2013)	May()	June()	July()	Aug(05-08-2013)	Sep()	Oct()	Nov()	Dec()	Jan()	Feb(18-02-2014)	March()	Av.	Maxi.	Mini.
1	Acidity .	mg/l	10				ND						30		20	30	10
2	Alkalinity .	mg/l	124				174						138		145.3	174	124
3	Ammo.Nitrogen	mg/l	BDL				0.043						0.15		0.1	0.15	0.043
4	B.O.D.	mg/l	1.1				2.7						2.1		2.0	2.7	1.1
5	Boron	mg/l	BDL				ND						BDL				
6	Calcium	mg/l	28.83				60.07						32.839		40.6	60.07	28.83
7	Calcium Hardn.	Mg/Lts	72				150						82		101.3	150	72
8	Chemical Oxygen Demand	mg/l	18.8				not working						19.76		19.3	19.76	18.8
9	Chloride	mg/l	19.99				15.42						18.32		17.9	19.99	15.42
10	Conductivity	micromho/cm	295				288						276		286.3	295	276
11	Dissolved Oxygen	mg/l	5				8						7.6		6.9	8	5
12	Magnesium	mg/l	13.59				15.54						12.627		13.9	15.54	12.627
13	Magnesium Hardness as	Mg/Lts	56				64						52		57.3	64	52
14	Nitrate	mg/l	1.35				1.01						1.6		1.3	1.6	1.01
15	Nitrite	mg/l	0.12				0.042						0.04		0.1	0.12	0.04
16	P-Alkanity	--	nil				nd						Nil				
17	pH	pH Units	8.32				8.89						7.33		8.2	8.89	7.33
18	Phosphate	mg/l	BDL				0.16						0.495		0.3	0.495	0.16
19	Potassium	mg/l	instrument out of order				ND						3.17		3.2	3.17	3.17
20	Sodium	mg/l	instrument out of order				ND						19.47		19.5	19.47	19.47
21	Sulphate	mg/l	5.75				8.7						4.25		6.2	8.7	4.25
22	S. Solids	mg/l	12				40						31		27.7	40	12
23	T. D. S.	mg/l	184				150						143		159.0	184	143
24	Total Hardn.	mg/l	128				214						134		158.7	214	128
25	T.K.N.	mg/l	instrument out of order				ND						instrument is out of				
26	Total Solids	mg/l	196				190						174		186.7	196	174
27	Turbidity	N.T.U.	5				8						10		7.7	10	5
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2014-2015

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake). Cat- A

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.14	5.5.14	25.6.14	12.7.14	12.8.14	11.9.14	20.10.14	NR	20.12.14	13.1.15	10.2.15	17.3.15	-	-	-
2	Appearance		sl.tur.	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	22	17	19	-	19.33	22	17
4	Turbidity	NTU	5	18	7	20	20	7	6	-	14	10	8	14	11.73	20	5
5	Colour	PCS	-	-	-	-	-	-	-	-	CL	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	OL	-	-	-	-	-	-
7	pH	pH Unit	7.84	8.23	8.7	8.82	7.81	7.39	8.05	-	7.96	8.19	8.1	8.21	8.12	8.82	7.39
8	Sp. Conductivity	µMhos/cm.	289	255	241	435	264	-	290	-	272	268	263	330	290.70	435	241
9	T. Solids	mg/1	171	129	165	370	149	259	267	-	192	172	165	226	205.91	370	129
10	D. Solids	mg/1	158	109	152	301	132	244	251	-	164	148	143	194	181.45	301	109
11	S. Solids	mg/1	13	20	13	69	17	15	16	-	28	24	22	32	24.45	69	13
12	Amm. Nitrogen	mg/1	-	0.005	-	-	0.09	0.102	0.134	-	-	0.032	0.038	-	0.07	0.134	0.005
13	Nitrite Nitrogen	mg/1	-	-	0.125	0.05	0.06	0.052	0.011	-	0.115	BDL	BDL	0.09	0.07	0.125	0.011
14	Nitrate Nitrogen	mg/1	-	-	2.5	3.25	3.2	0.032	0.041	-	1.8	2.2	2.6	3.25	2.10	3.25	0.032
15	Phosphate (PO_4)	mg/1	-	0.165	0.099	0.75	BDL	0.265	0.181	-	1.023	0.68	0.52	0.099	0.42	1.023	0.099
16	Chloride	mg/1	13	17.99	15.76	43.07	25.42	11.99	14.99	-	19	20.1	19.14	17.35	19.80	43.07	11.99
17	Sulphate (SO_4)	mg/1	4.25	5.75	-	-	-	2.27	-	-	11.75	7.58	7.36	2.25	5.89	11.75	2.25
18	T. Alkalinity	mg/1	122	108	114	180	110		100	-	124	128	124	104	121.40	180	100
19	T. Hardness	mg/1	136	114	102	170	98	100	134	-	114	124	120	122	121.27	170	98
20	CalciumHardness	mg/1	78	50	78	110	58	82	96	-	92	90	86	66	80.55	110	50
21	Magnesium H.	mg/1	58	64	24	60	40	18	38	-	22	34	34	56	40.73	64	18
22	D. Oxygen	mg/1	6.9	5.2	8.2	11.5	6.5	7.2	7.6	-	7.8	8.6	8.8	13.1	8.31	13.1	5.2
23	B.O.D.	mg/1	1.4	-	-	-	1.6	1.9	1.5	-	1.6	2.3	2.5	1.1	1.74	2.5	1.1
24	C.O.D.	mg/1	10	9.88	20	39.2	20	19.6	19.6	-	18.8	20	19.2	9.6	18.72	39.2	9.6
25	Sodium	mg/1	9.73	13.73	12.85	25.56	16.25	-	-	-	18.7	9.08	9.34	12.3	14.17	25.56	9.08
26	Potassium	mg/1	2.98	3.11	3.62	6.5	3.24	-	-	-	-	2.46	2.68	2.06	3.33	6.5	2.06
27	T. Coliform	MPN/100ml	110	17	80	220	1600	-	40	-	920	90	90	-	351.89	1600	17
28	F. Coliform	MPN/100ml	2	-	-	-	-	-	-	-	-	-	-	-	2.00	2	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cat. B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2014-2015

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake). Cat. - B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.14	5.5.14	25.6.14	12.7.14	12.8.14	11.9.14	NR	NR	20.12.14	13.1.15	10.2.15	17.3.15	-	-	-
2	Appearance		sl.tur.	sl.tur.	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	21	17	18	-	18.67	21	17
4	Turbidity	NTU	3	15	6	16	22	51	-	-	18	6	6	14	15.70	51	3
5	Colour	PCS	-	-	-	-	-	-	-	-	CL	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	OL	-	-	-	-	-	-
7	pH	pH Unit	7.94	8.28	8.74	8.53	7.83	8.32	-	-	8.12	8.16	8.1	7.97	8.20	8.74	7.83
8	Sp. Conductivity	µMhos/cm.	284	258	240	241	263	240	-	-	288	272	265	272	262.30	288	240
9	T. Solids	mg/1	193	139	177	152	158	198	-	-	218	182	176	133	172.60	218	133
10	D. Solids	mg/1	183	121	155	144	140	156	-	-	184	167	160	114	152.40	184	114
11	S. Solids	mg/1	10	18	22	8	18	42	-	-	34	15	16	19	20.20	42	8
12	Amm. Nitrogen	mg/1	-	-	-	-	0.055	-	-	-	0.075	0.242	0.228	-	0.15	0.242	0.055
13	Nitrite Nitrogen	mg/1	BDL	0.02	0.13	0.02	0.025	0.05	-	-	0.18	0.1	0.02	0.05	0.07	0.18	0.02
14	Nitrate Nitrogen	mg/1	2.55	1.75	1.7	2.4	3.25	1.4	-	-	1.9	2.46	2.25	2.75	2.24	3.25	1.4
15	Phosphate (PO_4)	mg/1	-	0.066	0.099	0.099	-	-	-	-	1.287	0.173	0.168	0.132	0.29	1.287	0.066
16	Chloride	mg/1	13	15.99	17.73	36	21.21	18.39	-	-	22	16.27	17.22	18	19.58	36.00	13
17	Sulphate (SO_4)	mg/1	4.75	-	18.75	4.75	6	-	-	-	15.5	8.5	8.2	3	8.68	18.75	3
18	T. Alkalinity	mg/1	120	104	126	110	104	98	-	-	128	142	134	116	118.20	142	98
19	T. Hardness	mg/1	134	108	122	114	104	92	-	-	134	134	128	126	119.60	134	92
20	Cal.Hardness	mg/1	70	44	76	78	58	54	-	-	110	102	98	80	77.00	110	44
21	Magnesium H.	mg/1	64	64	46	36	46	38	-	-	24	32	30	46	42.60	64	24
22	D. Oxygen	mg/1	7.2	7.2	8.8	8.2	5.5	6	-	-	7.4	-	7.4	9.8	7.50	9.8	5.5
23	B.O.D.	mg/1	1.2	1	-	-	2.6	1.1	-	-	2.6	2	1.8	1.2	1.69	2.6	1
24	C.O.D.	mg/1	10	9.88	20	19.76	20	30	-	-	37.6	18.89	19.2	9.6	19.49	37.6	9.6
25	Sodium	mg/1	9.61	10.02	13.26	16.74	16.11	14.33	-	-	23.8	8.2	8.36	11.09	13.15	23.8	8.2
26	Potassium	mg/1	3	3.1	2.62	3	3.33	2.6	-	-	-	3.09	3.23	3.76	3.08	3.76	2.6
27	T. Coliform	MPN/100ml	33	33	90	26	1600	-	-	-	1600	60	70	130	404.67	1600	26
28	F. Coliform	MPN/100ml	2	-	-	-	-	-	-	-	-	-	-	-	2.00	2	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. A																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2014-2015

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake). Cat. - B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.14	5.5.14	26.5.14	12.7.14	12.8.14	11.9.14	20.10.14	NR	20.12.14	13.1.15	10.2.15	17.3.15	-	-	-
2	Appearance		sl.tur	sl.tur	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	21	18	19	-	19.33	21	18
4	Turbidity	NTU	4	12	15	15	20	50	-	-	20	10	8	16	17.00	50	4
5	Colour	PCS	-	-	-	-	-	-	-	-	CL	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	OL	-	-	-	-	-	-
7	pH	pH Unit	7.94	8.32	8.87	8.71	7.83	7.24	7.98	-	7.88	7.89	7.95	7.78	8.04	8.87	7.24
8	Sp. Conductivity	µMhos/cm.	287	256	254	244	262	491	262	-	302	248	252	271	284.45	491	244
9	T. Solids	mg/1	148	127	143	149	160	310	171	-	234	165	162	138	173.36	310	127
10	D. Solids	mg/1	140	112	116	141	142	285	162	-	196	139	138	117	153.45	285	112
11	S. Solids	mg/1	8	15	27	8	18	25	9	-	38	26	24	21	19.91	38	8
12	Amm. Nitrogen	mg/1	BDL	-	-	-	0.05	-	BDL	-	0.11	0.23	0.25	-	0.16	0.25	0.05
13	Nitrite Nitrogen	mg/1	BDL	-	0.2	0.01	0.025	0.07	BDL	-	0.24	BDL	BDL	0.335	0.15	0.335	0.01
14	Nitrate Nitrogen	mg/1	1.9	1.65	2	2.55	3.3	7.9	2.1	-	2.2	-	2.2	3	2.88	7.90	1.65
15	Phosphate (PO_4)	mg/1	-	-	0.42	0.726	-	-	BDL	-	1.221	0.223	0.238	0.165	0.50	1.221	0.165
16	Chloride	mg/1	13.49	18.99	19.71	29.99	21.21	24.96	14.99	-	24	19.14	18.18	19	20.33	29.99	13.49
17	Sulphate (SO_4)	mg/1	-	4.75	19	6.25	6	40	3.11	-	12.75	8.9	9.2	3	11.30	40.00	3
18	T. Alkalinity	mg/1	128	106	130	108	110	192	120	-	118	148	144	138	131.09	192	106
19	T. Hardness	mg/1	132	108	114	118	104	218	138	-	142	152	136	132	135.82	218	104
20	Cal.Hardness	mg/1	68	42	76	104	58	128	76	-	108	104	94	80	85.27	128	42
21	Magnesium H.	mg/1	64	66	38	14	46	90	62	-	34	48	42	52	50.55	90	14
22	D. Oxygen	mg/1	6.2	6.7	7.2	7.7	6.8	4.4	6.3	-	7.2	8.4	8.6	9.8	7.21	9.8	4.4
23	B.O.D.	mg/1	1	1.1	-	-	1.4	1.9	1.2	-	2.9	2.8	2.6	1.2	1.79	2.9	1
24	C.O.D.	mg/1	9.24	19.76	20	19.76	20	30	9.8	-	47	30	28.8	9.6	22.18	47	9.24
25	Sodium	mg/1	11.04	10.35	13.13	17.08	16.16	13.21	10.82	-	21.4	9.22	9.64	10.6	12.97	21.4	9.22
26	Potassium	mg/1	2.87	3.19	2.53	2.99	3.25	7.55	2.62	-	Nil	2.82	2.92	3.75	3.45	7.55	2.53
27	T. Coliform	MPN/100ml	130	30	60	33	1600	-	34	-	2400	130	130	240	478.70	2400	30
28	F. Coliform	MPN/100ml	2	-	-	-	-	-	-	-	-	-	-	-	2.00	2	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2014-2015

Natural Water Resource : Upper Lake at Kamla Park,Bhopal.

Description of Sampling Station : Near w/s Intake well, Bhopal. Cat. - B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.14	5.5.14	25.6.14	12.7.14	12.8.14	11.9.14	20.10.14	NR	20.12.14	13.2.15	10.2.15	17.3.15	-	-	-
2	Appearance		sl.tur.	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	22	18	18	-	19.33	22	18
4	Turbidity	NTU	8	10	4	14	18	70	8	-	12	10	10	9	15.73	70	4
5	Colour	PCS	-	-	-	-	-	-	-	-	CL	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	OL	-	-	-	-	-	-
7	pH	pH Unit	7.84	8.3	8.82	8.44	7.91	8.19	7.14	-	7.85	8.42	8.31	7.93	8.10	8.82	7.14
8	Sp. Conductivity	µMhos/cm.	286	258	239	242	265	239	240	-	258	252	248	288	255.91	288	239
9	T. Solids	mg/1	188	146	157	147	150	204	210	-	176	190	182	124	170.36	210	124
10	D. Solids	mg/1	171	134	139	140	134	118	117	-	152	174	167	109	141.36	174	109
11	S. Solids	mg/1	17	12	18	7	16	86	93	-	24	16	15	15	29.00	93	7
12	Amm. Nitrogen	mg/1	-	-	-	-	0.04	-	BDL	-	-	BDL	-	-	0.04	0.04	0.04
13	Nitrite Nitrogen	mg/1	-	1.75	0.13	0.025	0.04	0.125	0.165	-	0.09	0.19	0.16	0.035	0.27	1.75	0.025
14	Nitrate Nitrogen	mg/1	2	-	2.3	1.95	3.15	3.4	3.8	-	1.2	1.68	1.8	2.45	2.37	3.8	1.2
15	Phosphate (PO_4)	mg/1	-	-	0.033	0.099	-	-	-	-	0.858	0.69	0.56	0.132	0.40	0.858	0.033
16	Chloride	mg/1	14	16.99	17.73	29.99	43.39	18.35	18.99	-	16	19.14	18.18	17	20.89	43.39	14
17	Sulphate (SO_4)	mg/1	5.5	4.25	16.25	5.75	7.75	-	-	-	13.75	8.22	8.38	2.5	8.04	16.25	2.5
18	T. Alkalinity	mg/1	120	110	126	114	106	104	110	-	112	124	118	120	114.91	126	104
19	T. Hardness	mg/1	130	114	108	102	94	106	114	-	126	124	126	142	116.91	142	94
20	CalciumHardness	mg/1	76	44	82	62	62	58	58	-	102	80	78	84	71.45	102	44
21	Magnesium H.	mg/1	54	70	26	40	32	48	56	-	24	44	48	58	45.45	70	24
22	D. Oxygen	mg/1	6.4	7.6	8.4	8.2	9.8	10	10.2	-	7.9	8.8	8.6	9.2	8.65	10.2	6.4
23	B.O.D.	mg/1	1.1	1.9	-	-	1.1	1.9	2.1	-	2.3	2.1	2.2	1.1	1.76	2.3	1.1
24	C.O.D.	mg/1	10	19.76	10	19.76	20	20	19.6	-	28.2	10	9.6	9.6	16.05	28.2	9.6
25	Sodium	mg/1	9.56	8.99	13.57	18.4	16.4	15.5	-	-	22.4	8.26	8.12	12.11	13.33	22.4	8.12
26	Potassium	mg/1	2.91	3.06	3.26	3.09	3.17	2.63	-	-	-	2.11	2.36	3.6	2.91	3.6	2.11
27	T. Coliform	MPN/100ml	80	50	70	33	1600	-	22	-	540	60	60	-	279.44	1600	22
28	F. Coliform	MPN/100ml	2	-	-	-	-	-	-	-	-	-	-	110	56.00	110	2
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2014-2015

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal. Cat. - B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.14	-	-	-	12.8.14	-	-	N.R.	-	-	10.2.15	-	-	-	-
2	Appearance		sl.tur.	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	18	-	18.00	18	18
4	Turbidity	NTU	4	-	-	-	14	-	-	-	-	-	9	-	9.00	14	4
5	Colour	PCS	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	7.93	-	-	-	7.31	-	-	-	-	-	8.21	-	7.82	8.21	7.31
8	Sp. Conductivity	µMhos/cm.	286	-	-	-	787	-	-	-	-	-	264	-	445.67	787	264
9	T. Solids	mg/1	140	-	-	-	469	-	-	-	-	-	220	-	276.33	469	140
10	D. Solids	mg/1	132	-	-	-	453	-	-	-	-	-	194	-	259.67	453	132
11	S. Solids	mg/1	8	-	-	-	16	-	-	-	-	-	26	-	16.67	26	8
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	0.025	-	0.03	0.025	0.025
13	Nitrite Nitrogen	mg/1	0.325	-	-	-	0.25	-	-	-	-	-	0.05	-	0.21	0.325	0.05
14	Nitrate Nitrogen	mg/1	1.55	-	-	-	5.7	-	-	-	-	-	1.95	-	3.07	5.70	1.55
15	Phosphate (PO_4)	mg/1	0.198	-	-	-	0.396	-	-	-	-	-	0.165	-	0.25	0.396	0.165
16	Chloride	mg/1	-	-	-	-	36.64	-	-	-	-	-	14.35	-	25.50	36.64	14.35
17	Sulphate (SO_4)	mg/1	13.49	-	-	-	36.5	-	-	-	-	-	15	-	21.66	36.50	13.49
18	T. Alkalinity	mg/1	118	-	-	-	304	-	-	-	-	-	136	-	186.00	304	118
19	T. Hardness	mg/1	124	-	-	-	252	-	-	-	-	-	124	-	166.67	252	124
20	CalciumHardness	mg/1	64	-	-	-	136	-	-	-	-	-	82	-	94.00	136	64
21	Magnesium H.	mg/1	60	-	-	-	116	-	-	-	-	-	42	-	72.67	116	42
22	D. Oxygen	mg/1	5.8	-	-	-	3.5	-	-	-	-	-	7.8	-	5.70	7.8	3.5
23	B.O.D.	mg/1	0.8	-	-	-	2.4	-	-	-	-	-	2.2	-	1.80	2.4	0.8
24	C.O.D.	mg/1	9.24	-	-	-	20	-	-	-	-	-	19.2	-	16.15	20	9.24
25	Sodium	mg/1	10.81	-	-	-	24.56	-	-	-	-	-	-	-	17.69	24.56	10.81
26	Potassium	mg/1	2.31	-	-	-	7.98	-	-	-	-	-	-	-	5.15	7.98	2.31
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-		-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2014-2015

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka,Bairagarh. Cat. - B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		6.4.14	-	-	-	12.8.14	-	-	N.R.	-	-	N.R.	-	-	-	-
2	Appearance		Sl.tur	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	4	-	-	-	15	-	-	-	-	-	-	-	9.50	15	4
5	Colour	PCS	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	pH	pH Unit	7.91	-	-	-	7.3	-	-	-	-	-	-	-	7.61	7.91	7.3
8	Sp. Conductivity	µMhos/cm.	286	-	-	-	788	-	-	-	-	-	-	-	537.00	788	286
9	T. Solids	mg/1	128	-	-	-	459	-	-	-	-	-	-	-	293.50	459	128
10	D. Solids	mg/1	122	-	-	-	441	-	-	-	-	-	-	-	281.50	441	122
11	S. Solids	mg/1	6	-	-	-	18	-	-	-	-	-	-	-	12.00	18	6
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	-			
13	Nitrite Nitrogen	mg/1	0.02	-	-	-	0.22	-	-	-	-	-	-	-	0.12	0.22	0.02
14	Nitrate Nitrogen	mg/1	1.65	-	-	-	5.35	-	-	-	-	-	-	-	3.50	5.35	1.65
15	Phosphate (PO_4)	mg/1	0.198	-	-	-	0.462	-	-	-	-	-	-	-	0.33	0.462	0.198
16	Chloride	mg/1	15.42	-	-	-	45.32	-	-	-	-	-	-	-	30.37	45.32	15.42
17	Sulphate (SO_4)	mg/1	-	-	-	-	37	-	-	-	-	-	-	-	37.00	37.00	37
18	T. Alkalinity	mg/1	118	-	-	-	304	-	-	-	-	-	-	-	211.00	304	118
19	T. Hardness	mg/1	120	-	-	-	226	-	-	-	-	-	-	-	173.00	226	120
20	CalciumHardness	mg/1	66	-	-	-	114	-	-	-	-	-	-	-	90.00	114	66
21	Magnesium H.	mg/1	54	-	-	-	112	-	-	-	-	-	-	-	83.00	112	54
22	D. Oxygen	mg/1	5.4	-	-	-	3.5	-	-	-	-	-	-	-	4.45	5.4	3.5
23	B.O.D.	mg/1	0.6	-	-	-	2.4	-	-	-	-	-	-	-	1.50	2.4	0.6
24	C.O.D.	mg/1	9.24	-	-	-	20	-	-	-	-	-	-	-	14.62	20	9.24
25	Sodium	mg/1	10.21	-	-	-	35.35	-	-	-	-	-	-	-	22.78	35.35	10.21
26	Potassium	mg/1	2.49	-	-	-	8.11	-	-	-	-	-	-	-	5.30	8.11	2.49
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-	-	-	-	-		-	-
29	T.K.N.	mg/1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2015-2016

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		21.4.15	21.5.2015	11.6.2015	7.7.2015	13.8.2015	21.9.2015	8.10.2015	18.11.2015		16.01.16	23.02.16	15.03.16	-	-	-
2	Appearance		-	-	-	-	-	-	-	-			Clear		-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-					-	-	-
4	Turbidity	NTU	16									0.6		8	8.20	16.00	0.6
5	Colour	PCS		colourless	colourless	colourless	colourless	colourless	colourless	colourless		colourless	CL	CL	-	-	-
6	Odour	T. No		odourless		odourless		odourless	odourless	odourless		odourless	OL	OL	-	-	-
7	pH	pH Unit	8.22	7.6	7.36	8.57	8.4	7.82	7.45	7.61		7.7	7.9	8.44	7.92	8.57	7.36
8	Sp. Conductivity	µMhos/cm.	588			257.2	340	265		271		294	304	259	322.28	588	257.2
9	T. Solids	mg/1	256	250	306	194	194	207	202	207		206	318	184	229.45	318	184
10	D. Solids	mg/1	234	180	282	190	172	185	182	189		200	204	158	197.82	282	158
11	S. Solids	mg/1	22	70	24	4	22	22	20	18		6	14	26	22.55	70	4
12	Amm. Nitrogen	mg/1		0.62	0.06	-	1.3	-	-	-		0.016	0.08	0.11	0.36	1.30	0.016
13	Nitrite Nitrogen	mg/1	BDL	0.59	BDL	-	0.14	-	-	-		BDL	0.02	0.125	0.22	0.59	0.02
14	Nitrate Nitrogen	mg/1	4.1	0.48	0.5	-	1.66	-	-	-		1	BDL	1.2	1.49	4.10	0.48
15	Phosphate (PO_4)	mg/1	0.165	-	-	-	BDL	-	-	-		0.016	0.022	0.363	0.14	0.36	0.016
16	Chloride	mg/1	58.98	9.9	24.99	28.83	30	15.23	15.98	14.28		27.65	27.65	17.99	24.68	58.98	9.9
17	Sulphate (SO_4)	mg/1	44	-	-		18	-	-	-		3.74	10	5.5	16.25	44.00	3.74
18	T. Alkalinity	mg/1	114	76	-	112.4	48	110	108	106		124	134	106	103.84	134	48
19	T. Hardness	mg/1	122	110	-	72	102	78	84	80		140	146	138	107.20	146	72
20	CalciumHardness	mg/1	96	90	-	56	60	58	66	58		112	110	118	82.40	118	56
21	Magnesium H.	mg/1	26	20	-	16	42	20	18	22		28	36	20	24.80	42	16
22	D. Oxygen	mg/1	4.8	7.5	-		6.8	7.6	7.7	7.4		8.4	8.5	7.8	7.39	8.50	4.8
23	B.O.D.	mg/1	1.7	1.6	2.4	0.74	1.9	1.1	1.1	1.3		1.4	1.5	1.4	1.47	2.40	0.74
24	C.O.D.	mg/1	28.8	16	12	3.64	16	15.5	19.2	17.45		17.64	17.64	19.04	16.63	28.80	3.64
25	Sodium	mg/1	48.46	-	10	22.26	4.16	-	-	-		14.2	-	-	19.82	48.46	4.16
26	Potassium	mg/1	4.46	-	2	5.83	2	-	-	-		1.6	-	-	3.18	5.83	1.6
27	T. Coliform	MPN/100ml	170	-	140	3	60	63	94	70		90	-	-	86.25	170	3
28	F. Coliform	MPN/100ml	-	-	27	-	-	-	-	-		-	-	-	27	27	27
29	T.K.N.	mg/1										-			-	-	-
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2015-2016

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		21.4.15	20.5.2015	11.6.2015	7.7.2015	13.8.2015	21.9.2015	8.10.2015	18.11.2015		16.01.16	23.02.16	15.03.16	-	-	-
2	Appearance		-	-	-	-	-	-	-	-		Clear	Clear		-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-					-	-	-
4	Turbidity	NTU	11				6					0.7		10	6.93	11.00	0.70
5	Colour	PCS		colourless	colourless	colourless	colourless	colourless	colourless	colourless		1	CL	CL	-	-	-
6	Odour	T. No		odourless		odourless		odourless	odourless	odourless		OL	OL	OL	-	-	-
7	pH	pH Unit	7.57	7.6	7.27	8.48	8.8	7.62	7.56	7.76		7.61	7.66	8.28	7.84	8.80	7.27
8	Sp. Conductivity	µMhos/cm.	415	-	-	270.2	338	270	-	280		270	274	286	300.40	415	270
9	T. Solids	mg/1	280	260	304	202	256	222	214	218		195	202	202	232.27	304	195
10	D. Solids	mg/1	260	200	286	190	246	191	192	198		184	188	172	209.73	286	172
11	S. Solids	mg/1	20	60	18	12	10	30	22	20		11	14	30	22.45	60	10
12	Amm. Nitrogen	mg/1		0.68	0.32	-	0.78	-	-	-		0.014	0.36	0.13	0.38	0.78	0.01
13	Nitrite Nitrogen	mg/1	BDL	0.59	BDL	-	0.04	-	-	-		BDL	BDL	0.145	0.26	0.59	0.04
14	Nitrate Nitrogen	mg/1	2.5	0.65	1.1	-	1.812	-	-	-		0.84	0.86	1.35	1.30	2.50	0.65
15	Phosphate (PO_4)	mg/1	0.264	-	-	-	BDL	-	-	-		0.02	0.04	0.429	0.19	0.43	0.02
16	Chloride	mg/1	32.98	9.98	22.49	32.68	26	16.18	16.98	15.23		36.86	36.86	19.99	24.20	36.86	9.98
17	Sulphate (SO_4)	mg/1	1	-	-	-	10	-	-	-		2.53	11.4	6.75	6.34	11.40	1.00
18	T. Alkalinity	mg/1	130	70	-	109.2	130	114	102	112		128	132	114	114.12	132	70
19	T. Hardness	mg/1	144	130	-	84	160	80	92	82		132	136	152	119.20	160	80
20	Cal.Hardness	mg/1	124	90	-	52	72	56	72	62		100	104	130	86.20	130	52
21	Magnesium H.	mg/1	20	40	-	32	88	24	20	20		32	32	22	33.00	88	20
22	D. Oxygen	mg/1	6.3	7.8	-	-	7	7.7	7.6	7.6		8.6	8.7	7.5	7.64	8.70	6.30
23	B.O.D.	mg/1	1.3	1.6	2.2	0.97	0.8	1.2	1	1.2		1.5	1.6	1.9	1.39	2.20	0.80
24	C.O.D.	mg/1	9.6	24	16	3.64	11	19.38	19.2	17.45		24.4	29.5	28.56	18.43	29.50	3.64
25	Sodium	mg/1	22.61	-	10	23.02	9.9	-	-	-		15.3	-	-	16.17	23.02	9.90
26	Potassium	mg/1	2.81	-	2	5.77	2.8	-	-	-		1.4	-	-	2.96	5.77	1.40
27	T. Coliform	MPN/100ml	280	-	150	5		94	84	110		130	-	-	121.86	280	5
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	-	-		-	-	-			
29	T.K.N.	mg/1													-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2015-2016

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		21.4.15	20.5.2015	11.6.2015	7.7.2015	13.8.2015	21.9.2015	8.10.2015	18.11.2015		16.01.16	23.02.16	15.03.16	-	-	-
2	Appearance		-	-	-	-	-	-	-	-		Clear	Clear	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-		-	-	-	-	-	-
4	Turbidity	NTU	12	-	-	-	4.4	-	-	-		0.84	-	18	8.81	18	0.84
5	Colour	PCS		colourless	colourless	colourless	colourless	colourless	colourless	colourless		1	CL	CL	-	-	-
6	Odour	T. No		odourless		odourless		odourless	odourless	odourless		OL	OL	OL	-	-	-
7	pH	pH Unit	7.76	7.4	7.42	8.79	9.0	7.91	7.62	7.8		7.65	7.7	8.73	7.98	9	7.4
8	Sp. Conductivity	µMhos/cm.	641	-	-	257.2	293	258	-	265		263	268	310	319.4	641	257.2
9	T. Solids	mg/1	158	250	308	192	224	206	198	208		197	202	232	215.9	308	158
10	D. Solids	mg/1	148	200	282	178	216	180	180	186		179	182	188	192.6	282	148
11	S. Solids	mg/1	10	50	26	14	8	26	18	22		18	20	44	23.3	50	8
12	Amm. Nitrogen	mg/1		0.65	0.64	-	0.72	-	-	-		0.022	0.66	0.6	0.5	0.72	0.022
13	Nitrite Nitrogen	mg/1	0.015	0.5	BDL	-	0.5	-	-	-		BDL	BDL	0.175	0.30	0.5	0.015
14	Nitrate Nitrogen	mg/1	2.7	0.4	1.34	-	2.016	-	-	-		0.88	0.9	1.55	1.40	2.70	0.4
15	Phosphate (PO_4)	mg/1	0.66		BDL	-	BDL	-	-	-		0.024	0.026	0.561	0.32	0.66	0.024
16	Chloride	mg/1	24.99	19.98	27.48	30.75	32	14.28	14.98	17.13		36.86	68.2	26.99	28.51	68.20	14.28
17	Sulphate (SO_4)	mg/1	0.25	-	20	-	8	-	-	-		3.92	20	8.25	10.07	20.00	0.25
18	T. Alkalinity	mg/1	120	78	-	109.6	130	106	106	110		128	132	118	113.76	132	78
19	T. Hardness	mg/1	184	120	-	76	124	74	90	78		132	138	164	118	184	74
20	Cal.Hardness	mg/1	124	100	-	48	54	54	68	60		96	100	136	84	136	48
21	Magnesium H.	mg/1	60	20	-	28	70	20	22	18		36	38	28	34	70	18
22	D. Oxygen	mg/1	6.3	7.6	-		6.8	7.5	7.4	7.7		8.5	8.6	7.1	7.50	8.6	6.3
23	B.O.D.	mg/1	1.9	1.8	2.6	1.04	1	1.3	1.3	1.4		1.4	1.5	3.3	1.69	3.3	1
24	C.O.D.	mg/1	19.2	24	12	3.64	10	19.38	19.2	21.82		25.4	17.64	47.6	19.99	47.6	3.64
25	Sodium	mg/1	13.27	-	14	22.2	8.7	-	-	-		22	-	-	16.03	22.2	8.7
26	Potassium	mg/1	2.79	-	3	6.08	3.35	-	-	-		2	-	-	3.44	6.08	2
27	T. Coliform	MPN/100ml	170	-	120	2	-	110	110	120		-	-	-	105.33	170	2
28	F. Coliform	MPN/100ml	-	-	25	-	-	-	-	-		-	-	-	25	25	25
29	T.K.N.	mg/1													-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2015-2016

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		21.4.15	20.5.2015	11.6.2015	7.7.2015	13.8.2015	21.9.2015	8.10.2015	18.11.2015		16.01.16	23.02.16	15.03.16	-	-	-
2	Appearance		-	-	-	-	-	-	-	-		Clear	Clear	-	-	-	-
3	Temperature	oC	-	-	-	-	-	-	-	-		-	-	-	-	-	-
4	Turbidity	NTU	19	-	-	-	20	-	-	-		1.1		14	13.53	20	1.1
5	Colour	PCS	-	colourless	colourless	colourless	colourless	colourless	colourless	colourless		1	CL	CL	-	-	-
6	Odour	T. No	-	odourless	odourless	odourless	odourless	odourless	odourless	odourless		OL	OL	OL	-	-	-
7	pH	pH Unit	8.3	7.4	7.82	8.33	8.3	7.79	7.51	7.57		7.67	7.78	8.36	7.89	8.36	7.4
8	Sp. Conductivity	µMhos/cm.	337	-	-	254.8	230	275		282		340	348	324	298.85	348	230
9	T. Solids	mg/1	178	280	310	252	188	216	202	210		238	246	242	232.909	310	178
10	D. Solids	mg/1	154	200	288	236	170	195	178	183		232	232	204	206.545	288	154
11	S. Solids	mg/1	21	80	22	16	18	21	24	17		6	12	38	25.00	80	6
12	Amm. Nitrogen	mg/1	-	0.64	0.32	-	1.86	-	-	-		0.019	0.36	0.18	0.56	1.86	0.019
13	Nitrite Nitrogen	mg/1	0.11	0.52	BDL	-	0.11	-	-	-		BDL	0.02	0.19	0.19	0.52	0.02
14	Nitrate Nitrogen	mg/1	3.1	0.48	2	-	1.56	-	-	-		1.06	BDL	1.7	1.65	3.1	0.48
15	Phosphate (PO_4)	mg/1	0.23	-	BDL	-	BDL	-	-	-		0.02	0.03	0.594	0.22	0.594	0.02
16	Chloride	mg/1	31.99	19.98	19.99	30.75	34	17.13	15.98	16.18		36.86	36.86	24.99	25.88	36.86	15.98
17	Sulphate (SO_4)	mg/1	7.75	-	10	-	18.14	-	-	-		3.74	10	9.5	9.86	18.14	3.74
18	T. Alkalinity	mg/1	108	80	-	114.4	40	118	104	114		120	126	124	104.84	126	40
19	T. Hardness	mg/1	130	120	-	68	100	82	88	84		164	172	178	118.60	178	68
20	CalciumHardness	mg/1	108	90	-	44	60	60	68	62		112	120	152	87.60	152	44
21	Magnesium H.	mg/1	22	30	-	24	40	22	20	22		52	52	26	31.00	52	20
22	D. Oxygen	mg/1	6.2	7.6	-	-	7.3	7.8	7.3	7.5		8.2	8.1	7.4	7.49	8.2	6.2
23	B.O.D.	mg/1	1.2	1.9	2.2	0.89	1.8	1	1.2	1.1		1.8	1.9	2.6	1.60	2.6	0.89
24	C.O.D.	mg/1	19.2	16	12	3.64	15	15.5	19.2	21.82		19.6	19.6	38.08	18.15	38.08	3.64
25	Sodium	mg/1	21.8	-	12	24.11	5.14	-	-	-		16.4	-	-	15.89	24.11	5.14
26	Potassium	mg/1	2.98	-	2	5.93	2.1	-	-	-		1.4	-	-	2.88	5.93	1.4
27	T. Coliform	MPN/100ml	140	-	120	0	50	84	79	94		90	-	-	82.125	140	0
28	F. Coliform	MPN/100ml	-	-	32	-	-	-	-	-		-	-	-	32	32	32
29	T.K.N.	mg/1	-	-		-	-	-	-	-		-	-	-	-	-	-
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2015-2016

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal. CAT-B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		21.4.15				13.8.2015			18.11.15			not		-	-	-
2	Appearance		-				-			-					-	-	-
3	Temperature	oC	-				-			-					-	-	-
4	Turbidity	NTU	28				3.2			-					15.60	28.00	3.20
5	Colour	PCS					CL			CL					-	-	-
6	Odour	T. No					OL			OL					-	-	-
7	pH	pH Unit	7.42				8.14			8.09					7.88	8.14	7.42
8	Sp. Conductivity	µMhos/cm.	556				375			288					406.33	556	288.00
9	T. Solids	mg/1	392				262			199					284.33	392	199.00
10	D. Solids	mg/1	360				250			179					263.00	360	179.00
11	S. Solids	mg/1	32				12			20					21.33	32	12.00
12	Amm. Nitrogen	mg/1	2.27				BDL			-					2.27	2.27	2.27
13	Nitrite Nitrogen	mg/1	0.22				BDL			-					0.22	0.22	0.22
14	Nitrate Nitrogen	mg/1	6.7				2.52			-					4.61	6.70	2.52
15	Phosphate (PO_4)	mg/1	0.396				BDL			-					0.40	0.40	0.40
16	Chloride	mg/1	35.98				62			13.32					37.10	62.00	13.32
17	Sulphate (SO_4)	mg/1	0.22				14.2								7.21	14.20	0.22
18	T. Alkalinity	mg/1	296				184			162					214	296	162
19	T. Hardness	mg/1	244				188			168					200	244	168
20	CalciumHardness	mg/1	124				96			100					106.67	124	96
21	Magnesium H.	mg/1	120				92			68					93.33	120	68
22	D. Oxygen	mg/1	4.8				-			7.8					6.30	7.80	4.80
23	B.O.D.	mg/1	2.2				2.4			1.9					2.17	2.40	1.90
24	C.O.D.	mg/1	19.6				14			11.62					15.07	19.60	11.62
25	Sodium	mg/1					14.3			-					14.30	14.30	14.30
26	Potassium	mg/1	4.12				2.44			-					3.28	4.12	2.44
27	T. Coliform	MPN/100ml	-				-			-							
28	F. Coliform	MPN/100ml	-				-			-					-	-	-
29	T.K.N.	mg/1	-				-			-					-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2015-2016

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka,Bairagarh. CAT-B

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Not approachable				13.08.15			18.11.15			not approachable		-	-	-
2	Appearance						-			-					-	-	-
3	Temperature	oC					-			-					-	-	-
4	Turbidity	NTU					3.6			-					3.6	3.6	3.6
5	Colour	PCS					CL			CL					-	-	-
6	Odour	T. No					OL			OL					-	-	-
7	pH	pH Unit					7.98			7.38					7.68	7.98	7.38
8	Sp. Conductivity	µMhos/cm.					367			284					325.5	367	284
9	T. Solids	mg/1					250			220					235	250	220
10	D. Solids	mg/1					236			188					212	236	188
11	S. Solids	mg/1					14			32					23	32	14
12	Amm. Nitrogen	mg/1					BDL			-							
13	Nitrite Nitrogen	mg/1					BDL			-							
14	Nitrate Nitrogen	mg/1					2.18			-					2.18	2.18	2.18
15	Phosphate (PO_4)	mg/1					BDL			-							
16	Chloride	mg/1					48			17.13					32.565	48.00	17.13
17	Sulphate (SO_4)	mg/1					16.2			-					16.2	16.20	16.2
18	T. Alkalinity	mg/1					144			142					143	144	142
19	T. Hardness	mg/1					152			142					147	152	142
20	CalciumHardness	mg/1					80			94					87	94	80
21	Magnesium H.	mg/1					72			48					60	72	48
22	D. Oxygen	mg/1					-			7.9					7.9	7.9	7.9
23	B.O.D.	mg/1					2.2			2.3					2.25	2.3	2.2
24	C.O.D.	mg/1					12			19.38					15.69	19.38	12
25	Sodium	mg/1					12.9			-					12.9	12.9	12.9
26	Potassium	mg/1					2.18			-					2.18	2.18	2.18
27	T. Coliform	MPN/100ml					-			-							
28	F. Coliform	MPN/100ml					-			-					-	-	-
29	T.K.N.	mg/1					-			-					-	-	-
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2016-2017

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.16	18.05.16	13.06.16	11.07.16	10.08.16	15.09.16	12.10.16	16.11.16	8.12.16	09.01.17	07.02.17	07.03.17	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC						-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	1.1	10	16	18	16	10	10	8	12	10	2	10	10.26	18	1.1
5	Colour	PCS	CL	CL	CL	CL	CL	CL	CL	Colorless	Colorless	Colorless	CL	CL	-	-	-
6	Odour	T. No	OL	OL	OL	OL	OL	OL	OL	Odourless	Odourless	Odourless	OL	OL	-	-	-
7	pH	pH Unit	7.81	7.76	8.12	7.91	6.75	8.32	8.26	7.77	7.62	7.74	8.32	7.64	7.84	8.32	6.75
8	Sp. Conductivity	µMhos/cm.	298	296	336	310	169	278	258	326	328	322	286	318	293.75	336	169
9	T. Solids	mg/1	216	208	198	298	106	201	230	246	240	228	222	332	227.08	332	106
10	D. Solids	mg/1	208	200	176	272	68	183	218	234	226	212	196	322	209.58	322	68
11	S. Solids	mg/1	8	8	22	26	34	18	12	12	14	16	26	10	17.17	34	8
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.18	-	-	-	-	0.012	0.096	0.18	0.012
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.198	-	-	-	-	BDL	0.198	0.198	0.198
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	0.82	0.82	0.82	0.82
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.165	-	-	-	-	0.04	0.10	0.165	0.04
16	Chloride	mg/1	29.99	28.99	30.55	25.62	15.76	32.78	32.52	29.89	31.53	33.98	13.79	27.96	27.78	33.98	13.79
17	Sulphate (SO_4)	mg/1	-	-	-	-	-	-	7	-	-	-	-	2.48	4.74	7.00	2.48
18	T. Alkalinity	mg/1	130	130	148	136	72	140	134	124	142	138	132	136	130.17	148	72
19	T. Hardness	mg/1	136	144	102	142	64	136	136	156	158	142	130	134	131.67	158	64
20	CalciumHardness	mg/1	106	114	64	110	44	108	102	112	132	110	90	98	99.17	132	44
21	Magnesium H.	mg/1	30	30	38	32	20	12	34	44	26	32	40	36	31.17	44	12
22	D. Oxygen	mg/1	8.2	7.8	7.3	8.1	8.2	7.5	7.5	8.2	7.5	7.8	8.2	7.7	7.83	8.2	7.3
23	B.O.D.	mg/1	1.3	2.3	1.6	1.2	1.7	1.5	1.6	1.5	1.6	1.6	1.9	1.6	1.61667	2.3	1.2
24	C.O.D.	mg/1	20	20	19.76	19.76	9.88	20.0	20	18.72	20	19.88	20	20	19.00	20	9.88
25	Sodium	mg/1	15.1	15.2	5.64	14.2	-	14.64	13.86	14.76	20	6.38	16.79	8.62	13.20	20	5.64
26	Potassium	mg/1	1.4	1.2	2.3	1.1	-	1.0	1.03	1	1.08	1.2	2.43	1.03	1.34	2.43	1
27	T. Coliform	MPN/100ml	90	120	50	110	90	94	110	90	94	110	94	110	96.83	120	50
28	F. Coliform	MPN/100ml		25	2	2	2	2	2	2	2	2	2	2	4.09091	25	2
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2016-2017

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.16	18.05.2016	13.6.2016	11.07.2016	10.08.2016	15.09.16	12.10.16	16.11.16	8.12.16	09.01.17	07.02.17	07.03.17	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC						-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	8	8	16	8	14	8	8	10	10	8	2	10	9.17	16	2
5	Colour	PCS	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	-	-	-
6	Odour	T. No	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	-	-	-
7	pH	pH Unit	7.63	7.68	8.16	8.56	7.13	8.28	8.18	7.81	7.96	7.86	8.5	7.86	7.97	8.56	7.13
8	Sp. Conductivity	µMhos/cm.	276	276	342	342	200	275	262	284	268	296	290	288	283.25	342	200
9	T. Solids	mg/1	198	198	260	262	86	196	232	196	216	224	204	222	207.83	262	86
10	D. Solids	mg/1	188	186	248	250	52	180	224	208	202	212	180	210	195.00	250	52
11	S. Solids	mg/1	10	12	12	12	34	16	8	12	14	12	24	12	14.83	34	8
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.15	-	-	-	-	0.02	0.09	0.15	0.02
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.165	-	-	-	-	BDL	0.17	0.17	0.165
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	0.84	0.84	0.84	0.84
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.231	-	-	-	-	0.22	0.23	0.231	0.22
16	Chloride	mg/1	33.98	28.99	27.59	26.99	11.82	30.85	28.57	31.81	30.55	41.98	11.82	28.92	27.82	41.98	11.82
17	Sulphate (SO_4)	mg/1		-	-	-	-	-	7	-	-	-	-	3.86	5.43	7.00	3.86
18	T. Alkalinity	mg/1	132	132	136	134	66	138	148	130	138	134	132	132	129.33	148	66
19	T. Hardness	mg/1	134	134	164	162	64	142	118	154	156	152	118	146	137.00	164	64
20	Cal.Hardness	mg/1	98	98	112	80	46	112	74	104	124	104	60	104	93.00	124	46
21	Magnesium H.	mg/1	36	36	52	82	18	30	44	50	32	48	58	42	44.00	82	18
22	D. Oxygen	mg/1	8.1	7.5	7.2	7.1	7.7	7.6	7.8	7.9	7.6	7.5	7.9	7.6	7.63	8.1	7.1
23	B.O.D.	mg/1	1.5	2.3	1.5	0.9	1.7	1.4	1.6	1.6	1.3	1.8	1.7	1.7	1.58	2.3	0.9
24	C.O.D.	mg/1	20	20	9.88	10	9.88	10.0	20	18.72	10	19.76	20	10	14.85	20	9.88
25	Sodium	mg/1	14.9	14.7	6.12	9.6	-	15.12	6.12	14.87	14.38	13.73	17.15	7.63	12.21	17.15	6.12
26	Potassium	mg/1	1.3	1.3	2.3	2.1	-	1.5	1.8	1.02	2.1	1.9	2.63	1.4	1.76	2.63	1.02
27	T. Coliform	MPN/100ml	90	140	90	90	130	120	110	130	94	90	110	90	107	140	90
28	F. Coliform	MPN/100ml	<2	25	2	2	2	2	2	2	2	2	2	2	4.09091	25	2
Cat. B			**B**	**B**	**B**	**C**	**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**		

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2016-2017

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.16	18.05.2016	13.6.2016	11.07.2016	10.08.201	15.09.16	12.10.16	16.11.16	08.12.16	09.01.17	07.02.17	07.03.17	-	-	-
2	Appearance			Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC						-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	9	7	18	6	15	12	10	12	8	10	3	12	10.17	18	3
5	Colour	PCS	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	-	-	-
6	Odour	T. No	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	-	-	-
7	pH	pH Unit	7.79	7.68	8.36	8.62	6.68	8.43	8.36	8.23	8.12	8.18	8.62	7.94	8.08	8.62	6.68
8	Sp. Conductivity	µMhos/cm.	273	274	318	298	173	268	258	286	268	272	293	228	267.42	318	173
9	T. Solids	mg/1	210	198	222	232	106	208	226	200	216	218	208	230	206.17	232	106
10	D. Solids	mg/1	196	186	212	222	80	142	216	210	202	206	186	218	189.67	222	80
11	S. Solids	mg/1	14	12	10	10	26	30	10	10	14	12	22	12	15.17	30	10
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.231	-	-	-	-	0.016	0.1235	0.231	0.016
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.19	-	-	-	-	BDL	0.19	0.19	0.19
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	0.86	0.86	0.86	0.86
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.231	-	-	-	-	0.02	0.13	0.231	0.02
16	Chloride	mg/1	35.98	29.99	31.53	30.99	9.8	32.78	30.55	32.78	33.5	32.98	17.73	30.85	29.12	35.98	9.8
17	Sulphate (SO_4)	mg/1		-	-	-	-	-	9	-	-	-	-	2.58	5.79	9.00	2.58
18	T. Alkalinity	mg/1	132	134	132	132	68	136	152	134	138	142	138	140	131.50	152	68
19	T. Hardness	mg/1	136	148	128	128	68	140	128	152	146	126	128	192	135.00	192	68
20	Cal.Hardness	mg/1	98	116	66	56	50	106	76	100	100	76	90	142	89.67	142	50
21	Magnesium H.	mg/1	38	32	62	72	18	34	52	52	46	50	38	50	45.33	72	18
22	D. Oxygen	mg/1	8.4	7.6	7	7.2	6.7	7.7	7.7	7.7	7.9	7.8	7.8	7.8	7.61	8.4	6.7
23	B.O.D.	mg/1	1.3	2.5	1.4	1	1.8	1.4	1.8	1.8	1.6	1.9	1.8	1.8	1.68	2.5	1
24	C.O.D.	mg/1	20	20	9.88	10	9.88	20.0	20	28.08	20	19.88	20	20	18.14	28.08	9.88
25	Sodium	mg/1	18.3	16.8	5.2	8.3	-	14.94	7.08	15.82	14.38	13.68	17.96	8.31	12.80	18.3	5.2
26	Potassium	mg/1	1.6	1.4	2.1	2.6	-	1.4	2	1.03	1.09	1.03	2.59	1.6	1.68	2.6	1.03
27	T. Coliform	MPN/100ml	110	110	130	60	110	130	130	110	90	130	90	130	110.83	130	60
28	F. Coliform	MPN/100ml	2	22	2	2	2	2	2	2	2	2	2	2	3.67	22	2
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2016-2017

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.16	18.05.2016	13.6.2016	11.07.2016	10.08.201	15.09.16	12.10.16	16.11.16	8.12.2016	09.01.2017	07.02.17	07.03.17	-	-	-
2	Appearance			Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC						-	-	-	-	-	-	-	-	-	-
4	Turbidity	NTU	1.2	10	18	26	16	10	8	6	10	8	3	8	10.35	26	1.2
5	Colour	PCS	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	CL	-	-	-
6	Odour	T. No	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	OL	-	-	-
7	pH	pH Unit	7.76	7.88	7.84	8.36	7.04	8.26	8.12	7.86	7.62	7.98	7.85	7.78	7.86	8.36	7.04
8	Sp. Conductivity	µMhos/cm.	336	282	232	232	181	266	268	346	312	284	280	292	275.92	346	181
9	T. Solids	mg/1	242	232	184	192	100	142	234	256	246	238	192	232	207.50	256	100
10	D. Solids	mg/1	234	224	166	166	72	126	224	248	234	228	180	224	193.83	248	72
11	S. Solids	mg/1	8	8	18	26	28	16	10	8	12	10	12	8	13.67	28	8
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.17	-	-	-	-	0.012	0.09	0.17	0.012
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.165	-	-	-	-	BDL	0.17	0.165	0.165
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	-	-	-	-	-	0.82	0.82	0.82	0.82
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.132	-	-	-	-	0.04	0.09	0.132	0.04
16	Chloride	mg/1	35.98	26.99	28.56	35.98	12.81	30.85	35.5	27.96	27.59	29.99	15.76	30.85	28.24	35.98	12.81
17	Sulphate (SO_4)	mg/1		-	-	-	-	-	7.75	-	-	-	-	2.48	5.12	7.75	2.48
18	T. Alkalinity	mg/1	124	124	138	146	70	140	132	118	122	124	120	128	123.83	146	70
19	T. Hardness	mg/1	168	168	104	112	70	136	134	160	184	140	130	142	137.33	184	70
20	CalciumHardness	mg/1	118	118	62	66	52	100	100	112	98	108	86	108	94.00	118	52
21	Magnesium H.	mg/1	50	50	42	46	18	36	34	48	36	32	44	34	39.17	50	18
22	D. Oxygen	mg/1	8.1	7.8	7.4	7.4	8.1	7.8	7.6	8	7.8	7.77	7.7	7.6	7.76	8.1	7.4
23	B.O.D.	mg/1	1.7	2.5	1.7	1.9	1.6	1.5	1.5	1.6	1.5	1.4	1.6	1.5	1.67	2.5	1.4
24	C.O.D.	mg/1	20	20	19.76	20	9.8	10.0	20	18.72	20	19.88	20	20	18.18	20	9.8
25	Sodium	mg/1	15.6	15.6	5.32	5.84	-	14.72	14.16	15.02	14.38	8.34	18.02	7.78	12.25	18.02	5.32
26	Potassium	mg/1	1.3	1.4	2.2	2	-	1.2	1.02	1.1	1.04	1.02	2.46	1.01	1.43	2.46	1.01
27	T. Coliform	MPN/100ml	90	150	50	50	90	110	130	94	110	90	90	130	98.67	150	50
28	F. Coliform	MPN/100ml		27	2	2	2	2	2	2	2	2	2	2	4.27	27	2
Cat. B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2016-2017

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Dry	-	-	-	-	-	12.10.16	-	-	09.1.17	-	-	-	-	-
2	Appearance			-	-	-	-	-		-	-		-	-	-	-	-
3	Temperature	oC		-	-	-	-	-	-	-	-		-	-	-	-	-
4	Turbidity	NTU	-	-	-	-	-	-	12	-	-	8	-	-	10.00	12	8
5	Colour	PCS	-	-	-	-	-	-	CL	-	-	Colourless	-	-	-	-	-
6	Odour	T. No	-	-	-	-	-	-	OL	-	-	Odourless	-	-	-	-	-
7	pH	pH Unit	-	-	-	-	-	-	8.38	-	-	7.98	-	-	8.18	8.38	7.98
8	Sp. Conductivity	µMhos/cm.	-	-	-	-	-	-	234	-	-	250	-	-	242	250	234
9	T. Solids	mg/1	-	-	-	-	-	-	186	-	-	218	-	-	202	218	186
10	D. Solids	mg/1	-	-	-	-	-	-	170	-	-	232	-	-	201	232	170
11	S. Solids	mg/1	-	-	-	-	-	-	16	-	-	14	-	-	15	16	14
12	Amm. Nitrogen	mg/1	-	-	-	-	-	-	0.231	-	-		-	-	0.231	0.231	0.231
13	Nitrite Nitrogen	mg/1	-	-	-	-	-	-	0.165	-	-		-	-	0.165	0.165	0.165
14	Nitrate Nitrogen	mg/1	-	-	-	-	-	-	-	-	-		-	-			
15	Phosphate (PO_4)	mg/1	-	-	-	-	-	-	0.85	-	-		-	-	0.85	0.85	0.85
16	Chloride	mg/1	-	-	-	-	-	-	30.55	-	-	30.99	-	-	30.77	30.99	30.55
17	Sulphate (SO_4)	mg/1	-	-	-	-	-	-	7.75	-	-		-	-	7.75	7.75	7.75
18	T. Alkalinity	mg/1	-	-	-	-	-	-	154	-	-	138	-	-	146	154	138
19	T. Hardness	mg/1	-	-	-	-	-	-	124	-	-	142	-	-	133	142	124
20	CalciumHardness	mg/1	-	-	-	-	-	-	76	-	-	100	-	-	88	100	76
21	Magnesium H.	mg/1	-	-	-	-	-	-	48	-	-	42	-	-	45	48	42
22	D. Oxygen	mg/1	-	-	-	-	-	-	7.6	-	-	7.7	-	-	7.65	7.7	7.6
23	B.O.D.	mg/1	-	-	-	-	-	-	1.5	-	-	2	-	-	1.75	2	1.5
24	C.O.D.	mg/1	-	-	-	-	-	-	20	-	-	20	-	-	20	20	20
25	Sodium	mg/1	-	-	-	-	-	-	6.98	-	-	13.3	-	-	10.14	13.3	6.98
26	Potassium	mg/1	-	-	-	-	-	-	2.2	-	-	2.03	-	-	2.115	2.2	2.03
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	110	-	-	90	-	-			
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	2	-	-	2	-	-	-	-	-
29	T.K.N.	mg/1	-	-	-	-	-	-		-	-	-	-	-	-	-	-
Cat. B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2016-2017

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Dry	-	-	11.07.16	-	14.09.16	12.10.16			09.1.17	not approachable		-	-	-
2	Appearance														-	-	-
3	Temperature	oC							-						-	-	-
4	Turbidity	NTU				6		10	14			10			10	14	6
5	Colour	PCS				Colourless		Colourless	CL			Colourless			-	-	-
6	Odour	T. No				Odourless		Odourless	OL			Odourless			-	-	-
7	pH	pH Unit				8.31		8.47	8.42			7.79			8.25	8.47	7.79
8	Sp. Conductivity	µMhos/cm.				248		230	246			294			254.5	294	230
9	T. Solids	mg/1				192		232	206			284			228.5	284	192
10	D. Solids	mg/1				178		218	188			268			213.00	268	178
11	S. Solids	mg/1				14		12	18			16			15	18	12
12	Amm. Nitrogen	mg/1				-		-	0.18						0.18	0.18	0.18
13	Nitrite Nitrogen	mg/1				-		-	0.195						0.195	0.195	0.195
14	Nitrate Nitrogen	mg/1				-		-	-								
15	Phosphate (PO_4)	mg/1				-		-	0.792						0.79	0.792	0.792
16	Chloride	mg/1				32.98		31.99	34.49			31.99			32.86	34.49	31.99
17	Sulphate (SO_4)	mg/1							9						9.00	9.00	9
18	T. Alkalinity	mg/1				148		112	158			142			140.00	158	112
19	T. Hardness	mg/1				116		110	130			148			126.00	148	110
20	CalciumHardness	mg/1				68		76	78			104			81.50	104	68
21	Magnesium H.	mg/1				48		34	52			44			44.50	52	34
22	D. Oxygen	mg/1				7.6		7.4	7.5			7.3			7.45	7.6	7.3
23	B.O.D.	mg/1				1.8		1.6	1.9			2.2			1.88	2.2	1.6
24	C.O.D.	mg/1				20		10	20			20			17.50	20	10
25	Sodium	mg/1				5.36		10.4	7.85			14.32			9.48	14.32	5.36
26	Potassium	mg/1				1.9		1.1	2.4			2.04			1.86	2.4	1.1
27	T. Coliform	MPN/100ml				90		40	-			90			73.33	90	40
28	F. Coliform	MPN/100ml				2		2				2			2.00	2	2
29	T.K.N.	mg/1										-			-	-	-
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2017-2018

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		10.04.17	03.05.2017	7.06.17	04.07.17	8.8.17	7.09.17	06.10.2017	3.11.2017	6.12.17	4.1.18	9.2.18	9.3.18	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	28	28	29	24	23	26	27	26	19	18	17	21	23.83	29	17
4	Turbidity	NTU	8	10	12	38	16	14	16	-	'Instrument out of order	-	-	16	16.25	38.0	8.0
5	Colour	PCS	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	-	-	-
6	Odour	T. No	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	-	-	-
7	pH	pH Unit	7.76	7.82	7.96	7.74	7.68	7.48	7.84	8.12	8.55	8.07	8.64	7.82	7.96	8.6	7.5
8	Sp. Conductivity	µMhos/cm.	286	288	296	292	288	286	296	226	218	232	270	312	274.17	312.0	218.0
9	T. Solids	mg/1	206	206	212	228	220	198	252	242	112	246	260	250	219.33	260.0	112.0
10	D. Solids	mg/1	198	196	200	194	188	186	238	226	88	224	216	202	196.33	238.0	88.0
11	S. Solids	mg/1	8	10	12	34	32	12	14	16	24	22	44	48	23.00	48.0	8.0
12	Amm. Nitrogen	mg/1	-	0.20	0.25	0.2	0.15	0.2	0.155	0.285	-	0.4	BDL	0.25	0.23	0.4	0.2
13	Nitrite Nitrogen	mg/1	-	0.196	0.198	0.9	0.85	0.75	0.07	BDL	-	0.875	3.64	0.07	0.84	3.6	0.1
14	Nitrate Nitrogen	mg/1	-	-	-	BDL	1.36	1.35	0.975	1.45	-	BDL	BDL	1.2	1.27	1.5	1.0
15	Phosphate (PO_4)	mg/1	-	0.170	0.175	0.17	0.170	0.165	1.48	BDL	-	BDL	BDL	BDL	0.39	1.5	0.2
16	Chloride	mg/1	26.99	27.99	28.99	30.55	27.59	25.62	26.6	21.68	20.99	20.43	20.43	32.98	25.90	33.0	20.4
17	Sulphate (SO_4)	mg/1	-	7	6.0	6.26	6.2	6.15	14	6.75	-	6.28	4.6	6.26	6.95	14.0	4.6
18	T. Alkalinity	mg/1	134	138	142	144	138	134	140	144	136	146	92	136	135.33	146.0	92.0
19	T. Hardness	mg/1	126	132	140	136	128	132	142	88	134	94	112	132	124.67	142.0	88.0
20	CalciumHardness	mg/1	88	92	94	98	92	98	104	44	96	46	64	94	84.17	104.0	44.0
21	Magnesium H.	mg/1	38	40	46	38	36	34	38	44	38	48	48	38	40.50	48.0	34.0
22	D. Oxygen	mg/1	8.1	7.9	7.5	8.1	7.9	7.7	7.9	8.0	8.4	8.1	8.3	7.6	7.96	8.4	7.5
23	B.O.D.	mg/1	1.5	2.2	2.0	1.7	1.9	1.8	2.1	2.1	1.5	2.2	1.9	1.5	1.87	2.2	1.5
24	C.O.D.	mg/1	19.76	20.0	20.0	20.5	19.2	19.2	20.0	19.2	20	20	20	9.6	18.96	20.5	9.6
25	Sodium	mg/1	15.10	14.84	15.21	14.38	16.68	14.21	14.6	9.64	15.62	9.66	25.62	22.64	15.68	25.6	9.6
26	Potassium	mg/1	1.03	1.04	1.06	1.05	1.02	1.01	1.2	2.62	3.21	2.11	5.71	1.6	1.89	5.7	1.0
27	T. Coliform	MPN/100ml	110	130	130	110	230	110	140	130	>1600	110	>1600	240	144.00	240.0	110.0
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	-	2	-	2			
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2017-2018

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		10.04.17	03.05.2017	13.6.2017	04.07.17	8.8.17	7.09.17	06.10.17	03.11.17	6.12.17	4.1.18	9.2.18	9.3.18	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	29	28	29	28	23	26	27	25	21	20	17	21	24.50	29	17
4	Turbidity	NTU	8	10	12	14	14	12	16	-	-	-	-	16	12.75	16	8
5	Colour	PCS	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	-	-	-
6	Odour	T. No	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	-	-	-
7	pH	pH Unit	7.96	7.94	7.82	8.02	7.67	7.52	7.86	8.24	7.87	7.83	8.64	7.82	7.93	8.64	7.52
8	Sp. Conductivity	µMhos/cm.	276	278	284	294	252	256	296	216	211	214	270	312	263.25	312	211
9	T. Solids	mg/1	230	232	240	198	244	216	246	236	234	238	260	250	235.33	260	198
10	D. Solids	mg/1	218	222	228	172	208	206	232	224	220	222	216	202	214.17	232	172
11	S. Solids	mg/1	12	10	12	26	36	10	14	12	14	16	44	48	21.17	48	10
12	Amm. Nitrogen	mg/1	-	-	0.24	0.12	0.2	0.25	0.07	0.12	0.2	0.08	BDL	0.25	0.17	0.25	0.07
13	Nitrite Nitrogen	mg/1	-	-	0.15	0.135	0.15	0.15	0.06	BDL	0.15	1.26	3.64	0.07	0.64	3.64	0.06
14	Nitrate Nitrogen	mg/1	-	-	BDL	1.1	1.26	BDL	1.85	1.1	1.2	0.18	BDL	1.2	1.13	1.85	0.18
15	Phosphate (PO_4)	mg/1	-	-	0.25	0.462	0.2	0.2	0.198	BDL	0.3	0.2	BDL	BDL	0.26	0.462	0.198
16	Chloride	mg/1	31.53	32.98	34.98	20.99	26.6	25.62	24.63	20.66	30.0	26.93	20.43	32.98	27.36	34.98	20.43
17	Sulphate (SO_4)	mg/1	-	-	5.2	9	5.34	5.26	13.75	6.75	6.26	6.28	4.6	6.26	6.87	13.75	4.6
18	T. Alkalinity	mg/1	134	136	138	122	129	126	7.86	138	134	138	92	136	119.24	138	7.86
19	T. Hardness	mg/1	198	202	210	136	194	182	134	90	88	90	112	132	147.33	210	88
20	Cal.Hardness	mg/1	152	154	158	114	156	148	96	42	44	42	64	94	105.33	158	42
21	Magnesium H.	mg/1	46	48	52	22	38	34	38	48	44	48	48	38	42.00	52	22
22	D. Oxygen	mg/1	7.8	7.7	7.6	7.6	7.7	7.8	8.2	8.0	8.1	8.2	8.3	7.6	7.88	8.3	7.6
23	B.O.D.	mg/1	2.4	2.5	2.6	2.2	2.1	1.8	1.8	2.0	2.2	2.1	1.9	1.5	2.09	2.6	1.5
24	C.O.D.	mg/1	9.22	10.0	20	29.64	9.6	9.6	20.0	9.2	10.0	10	20	9.6	13.91	29.64	9.2
25	Sodium	mg/1	13.26	14.21	14.38	18.4	12.64	6.48	15.7	9.54	9.74	9.78	25.62	22.64	14.37	25.62	6.48
26	Potassium	mg/1	1.8	1.6	1.5	2.1	1.6	1.4	1.3	1.58	1.01	1.04	5.71	1.6	1.85	5.71	1.01
27	T. Coliform	MPN/100ml	140	130	140	150	230	130	110	110	110	130	1600	240	268.33	1600	110
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	2	2	-	2	2	2	2
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2017-2018

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		10.04.17	3.05.2017	13.6.2017	04.07.17	08.08.17	07.09.17	06.10.17	3.11.17	6.12.17	4.1.18	9.2.18	9.3.18	-	-	-
2	Appearance		Clear	Clear	Clear		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	28	28	29	25	24	23	27	25	17	19	16	19	23.33	29.0	16.0
4	Turbidity	NTU	10	12	12	16	14	12	14	10	Instrument out of order	-	-	14	12.67	16.0	10.0
5	Colour	PCS	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	-	-	-
6	Odour	T. No	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	-	-	-
7	pH	pH Unit	7.88	7.76	7.82	8.13	7.76	7.78	7.96	7.76	8	7.8	8.59	7.91	7.93	8.6	7.8
8	Sp. Conductivity	µMhos/cm.	266	272	284	302	264	268	286	266	241	264	269	330.6	276.05	330.6	241.0
9	T. Solids	mg/1	234	238	244	218	260		258	242	174	236	256	248	237.09	260.0	174.0
10	D. Solids	mg/1	222	228	232	184	226	228	244	226	122	222	210	231	214.58	244.0	122.0
11	S. Solids	mg/1	10	10	12	34	34	12	14	10	52	14	46	17	22.08	52.0	10.0
12	Amm. Nitrogen	mg/1	-	0.250	0.266	0.13	0.2	0.25	0.025	0.252	-	0.14	BDL	0.25	0.20	0.3	0.0
13	Nitrite Nitrogen	mg/1	-	0.20	0.25	0.145	0.15	0.2	0.06	0.22	-	0.24	BDL	0.2	0.19	0.3	0.1
14	Nitrate Nitrogen	mg/1	-	-		1.25	1.62	1.86	1.96	1.0	-	1.8	3	1.84	1.79	3.0	1.0
15	Phosphate (PO_4)	mg/1	-	0.24	0.3	0.528	0.1	0.3	1.45	BDL	-	0.3	BDL	0.3	0.44	1.5	0.1
16	Chloride	mg/1	33.5	34.49	35.98	23.99	26.6	23.65	37.44	23.63	23.99	24	20.43	35.98	28.64	37.4	20.4
17	Sulphate (SO_4)	mg/1	-	10.0	10.8	10.5	6.94	7.21	13.25	6.75	-	7.11	9.2	9.25	9.10	13.3	6.8
18	T. Alkalinity	mg/1	138	140	146	126	142	146	136	148	144	140	118	126	137.50	148.0	118.0
19	T. Hardness	mg/1	192	196	202	142	198	206	138	208	134	192	134	132	172.83	208.0	132.0
20	Cal.Hardness	mg/1	148	150	154	116	164	164	102	166	94	146	64	78	128.83	166.0	64.0
21	Magnesium H.	mg/1	44	46	48	26	34	42	36	44	40	46	70	54	44.17	70.0	26.0
22	D. Oxygen	mg/1	7.9	7.7	7.4	7.3	7.5	7.8	8.3	8.2	8.8	8.1	6.9	8.4	7.86	8.8	6.9
23	B.O.D.	mg/1	2.5	2.6	2.7	2.4	2.5	2.4	2.0	2.0	1.5	2	1.6	1.5	2.14	2.7	1.5
24	C.O.D.	mg/1	19.76	20.0	20	29.64	19.2	20.0	20.0	19.2	20	20	20	19.76	20.63	29.6	19.2
25	Sodium	mg/1	14.31	13.21	13.42	16.6	13.62	12.86	14.8	1.52	15.06	1.44	26	12.42	12.94	26.0	1.4
26	Potassium	mg/1	1.6	1.02	1.04	2.2	1.01	1.02	1.4	1.42	2.09	1.46	5.83	1.82	1.83	5.8	1.0
27	T. Coliform	MPN/100ml	130	150	150	220	350	220	150	140	1600	140	1600	1600	537.50	1600.0	130.0
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	2	2	2	2			
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2017-2018

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		10.04.17	03.05.2017	13.06.2017	04.07.17	08.08.17	7.9.17	06.10.17	03.11.17	6.12.17	4.1.18	9.2.18	9.3.18	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	28	29	29	29	24	24	27	25	18	19	-	20	24.73	29.0	18.0
4	Turbidity	NTU	10	12	12	12	34	14	14	-	'Instrument out of order	-	-	-	15.43	34.0	10.0
5	Colour	PCS	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	C. Less	-	-	-
6	Odour	T. No	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	O. less	-	-	-
7	pH	pH Unit	7.66	7.74	7.84	7.67	7.84	7.72	7.78	7.86	8.35	7.82	8.33	7.92	7.88	8.4	7.7
8	Sp. Conductivity	µMhos/cm.	281	288	292	290	284	278	284	274	223	278	274	334.5	281.71	334.5	223.0
9	T. Solids	mg/1	240	244	252	306	466	238	242	236	146	242	274	246	261.00	466.0	146.0
10	D. Solids	mg/1	228	232	240	270	242	228	228	222	138	226	232	217	225.25	270.0	138.0
11	S. Solids	mg/1	12	12	12	36	224	10	14	14	8	16	42	29	35.75	224.0	8.0
12	Amm. Nitrogen	mg/1	-	0.20	0.25	0.3	0.4	0.25	0.025	0.135	-	0.24	BDL	0.23	0.23	0.4	0.0
13	Nitrite Nitrogen	mg/1	-	0.170	0.175	0.7	0.75	0.68	0.050	BDL	-	1.54	3.51	0.6	0.91	3.5	0.1
14	Nitrate Nitrogen	mg/1	-	-	3.64	0.2	1.68	1.54	1.95	1.45	-	BDL	BDL	1.52	1.71	3.6	0.2
15	Phosphate (PO_4)	mg/1	-	0.135	0.14	0.35	0.48	0.3	0.162	BDL	-	BDL	BDL	0.14	0.24	0.5	0.1
16	Chloride	mg/1	33.74	33.50	34.98	30.55	28.57	26.6	27.59	21.68	19.99	22.29	20.43	27.6	27.29	35.0	20.0
17	Sulphate (SO_4)	mg/1	-	7.80	7.92	6.54	7.61	7.52	15.0	7.25	-	6.52	5.6	6.25	7.80	15.0	5.6
18	T. Alkalinity	mg/1	130	134	142	136	134	138	142	148	126	150	102	138	135.00	150.0	102.0
19	T. Hardness	mg/1	124	144	148	142	130	120	128	198	136	208	110	132	143.33	208.0	110.0
20	CalciumHardness	mg/1	90	108	112	108	90	92	94	150	90	168	64	94	105.00	168.0	64.0
21	Magnesium H.	mg/1	34	36	36	34	32	28	34	48	46	40	-	38	36.91	48.0	28.0
22	D. Oxygen	mg/1	7.9	7.6	7.5	7.9	7.8	7.9	8.1	8.0	8.3	8.1	7.2	7.6	7.83	8.3	7.2
23	B.O.D.	mg/1	7.9	7.6	7.5	7.9	2.0	1.8	1.9	1.9	1.6	2	1.5	1.5	3.76	7.9	1.5
24	C.O.D.	mg/1	19.76	20.0	20	90.2	10.0	10.0	20.0	9.6	20	10	20	19.2	22.40	90.2	9.6
25	Sodium	mg/1	14.38	14.78	15.12	13.32	12.38	11.46	14.61	9.62	12.96	12.36	26.23	14.76	14.33	26.2	9.6
26	Potassium	mg/1	1.02	1.4	1.6	1.3	1.1	1.2	1.2	2.46	2.45	1.2	5.62	1.26	1.82	5.6	1.0
27	T. Coliform	MPN/100ml	90	110	130	120	130	220	130	110	1600	120	1600	110	372.50	1600.0	90.0
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	-	2	-	2			
Cat. B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2017-2018

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		10.04.2017			4.7.17			NA			4.1.18			-	-	-
2	Appearance					-						Clear			-	-	-
3	Temperature	oC	28									20			24.00	28.0	20.0
4	Turbidity	NTU	12			-						-			12.00	12.0	12.0
5	Colour	PCS	C. Less			C. Less						C. Less			-	-	-
6	Odour	T. No	O. less			O. less						O. less			-	-	-
7	pH	pH Unit	7.84			7.80						7.82			7.82	7.8	7.8
8	Sp. Conductivity	µMhos/cm.	328			330						336			331.33	336.0	328.0
9	T. Solids	mg/1	238			242						260			246.67	260.0	238.0
10	D. Solids	mg/1	224			220						232			225.33	232.0	220.0
11	S. Solids	mg/1	14			22						28			21.33	28.0	14.0
12	Amm. Nitrogen	mg/1				-						-					
13	Nitrite Nitrogen	mg/1				-						-					
14	Nitrate Nitrogen	mg/1				-						-					
15	Phosphate (PO_4)	mg/1				-						-					
16	Chloride	mg/1	32.52			32.52						30.65			31.90	32.5	30.7
17	Sulphate (SO_4)	mg/1				-						-					
18	T. Alkalinity	mg/1	114			122						132			122.67	132.0	114.0
19	T. Hardness	mg/1	148			158						162			156.00	162.0	148.0
20	CalciumHardness	mg/1	112			118						120			116.67	120.0	112.0
21	Magnesium H.	mg/1	36			40						42			39.33	42.0	36.0
22	D. Oxygen	mg/1	7.7			7.6						7.5			7.60	7.7	7.5
23	B.O.D.	mg/1	2.1			2.2						2.3			2.20	2.3	2.1
24	C.O.D.	mg/1	19.28			19.28						28.8			22.45	28.8	19.3
25	Sodium	mg/1	14.28			14.18						14.22			14.23	14.3	14.2
26	Potassium	mg/1	1.03			1.03						1.06			1.04	1.1	1.0
27	T. Coliform	MPN/100ml	110			120						120			116.67	120.0	110.0
28	F. Coliform	MPN/100ml	2			2						2			2.00	2.0	2.0
29	T.K.N.	mg/1													-	-	-
Cat. B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2017-2018

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		10.04.2017			04.7.17			6.10.17			4.1.18			-	-	-
2	Appearance								Clear			Clear			-	-	-
3	Temperature	oC	28						27			16			23.67	28	16
4	Turbidity	NTU	12			-			-			12			12	12	12
5	Colour	PCS				C. Less			C. Less			C. Less			-	-	-
6	Odour	T. No				O. less			O. less			O. less			-	-	-
7	pH	pH Unit	7.76			7.62			8.12			8.14			7.91	8.14	7.62
8	Sp. Conductivity	µMhos/cm.	332			340			162			286			280	340	162
9	T. Solids	mg/1	242			248			126			222			209.5	248	126
10	D. Solids	mg/1	228			220			90			204			185.50	228	90
11	S. Solids	mg/1	14			28			36			18			24	36	14
12	Amm. Nitrogen	mg/1				-			BDL			0.145			0.145	0.145	0.145
13	Nitrite Nitrogen	mg/1				-			0.2			0.145			0.17	0.20	0.145
14	Nitrate Nitrogen	mg/1				-			1.2			2.95			2.08	2.95	1.2
15	Phosphate (PO_4)	mg/1				-			BDL			0.759			0.76	0.759	0.759
16	Chloride	mg/1	34.49			34.69			32.52			35.47			34.29	35.47	32.52
17	Sulphate (SO_4)	mg/1				-			6.5			10			8.25	10.00	6.5
18	T. Alkalinity	mg/1	148			160			162			152			155.50	162	148
19	T. Hardness	mg/1	152			164			126			124			141.50	164	124
20	CalciumHardness	mg/1	114			120			90			72			99.00	120	72
21	Magnesium H.	mg/1	38			44			36			52			42.50	52	36
22	D. Oxygen	mg/1	7.6			7.7			7.9			7.6			7.70	7.9	7.6
23	B.O.D.	mg/1	2.3			2.4			1.58			2.4			2.17	2.4	1.58
24	C.O.D.	mg/1	19.28			19.88			20			19.2			19.59	20	19.2
25	Sodium	mg/1	15.02			16.2			12.2			-			14.47	16.2	12.2
26	Potassium	mg/1	1.02			1.02			1.2			14.2			4.36	14.2	1.02
27	T. Coliform	MPN/100ml	130			140			70			-			113.33	140	70
28	F. Coliform	MPN/100ml	2			2			2			-			-	-	-
29	T.K.N.	mg/1													-	-	-
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2018-2019

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		02.04.18	16.05.18	18.06.18	06.07.18	06.08.18	06.09.18	08.10.18	14.11.18	05.12.18	16.01.19	11.02.19	07.03.19	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	28	30	31	28	26	24	28	25	-	19	20	22	25.55	31.0	19.0
4	Turbidity	NTU	8.9	16.1	18.2	16	21.3	32.8	26.4	21.3	16.6	16	18.4	22.3	19.53	32.8	8.9
5	Colour	PCS	Colorless	Colourless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	-	-	-
6	Odour	T. No	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	7.62	7.83	8.12	7.94	8.23	8.2	8.28	8.29	8.19	7.6	8.11	8.24	8.05	8.3	7.6
8	Sp. Conductivity	µMhos/cm.	298	299	318.6	311	269.2	326.1	173.6	168	262	290	279.2	296.7	274.28	326.1	168.0
9	T. Solids	mg/1	204	234	242	238	260	200	172	184	194	246	218	234	218.83	260.0	172.0
10	D. Solids	mg/1	160	192	194	220	instrument	172	140	146	162	200	176	186	177.09	220.0	140.0
11	S. Solids	mg/1	44	42	48	18	15	28	22	38	32	46	42	48	35.25	48.0	15.0
12	Amm. Nitrogen	mg/1	0.25	0.23	0.245	0.122	BDL	BDL	BDL	BDL	0.105	BDL	BDL	BDL	0.19	0.3	0.1
13	Nitrite Nitrogen	mg/1	0.01	0.01	0.12	BDL	0.05	0.05	0.04	0.050	0.155	0.01	0.5	0.02	0.09	0.5	0.0
14	Nitrate Nitrogen	mg/1	3.42	3.36	3.3	1.65	4.54	4.38	2.86	4.38	1.25	5.01	2.5	4.84	3.46	5.0	1.3
15	Phosphate (PO_4)	mg/1	0.162	0.173	0.92	0.66	0.07	0.06	0.08	0.06	0.396	0.37	0.49	0.28	0.31	0.9	0.1
16	Chloride	mg/1	32.98	29.56	34.49	44.34	19.71	24.63	19	19.0	23.99	26	18.72	22.66	26.26	44.3	18.7
17	Sulphate (SO_4)	mg/1	5.4	4.96	7.0	9.28	22.74	23.18	8.28	8.31	10	12.46	10	7.62	10.77	23.2	5.0
18	T. Alkalinity	mg/1	118	96	130	100	114	140	98	102	112	120	126	136	116.00	140.0	96.0
19	T. Hardness	mg/1	128	118	134	128	142	148	92	98	118	132	138	158	127.83	158.0	92.0
20	CalciumHardness	mg/1	74	50	76	90	90	70	64	66	88	80	92	106	78.83	106.0	50.0
21	Magnesium H.	mg/1	54	68	58	38	52	78	28	32	30	52	46	52	49.00	78.0	28.0
22	D. Oxygen	mg/1	6.5	8.3	7.9	8.3	6.4	8.5	7.9	8.0	7.3	8.2	7.9	7.6	7.73	8.5	6.4
23	B.O.D.	mg/1	1.9	5.2	3.1	1.8	1.1	1.9	2.6	2.0	1.6	2.4	2.2	2.6	2.37	5.2	1.1
24	C.O.D.	mg/1	18.16	40	28.8	20	20.0	10	10	10.0	19.76	19.2	19.2	19.2	19.53	40.0	10.0
25	Sodium	mg/1	23.4	22.6	21.4	28.93	12.41	14.58	17.73	14.58	16.1	0.61	18.26	0.74	15.95	28.9	0.6
26	Potassium	mg/1	10.89	9.77	9.36	2.17	2.21	2.38	4.42	2.38	1.46	0.01	1.94	0.03	3.92	10.9	0.0
27	T. Coliform	MPN/100ml	240	280	350	110	140	170	280	170	170	170	150	170	200.00	350.0	110.0
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	2	2	2	2	2.00	2.0	2.0
29	Boron	mg/1	BDL	BDL	0.85	BDL	BDL	BDL	BDL	BDL	BDL	-	BDL	0.01	0.43	0.9	0.0
30	T.K.N.	mg/1	-	-	-	2.3	1.4	1.4	1.4	1.4	1.4	-			1.55	2.3	1.4
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2018-2019

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		02.04.18	16.05.18	18.06.18	06.07.18	06.08.18	06.09.18	08.10.18	14.11.18	05.12.18	16.01.19	11.02.19	07.03.19	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	27	29	31 ºC	29 ºC	25 ºC	24	28	26	-	19	20	23	24.50	29.0	19.0
4	Turbidity	NTU	26	16.9	22	14.7	19.5	29.8	21.2	27.3	10.2	20	18.4	22.6	20.72	29.8	10.2
5	Colour	PCS	Colorless	Colourless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	-	-	-
6	Odour	T. No	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	7.3	8.13	8.24	7.87	8.23	7.82	8.26	8.38	8.06	7.71	8.24	8.12	8.03	8.4	7.3
8	Sp. Conductivity	µMhos/cm.	280	293	304	526	270.7	299.2	138.4	176.2	199	281	264.4	278.3	275.85	526.0	138.4
9	T. Solids	mg/1	228	224	214	350	186	188	188	242	142	264	206	218	220.83	350.0	142.0
10	D. Solids	mg/1	190	188	182	332	instrument	160	140	210	124	213	162	170	188.27	332.0	124.0
11	S. Solids	mg/1	38	36	32	18	14	28	48	32	18	51	44	48	33.92	51.0	14.0
12	Amm. Nitrogen	mg/1	0.24	0.23	0.38	0.16	BDL	BDL	0.25	0.12	0.08	BDL	0.15	BDL	0.20	0.4	0.1
13	Nitrite Nitrogen	mg/1	0.06	0.04	0.12	0.09	0.05	0.05	0.12	0.080	0.12	0.56	0.12	0.64	0.17	0.6	0.0
14	Nitrate Nitrogen	mg/1	5.9	5.71	4.35	0.97	5.74	4.54	2.54	2.88	0.95	4.13	4.25	5.24	3.93	5.9	1.0
15	Phosphate (PO_4)	mg/1	0.17	0.15	0.42	1.46	0.08	0.07	0.06	0.04	0.198	0.03	0.49	0.04	0.27	1.5	0.0
16	Chloride	mg/1	25	27.59	29.56	48.28	22.66	22.66	30	21.0	16.99	24	23.65	24.51	26.33	48.3	17.0
17	Sulphate (SO_4)	mg/1	1.25	1.21	5	14	27.3	28.39	22.74	23.18	7	12.87	13.75	14.2	14.24	28.4	1.2
18	T. Alkalinity	mg/1	124	90	128	98	100	130	90	76	102	132	128	136	111.17	136.0	76.0
19	T. Hardness	mg/1	132	84	132	216	154	124	122	98	74	142	140	144	130.17	216.0	74.0
20	CalciumHardness	mg/1	78	56	96	176	90	52	94	44	56	96	92	92	85.17	176.0	44.0
21	Magnesium H.	mg/1	54	28	36	40	64	72	-	54	18	46	48	52	46.55	72.0	18.0
22	D. Oxygen	mg/1	7.5	8.4	8.2	8.1	6.1	7.8	8.4	8.6	7.8	8.4	8.1	7.9	7.94	8.6	6.1
23	B.O.D.	mg/1	2.1	2.4	2.6	1.8	2.3	2.7	2.9	2.4	1.3	2.1	2.4	2.8	2.32	2.9	1.3
24	C.O.D.	mg/1	18.16	20	19.2	20	20	20	20	20.0	19.76	19.2	19.2	19.2	19.56	20.0	18.2
25	Sodium	mg/1	27.67	26.55	28.91	21.77	13.56	13.69	19.77	13.58	14.3	0.89	18.6	0.78	16.67	28.9	0.8
26	Potassium	mg/1	5.7	5.23	6.21	3.96	2.14	2.89	4.59	4.33	1.07	0.03	2.72	0.02	3.24	6.2	0.0
27	T. Coliform	MPN/100ml	220	240	280	94	70	94	280	280	120	140	150	210	181.50	280.0	70.0
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	2	2	2	2	2.00	2.0	2.0
29	Boron	mg/1	BDL	0.6	2	0.03	0.8	0.7	BDL	BDL	BDL	-	BDL	BDL	0.83	2.0	0.0
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2018-2019

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		02.04.18	16.05.18	18.06.18	06.07.18	06.08.18	06.09.18	08.10.18	14.11.18	05.12.18	17.01.19	11.02.19	07.03.19	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	27	30	30 ºC	28	25	25	28	26	-	17	20	22	24.8	30	17
4	Turbidity	NTU	8	10.8	16.2	14.6	24.8	27.3	22.7	27.3	12.7	9.8	8.12	10.36	16.06	27.3	8
5	Colour	PCS	Colorless	Colourless	Colourless	Colourless	Colourless	Colourless	Colourless	Colourless	Colourless	Colourless	Colourless	Colorless	-	-	-
6	Odour	T. No	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odorless	-	-	-
7	pH	pH Unit	6.85	8.62	8.28	7.87	8.27	7.9	8.57	8.38	7.72	7.66	7.87	8.02	8.00	8.62	6.85
8	Sp. Conductivity	µMhos/cm.	285	292	288.2	302	276.5	296.2	170.1	176.2	226	293	268.4	274.6	262.35	302	170.1
9	T. Solids	mg/1	240	224	200	208	190	197	230	242	162	224	202	206	210.42	242	162
10	D. Solids	mg/1	214	196	168	190	Instrument out of	167	198	210	138	172	168	168	180.82	214	138
11	S. Solids	mg/1	26	28	32	18	16	30	32	32	24	52	34	38	30.17	52	16
12	Amm. Nitrogen	mg/1	0.12	0.11	0.12	0.44	BDL	BDL	0.25	0.12	0.09	BDL	BDL	BDL	0.18	0.44	0.09
13	Nitrite Nitrogen	mg/1	0.04	0.04	0.1	0.17	0.05	0.08	0.12	0.080	0.135	0.32	0.75	0.54	0.20	0.75	0.04
14	Nitrate Nitrogen	mg/1	8.28	7.38	4.35	1.8	4.81	4.65	2.88	2.88	1.05	3.98	3.5	3.72	4.11	8.28	1.05
15	Phosphate (PO_4)	mg/1	0.198	0.18	0.39	1.81	0.05	0.04	0.66	0.04	0.297	0.07	0.66	0.05	0.37	1.81	0.04
16	Chloride	mg/1	25	26.6	28.57	32.52	19.71	21.68	21	21.0	18.99	26	24.63	25.46	24.26	32.52	18.99
17	Sulphate (SO_4)	mg/1	3.64	3.52	6.75	12.23	22.34	26.04	23.18	23.18	8.5	12.4	13.75	10.21	13.81	26.04	3.52
18	T. Alkalinity	mg/1	96	84	102	102	114	156	72	76	98	176	158	164	116.50	176	72
19	T. Hardness	mg/1	192	94	102	144	152	194	92	98	104	184	152	158	138.83	194	92
20	Cal.Hardness	mg/1	178	54	66	78	84	72	42	44	80	112	88	102	83.33	178	42
21	Magnesium H.	mg/1	14	40	36	66	68	122	50	54	24	72	64	56	55.50	122	14
22	D. Oxygen	mg/1	6.7	8.1	7.8	8.1	6.4	7.9	8.9	8.6	7.6	8.4	8.2	8.4	7.93	8.9	6.4
23	B.O.D.	mg/1	2.2	2.4	2.6	1.9	2.1	2.3	2.6	2.4	1.4	2.7	2.5	2.3	2.28	2.7	1.4
24	C.O.D.	mg/1	18.16	30	28.8	20	20	20	20	20.0	19.76	28.8	19.2	19.2	21.99	30	18.16
25	Sodium	mg/1	27.91	25.86	27.9	26.01	12.55	13.58	18.78	13.58	14.9	0.98	16.3	1.21	16.63	27.91	0.98
26	Potassium	mg/1	9.54	9.2	8.21	2.38	2.17	2.84	4.33	4.33	1.32	0.02	2.96	0.04	3.95	9.54	0.02
27	T. Coliform	MPN/100ml	540	920	1600	110	140	120	280	280	150	220	120	150	385.83	1600	110
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	2	2	2	2	2.00	2	2
29	Boron	mg/1	BDL	BDL	0.2	0.03	BDL	BDL	BDL	BDL	BDL	-	BDL	0.01	0.08	0.2	0.01
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2018-2019

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		02.04.18	16.05.18	18.06.18	06.07.18	06.08.18	06.09.18	08.10.18	14.11.18	05.12.18	17.01.19	11.02.19	07.03.19	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	Clear	Clear	-	-	-
3	Temperature	oC	29	30	31 ºC	29 ºC	26 ºC	25	28	25	-	18	20	22	-	-	-
4	Turbidity	NTU	8.6	9.7	10.2	12	31.3	34.8	17.3	21.3	18.4	38	18.7	20.8	20.09	38.0	8.6
5	Colour	PCS	Colorless	Colourless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	-	-	-
6	Odour	T. No	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	7.15	7.86	7.96	7.89	8.22	8.11	8.44	8.36	8.23	7.87	8.26	8.14	8.04	8.4	7.2
8	Sp. Conductivity	µMhos/cm.	280	297	286	528	254.4	298.5	169.6	173.6	284	290	265.2	278.9	283.77	528.0	169.6
9	T. Solids	mg/1	212	210	212	358	224	204	218	228	208	268	212	220	231.17	358.0	204.0
10	D. Solids	mg/1	190	186	184	336	instrument	180	180	186	172	204	166	168	195.64	336.0	166.0
11	S. Solids	mg/1	22	24	28	22	18	24	38	42	36	64	46	52	34.67	64.0	18.0
12	Amm. Nitrogen	mg/1	0.26	0.21	0.215	0.23	BDL	BDL	0.215	0.23	0.135	BDL	0.15	BDL	0.21	0.3	0.1
13	Nitrite Nitrogen	mg/1	0.04	0.06	0.18	0.18	0.05	0.05	0.18	0.18	0.17	0.6	0.15	0.54	0.20	0.6	0.0
14	Nitrate Nitrogen	mg/1	5.28	5.25	2.95	1.86	4.47	4.68	2.88	2.88	1.45	7.19	3.5	6.34	4.06	7.2	1.5
15	Phosphate (PO_4)	mg/1	0.24	0.28	0.49	0.23	BDL	0.04	0.49	0.49	0.528	0.04	0.82	0.62	0.39	0.8	0.0
16	Chloride	mg/1	27	29.56	31.53	31.53	19.71	21.68	20	20.0	27.99	28	27.59	28.28	26.07	31.5	19.7
17	Sulphate (SO_4)	mg/1	4.77	4.67	6.75	9.3	24.55	26.04	26.04	24.55	11.5	12.8	12.5	12.21	14.64	26.0	4.7
18	T. Alkalinity	mg/1	110	84	98	110	94	156	82	88	120	130	144	158	114.50	158.0	82.0
19	T. Hardness	mg/1	118	96	116	198	132	194	132	140	128	142	148	156	141.67	198.0	96.0
20	Cal.Hardness	mg/1	70	50	72	164	70	72	74	80	92	76	94	98	84.33	164.0	50.0
21	Magnesium H.	mg/1	48	46	44	34	62	122	58	60	36	66	54	58	57.33	122.0	34.0
22	D. Oxygen	mg/1	6.5	8.2	7.7	8.5	6.6	7.9	9.2	8.8	7.1	8.1	8.2	7.9	7.89	9.2	6.5
23	B.O.D.	mg/1	1.7	2.9	2.8	1.9	1	2.3	2.5	2.6	1.9	2.7	2.6	2.9	2.32	2.9	1.0
24	C.O.D.	mg/1	45.4	30	38.4	20	20	20	10	10.0	19.76	288	19.2	28.8	45.80	288.0	10.0
25	Sodium	mg/1	25.97	26.11	28.4	23.96	12.43	13.58	18.66	13.58	18.4	1.25	16.84	1.46	16.72	28.4	1.3
26	Potassium	mg/1	9.65	8.54	7.92	4.32	2.03	2.84	4.41	2.84	1.67	0.09	2.62	0.13	3.92	9.7	0.1
27	T. Coliform	MPN/100ml	920	1600	>1600	110	140	170	240	240	230	240	210	220	392.73	1600.0	110.0
28	F. Coliform	MPN/100ml	2	2	2	2	2	2	2	2	2	2	2	2	2.00	2.0	2.0
29	Boron	mg/1	BDL	BDL	-	BDL	BDL	BDL	0.6	BDL	BDL		BDL	BDL	0.60	0.6	0.6
Cat. B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2018-2019

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Dry	Not in Package	Not in Package	06.07.2018	Not in Package	Not in Package	06.10.18	Not in Package	Not in Package	Dry	Not in Package	Not in Package	-	-	-
2	Appearance					Clear			Clear						-	-	-
3	Temperature	oC				28			28						-	-	-
4	Turbidity	NTU				15			7.4						11.20	15.0	7.4
5	Colour	PCS				Colorless			Colorless						-	-	-
6	Odour	T. No	-	-	-	Odourless	-	-	Odourless	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	7.96	-	-	7.97	-	-	-	-	-	7.97	8.0	8.0
8	Sp. Conductivity	µMhos/cm.	-	-	-	328	-	-	188	-	-	-	-	-	258.00	328.0	188.0
9	T. Solids	mg/1	-	-	-	226	-	-	132	-	-	-	-	-	179.00	226.0	132.0
10	D. Solids	mg/1	-	-	-	210	-	-	104	-	-	-	-	-	157.00	210.0	104.0
11	S. Solids	mg/1	-	-	-	16	-	-	28	-	-	-	-	-	22.00	28.0	16.0
12	Amm. Nitrogen	mg/1	-	-	-	0.138	-	-	BDL	-	-	-	-	-	0.14	0.1	0.1
13	Nitrite Nitrogen	mg/1	-	-	-	0.15	-	-	0.2	-	-	-	-	-	0.18	0.2	0.2
14	Nitrate Nitrogen	mg/1	-	-	-	2.64	-	-	1.2	-	-	-	-	-	1.92	2.6	1.2
15	Phosphate (PO_4)	mg/1	-	-	-	0.78	-	-	BDL	-	-	-	-	-	0.78	0.8	0.8
16	Chloride	mg/1	-	-	-	33.5	-	-	20.69	-	-	-	-	-	27.10	33.5	20.7
17	Sulphate (SO_4)	mg/1	-	-	-	16	-	-	5.45	-	-	-	-	-	10.73	16.0	5.5
18	T. Alkalinity	mg/1	-	-	-	94	-	-	152	-	-	-	-	-	123.00	152.0	94.0
19	T. Hardness	mg/1	-	-	-	132	-	-	104	-	-	-	-	-	118.00	132.0	104.0
20	CalciumHardness	mg/1	-	-	-	70	-	-	92	-	-	-	-	-	81.00	92.0	70.0
21	Magnesium H.	mg/1	-	-	-	62	-	-	12	-	-	-	-	-	37.00	62.0	12.0
22	D. Oxygen	mg/1	-	-	-	8.2	-	-	7.4	-	-	-	-	-	7.80	8.2	7.4
23	B.O.D.	mg/1	-	-	-	2.3	-	-	2.1	-	-	-	-	-	2.20	2.3	2.1
24	C.O.D.	mg/1	-	-	-	20	-	-	20	-	-	-	-	-	20.00	20.0	20.0
25	Sodium	mg/1	-	-	-	27.84	-	-	13.9	-	-	-	-	-	20.87	27.8	13.9
26	Potassium	mg/1	-	-	-	2.58	-	-	1.62	-	-	-	-	-	2.10	2.6	1.6
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	70	-	-	-	-	-	70.00	70.0	70.0
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	2	-	-	-	-	-	2.00	2.0	2.0
29	Boron	mg/l	-	-	-	-	-	-	-	-	-	-	-	-			
30	Flouride	mg/l	-	-	-		-	-	-	-	-	-	-	-			
31	P. Alakalinity	mg/l	-	-	-	-	-	-	Absent	-	-	-	-	-	-	-	-
32	F. D. S.	mg/l	-	-	-	-	-	-	-	-	-	-	-	-			
33	T.K.N.		-	-	-		-	-		-	-	-	-	-			
Cat. B			-	-	-												

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2018-2019

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Dry	Not in Package	Not in Package	06.07.2018	Not in Package	Not in Package	06.10.18	Not in Package	Not in Package	Dry	Not in Package	Not in Package	-	-	-
2	Appearance					Clear			Clear						-	-	-
3	Temperature	oC				28			28						28.00	28.0	28.0
4	Turbidity	NTU				15			7.6						11.30	15.0	7.6
5	Colour	PCS				Colorless			Colorless						-	-	-
6	Odour	T. No	-	-	-	Odourless	-	-	Odourless	-	-	-	-	-	-	-	-
7	pH	pH Unit	-	-	-	7.53	-	-	8.02	-	-	-	-	-	7.78	8.0	7.5
8	Sp. Conductivity	µMhos/cm.	-	-	-	443	-	-	196.5	-	-	-	-	-	319.75	443.0	196.5
9	T. Solids	mg/1	-	-	-	336	-	-	144	-	-	-	-	-	240	336	144
10	D. Solids	mg/1	-	-	-	320	-	-	110	-	-	-	-	-	215	320	110
11	S. Solids	mg/1	-	-	-	16	-	-	34	-	-	-	-	-	25	34	16
12	Amm. Nitrogen	mg/1	-	-	-	0.143	-	-	BDL	-	-	-	-	-	0	0	0
13	Nitrite Nitrogen	mg/1	-	-	-	0.121	-	-	0.2	-	-	-	-	-	0	0	0
14	Nitrate Nitrogen	mg/1	-	-	-	2.92	-	-	1.3	-	-	-	-	-	2.11	2.9	1.3
15	Phosphate (PO_4)	mg/1	-	-	-	0.77	-	-	BDL	-	-	-	-	-	0.77	0.8	0.8
16	Chloride	mg/1	-	-	-	42.37	-	-	20.69	-	-	-	-	-	31.53	42.4	20.7
17	Sulphate (SO_4)	mg/1	-	-	-	13	-	-	5.85	-	-	-	-	-	9.43	13.0	5.9
18	T. Alkalinity	mg/1	-	-	-	146	-	-	168	-	-	-	-	-	157	168.0	146.0
19	T. Hardness	mg/1	-	-	-	174	-	-	118	-	-	-	-	-	146	174.0	118.0
20	CalciumHardness	mg/1	-	-	-	102	-	-	96	-	-	-	-	-	99	102.0	96.0
21	Magnesium H.	mg/1	-	-	-	72	-	-	22	-	-	-	-	-	47	72.0	22.0
22	D. Oxygen	mg/1	-	-	-	7.9	-	-	7.3	-	-	-	-	-	7.60	7.9	7.3
23	B.O.D.	mg/1	-	-	-	2.1	-	-	2.2	-	-	-	-	-	2.15	2.2	2.1
24	C.O.D.	mg/1	-	-	-	20	-	-	20	-	-	-	-	-	20.00	20.0	20.0
25	Sodium	mg/1	-	-	-	37.48	-	-	14.1	-	-	-	-	-	25.79	37.5	14.1
26	Potassium	mg/1	-	-	-	5.54	-	-	1.79	-	-	-	-	-	3.67	5.5	1.8
27	T. Coliform	MPN/100ml	-	-	-	-	-	-	70	-	-	-	-	-	70.00	70.0	70.0
28	F. Coliform	MPN/100ml	-	-	-	-	-	-	2	-	-	-	-	-	2.00	2.0	2.0
29	Boron	mg/l	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	Flouride	mg/l	-	-	-		-	-	-	-	-	-	-	-			
31	P. Alakalinity	mg/l	-	-	-	-	-	-	Absent	-	-	-	-	-			
32	F. D. S.	mg/l	-	-	-	-	-	-	-	-	-	-	-	-			
Cat. B			-	-	-		-	-		-	-	-	-	-			

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2019-20

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.19	06.05.19	17.06.19	04.07.19	01.08.19	06.09.19	14.10.19	22.11.19	24.12.19	03.01.20	20.02.20	02.03.20	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	31 ºC		29	25	23	24	27	25	22	18	24	25	24.20	29.0	18.0
4	Turbidity	NTU	24.3	19.8	9.3	13.3	22.7	18.2	18.2	18.2	16.3	6.3	10.6	9.3	15.54	24.3	6.3
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Sl. Muddy	Colorles	Colorless	Colorless	Colorless	Colourless	Colourless	Colorless	-	-	-
6	Odour	T. No	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	8.26	8.64	7.84	8.38	7.38	7.48	7.92	8.16	7.84	7.82	7.87	7.74	7.94	8.6	7.4
8	Sp. Conductivity	µMhos/cm.	294.6	270.7	282	524.5	328.2	278.2	284.7	289.3	278.9	227.3	284.1	276.2	301.56	524.5	227.3
9	T. Solids	mg/1	216	194	184	364	282	228	228	212	204	180	190	212	224.50	364.0	180.0
10	D. Solids	mg/1	172	176	166	348	228	184	196	186	180	158	168	184	195.50	348.0	158.0
11	S. Solids	mg/1	44	18	18	16	54	44	32	26	24	22	22	28	29.00	54.0	16.0
12	Amm. Nitrogen	mg/1	BDL	BDL	0.23	BDL	0.72	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.48	0.7	0.2
13	Nitrite Nitrogen	mg/1	0.15	0.04	0.26	0.02	1.14	BSL	0.28	BDL	BDL	BDL	BDL	BDL	0.32	1.1	0.0
14	Nitrate Nitrogen	mg/1	4.5	4.74	1.3	4.3	4.72	1.84	2.74	2.24	2.16	2.86	2.16	2.04	2.97	4.7	1.3
15	Phosphate (PO_4)	mg/1	0.82	0.06	0.396	0.03	0.34	0.06	0.72	1.26	1.19	0.7	0.08	0.07	0.48	1.3	0.0
16	Chloride	mg/1	26.6	26.68	23.65	40.11	31.99	26.99	24.99	31.99	29.35	26.99	14.78	12.81	26.41	40.1	12.8
17	Sulphate (SO_4)	mg/1	17.5	25.34	8	22.46	16.7	12.8	10.4	18.2	18.11	14.7	17.3	16.85	16.53	25.3	8.0
18	T. Alkalinity	mg/1	144	166	92	118	186	172	172	184	182	176	118	132	153.50	186.0	92.0
19	T. Hardness	mg/1	152	154	124	192	176	162	164	174	176	166	168	146	162.83	192.0	124.0
20	CalciumHardness	mg/1	94	88	92	156	138	128	126	136	144	130	136	120	124.00	156.0	88.0
21	Magnesium H.	mg/1	58	66	32	36	38	34	38	38	32	36	32	26	38.83	66.0	26.0
22	D. Oxygen	mg/1	7.6	7.1	7.7	8	8.2	8	-	7.9	8.0	9.2	8.2	8.4	8.03	9.2	7.1
23	B.O.D.	mg/1	2.9	2.3	2.1	1.6	1.5	1.6	-	1.4	1.3	1.8	1.3	1.2	1.73	2.9	1.2
24	C.O.D.	mg/1	19.2	20	19.76	19.6	19.2	19.2	9.6	29.4	20.0	19.36	19.76	19.2	19.52	29.4	9.6
25	Sodium	mg/1	24.6	11.56	14.18	9.54	32.8	10.72	28.4	28.2	26.74	22.6	16.3	15.43	20.09	32.8	9.5
26	Potassium	mg/1	2.6	1.14	1.49	1.33	1.2	2.84	1.7	2.1	2.11	2.62	2.5	2.37	2.00	2.8	1.1
27	T. Coliform	MPN/100ml	280	70	240	63	84	79	94	84	79	79	70	63	107.08	280.0	63.0
28	F. Coliform	MPN/100ml	2	1.8	2	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.83	2.0	1.8
29	Boron	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
Category-B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2019-20

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.19	06.05.19	17.06.19	04.07.19	01.08.19	06.09.19	14.10.19	22.11.19	24.12.19	03.01.20	20.02.20	02.03.20	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	33 ºC	-	29	24	24	24	27	24	21	18	25	25	24.10	29	18
4	Turbidity	NTU	12.7	21.8	16.7	9.64	26.2	12.8	16.2	16.2	15.3	11.8	13.5	12.3	15.43	26.2	9.64
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Sl. Muddy	Colorless	Colorless	Colorless	Colorless	Colourless	Colourless	Colorless	-	-	-
6	Odour	T. No	Odourless	Odorless	Odorless	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	8.13	8.54	8.01	8.3	7.38	7.92	7.82	8.06	7.98	7.86	7.79	7.82	7.97	8.54	7.38
8	Sp. Conductivity	µMhos/cm.	288.2	271.5	388.2	296.8	287.3	248.2	268.5	256.3	242.8	224.5	272.1	265.4	275.82	388.2	224.5
9	T. Solids	mg/1	238	194	248	204	218	194	200	186	180	164	192	204	201.83	248	164
10	D. Solids	mg/1	196	178	214	182	182	158	168	164	158	146	154	170	172.50	214	146
11	S. Solids	mg/1	42	16	34	22	36	36	32	22	22	18	38	34	29.33	42	16
12	Amm. Nitrogen	mg/1	BDL	BDL	0.27	BDL	0.38	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.33	0.38	0.27
13	Nitrite Nitrogen	mg/1	0.075	0.05	0.32	0.04	1.76	BDL	0.38	BDL	BDL	BDL	BDL	BDL	0.44	1.76	0.04
14	Nitrate Nitrogen	mg/1	3.5	4.53	1.8	3.48	5.28	3.7	3.71	3.50	1.64	1.48	4.76	4.64	3.50	5.28	1.48
15	Phosphate (PO_4)	mg/1	0.66	0.05	0.594	0.03	0.62	0.62	0.62	0.37	1.09	1.0	0.09	0.08	0.49	1.09	0.03
16	Chloride	mg/1	28.57	16.71	31.53	33.26	26.99	21.99	17.99	19	20.54	18.99	18.72	17.73	22.67	33.26	16.71
17	Sulphate (SO_4)	mg/1	13.75	18.34	11.5	16.48	20.7	18.2	18.2	12.23	13.78	12.69	22.81	24.8	16.96	24.80	11.5
18	T. Alkalinity	mg/1	174	176	134	110	178	152	172	158	160	156	124	128	151.83	178	110
19	T. Hardness	mg/1	164	152	182	150	156	164	162	152	152	144	164	160	158.50	182	144
20	Cal.Hardness	mg/1	102	84	140	78	120	130	130	112	118	112	134	126	115.50	140	78
21	Magnesium H.	mg/1	62	68	42	72	36	34	32	40	34	32	30	34	43.00	72	30
22	D. Oxygen	mg/1	8.1	7.4	7.3	8	8.1	8.2	-	8.1	8.3	8.5	8.4	8.6	8.09	8.6	7.3
23	B.O.D.	mg/1	2.6	2	2.4	2.1	1.8	1.7	-	1.5	1.4	1.3	1.9	1.7	1.85	2.6	1.3
24	C.O.D.	mg/1	19.2	20	19.76	19.6	9.6	9.6	19.2	20.0	20.0	20	19.76	19.2	17.99	20	9.6
25	Sodium	mg/1	16.4	12.55	16.08	11.99	28.3	16.2	32.4	27.9	22.16	19.64	12.07	12.85	19.05	32.4	11.99
26	Potassium	mg/1	1.7	2.17	1.78	1.84	1.06	2.7	2.1	1.1	1.78	1.35	2.8	2.72	1.93	2.8	1.06
27	T. Coliform	MPN/100ml	170	130	430	110	150	130	110	120	110	94	120	110	148.67	430	94
28	F. Coliform	MPN/100ml	2	1.8	2	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.83	2	1.8
29	Boron	mg/1	BDL	BDL	-	2.1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	2.1	2.1	2.1
Category--B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2019-20

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal)

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.19	06.05.19	17.06.19	04.07.19	01.08.19	06.09.19	14.10.19	22.11.19	24.12.19	03.01.20	20.02.20	02.03.20	-	-	-
2	Appearance		-	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	32 ºC	-	28	25	25	23	26	25	21	19	25	24	24.10	28.0	19.0
4	Turbidity	NTU	24.2	23.3	12.2	21.3	24.7	22.4	18.2	17.4	10.6	8.4	15.3	14.1	17.68	24.7	8.4
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Sl. Muddy	Colorless	Colorless	Colorless	Colorless	Colourless	Colourless	Colorless	-	-	-
6	Odour	T. No	Odourless	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	8.03	8.34	8.14	8.31	7.38	7.72	8.14	8.26	7.89	8.01	7.81	7.88	7.99	8.3	7.4
8	Sp. Conductivity	µMhos/cm.	298.3	244.4	298.8	316.4	336.2	264.7	298.2	254.4	294.3	286.8	283.2	276.4	287.68	336.2	244.4
9	T. Solids	mg/1	264	224	238	264	258	214	236	224	212	212	202	210	229.83	264.0	202.0
10	D. Solids	mg/1	206	190	214	228	212	176	198	168	182	186	170	182	192.67	228.0	168.0
11	S. Solids	mg/1	58	34	24	36	46	38	38	56	30	26	32	28	37.17	58.0	24.0
12	Amm. Nitrogen	mg/1	BDL	BDL	BDL	BDL	0.84	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.84	0.8	0.8
13	Nitrite Nitrogen	mg/1	0.02	0.05	0.03	0.08	1.27	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.29	1.3	0.0
14	Nitrate Nitrogen	mg/1	5.5	4.36	4.04	3.85	6.28	2.7	3.2	2.95	2.99	1.48	3.38	3.25	3.67	6.3	1.5
15	Phosphate (PO_4)	mg/1	0.52	BDL	0.01	0.02	0.82	0.74	1.24	0.82	1.86	1.3	0.08	0.06	0.68	1.9	0.0
16	Chloride	mg/1	31.53	16.71	14.78	23.48	25.99	18.99	25.99	21.68	30.33	26.99	14.78	15.76	22.25	31.5	14.8
17	Sulphate (SO_4)	mg/1	17.5	23.43	14.14	16.14	24.7	18.2	22.4	22.74	32.42	30.68	29.7	28.4	23.37	32.4	14.1
18	T. Alkalinity	mg/1	168	136	128	126	184	164	184	158	180	176	132	136	156.00	184.0	126.0
19	T. Hardness	mg/1	168	134	116	120	172	162	178	156	168	162	170	168	156.17	178.0	116.0
20	Cal.Hardness	mg/1	106	70	66	70	140	126	134	96	132	130	134	134	111.50	140.0	66.0
21	Magnesium H.	mg/1	62	64	50	50	32	36	44	60	36	32	36	34	44.67	64.0	32.0
22	D. Oxygen	mg/1	7.7	7.6	8.1	7.8	7.9	7.8	-	7.9	8.0	8.4	8.3	8.5	8.00	8.5	7.6
23	B.O.D.	mg/1	2.8	1.3	1.4	3.6	2.4	2.2	-	2.0	1.8	1.5	2.2	2.3	2.14	3.6	1.3
24	C.O.D.	mg/1	28.8	20	19.76	39.2	29.4	29.4	29.4	19.76	20.0	20	29.64	19.2	25.38	39.2	19.2
25	Sodium	mg/1	24.6	11.43	12.34	11.98	24.7	18.2	28.4	23.96	32.64	31.03	13.16	12.18	20.39	32.6	11.4
26	Potassium	mg/1	2.6	1.03	1.12	23.48	2.1	2.4	2.1	2.03	2.93	2.84	2.42	2.38	3.95	23.5	1.0
27	T. Coliform	MPN/100ml	240	140	280	350	280	220	120	210	170	150	150	140	204.17	350.0	120.0
28	F. Coliform	MPN/100ml	2	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.82	2.0	1.8
29	Boron	mg/1	-	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
	Cat. -B		**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**	**B**			

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2019-20

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		09.04.19	06.05.19	17.06.19	04.07.19	01.08.19	06.09.19	14.10.19	22.11.19	24.12.19	03.01.20	20.02.20	02.03.20	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	30		28	25	25	23	25	25	21	18	25	24	24.45	30.0	18.0
4	Turbidity	NTU	24.6	23.3	12.9	17.3	16.4	16.4	14.7	15.2	13.9	6.9	11.2	10.4	15.27	24.6	6.9
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Sl. Muddy	Colorless	Colorless	Colorless	Colorless	Colourless	Colourless	Colorless	-	-	-
6	Odour	T. No	Odorless	Odorless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	-	-
7	pH	pH Unit	7.94	8.35	7.96	8.34	7.74	7.79	7.52	7.76	8.02	7.62	7.86	7.92	7.90	8.4	7.5
8	Sp. Conductivity	µMhos/cm.	312.4	289.2	298.6	254.8	314.6	286.4	278.2	226.4	262.7	236.2	276.1	265.5	275.09	314.6	226.4
9	T. Solids	mg/1	250	184	206	196	240	202	202	189	180	192	188	198	202.25	250.0	180.0
10	D. Solids	mg/1	208	166	182	174	204	164	176	143	158	168	160	174	173.08	208.0	143.0
11	S. Solids	mg/1	42	18	24	22	36	38	26	46	22	24	28	24	29.17	46.0	18.0
12	Amm. Nitrogen	mg/1	BDL	BDL	0.24	BDL	0.34	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.29	0.3	0.2
13	Nitrite Nitrogen	mg/1	0.05	0.05	0.29	0.02	1.26	0.64	0.34	0.05	BDL	0.7	BDL	BDL	0.38	1.3	0.0
14	Nitrate Nitrogen	mg/1	4.5	5.54	1.55	4.68	4.87	2.84	2.3	1.65	2.44	2.86	4.66	3.92	3.48	5.5	1.6
15	Phosphate (PO_4)	mg/1	0.66	0.06	0.462	0.06	0.38	0.38	0.31	0.28	0.77	0.2	0.07	0.06	0.31	0.8	0.1
16	Chloride	mg/1	23.65	19.65	29.56	42.07	25.99	26.99	21.99	18.78	26.41	23.99	23.65	21.68	25.37	42.1	18.8
17	Sulphate (SO_4)	mg/1	10	21.74	9.5	24.04	16.4	14.6	12.2	8.28	15.98	10.4	18.26	17.54	14.91	24.0	8.3
18	T. Alkalinity	mg/1	146	136	94	104	172	184	168	136	166	164	124	120	142.83	184.0	94.0
19	T. Hardness	mg/1	164	134	138	144	126	164	158	138	156	162	160	150	149.50	164.0	126.0
20	CalciumHardness	mg/1	106	90	102	98	164	126	122	103	122	128	128	106	116.25	164.0	90.0
21	Magnesium H.	mg/1	58	44	36	46	126	38	36	35	34	34	32	44	46.92	126.0	32.0
22	D. Oxygen	mg/1	7.4	7.4	7.5	7.8	38	8.1	-	8.3	8.4	8.7	8.1	8.3	10.73	38.0	7.4
23	B.O.D.	mg/1	2.4	1.3	2.3	1.3	1.8	1.2	-	1.2	1.1	1.4	1.5	1.4	1.54	2.4	1.1
24	C.O.D.	mg/1	19.2	20	19.76	9.8	19.2	1.4	9.6	10.0	20.0	9.6	19.76	19.2	14.79	20.0	1.4
25	Sodium	mg/1	21.6	12.41	15.26	12.34	28.4	19.2	14.2	12.42	23.68	15.7	20.8	19.34	17.95	28.4	12.3
26	Potassium	mg/1	1.1	1.21	1.62	2.14	1.02	14.8	1.5	1.94	2.46	1.86	1.9	1.68	2.77	14.8	1.0
27	T. Coliform	MPN/100ml	210	140	350	130	120	1.2	84	70	1600	79	110	110	250.35	1600.0	1.2
28	F. Coliform	MPN/100ml	2	1.8	2	1.8	1.8	70	1.8	1.8	1.8	1.8	1.8	1.8	7.52	70.0	1.8
29	Boron	mg/1	BDL	BDL	-	BDL	BDL	1.8	BDL	BDL	BDL	BDL	BDL	BDL	1.80	1.8	1.8
30	T.K.N.	mg/1	BDL	BDL	-	BDL	BDL	BDL	BDL	BDL	BDL						
Category-B			B	B	B	B	B	B	B	B	B	B	B	B			

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2019-20

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Stagnant water	Not in package	Not in package	05.07.19	Not in package	Not in package	14.10.19	Not in package	Not in package	03.01.20	Not in package	Not in package	-	-	-
2	Appearance					Clear			Clear			Clear			-	-	-
3	Temperature	oC				24			26			19			23.00	26.0	19.0
4	Turbidity	NTU				11.3			20.4			16.4			16.03	20.4	11.3
5	Colour	PCS				Colorless			Colorless			Colorless			-	-	-
6	Odour	T. No				Odourless			Odourless			Odourless			-	-	-
7	pH	pH Unit				7.17			7.62			7.88			7.56	7.9	7.2
8	Sp. Conductivity	µMhos/cm.				343.9			327.4			297.5			322.93	343.9	297.5
9	T. Solids	mg/1				256			254			234			248.00	256.0	234.0
10	D. Solids	mg/1				238			212			198			216.00	238.0	198.0
11	S. Solids	mg/1				18			42			36			32.00	42.0	18.0
12	Amm. Nitrogen	mg/1				0.12			BDL			BDL			0.12	0.1	0.1
13	Nitrite Nitrogen	mg/1				0.13			0.72			0.33			0.39	0.7	0.1
14	Nitrate Nitrogen	mg/1				2.58			3.72			3.8			3.37	3.8	2.6
15	Phosphate (PO_4)	mg/1				0.04			1.21			0.94			0.73	1.2	0.0
16	Chloride	mg/1				29.13			28.99			21.99			26.70	29.1	22.0
17	Sulphate (SO_4)	mg/1				16.48			22.4			19.65			19.51	22.4	16.5
18	T. Alkalinity	mg/1				116			192			186			164.67	192.0	116.0
19	T. Hardness	mg/1				152			182			174			169.33	182.0	152.0
20	CalciumHardness	mg/1				96			140			138			124.67	140.0	96.0
21	Magnesium H.	mg/1				56			42			36			44.67	56.0	36.0
22	D. Oxygen	mg/1				8			-			8.2			8.10	8.2	8.0
23	B.O.D.	mg/1				2.8			-			2.1			2.45	2.8	2.1
24	C.O.D.	mg/1				9.8			29.4			20			19.73	29.4	9.8
25	Sodium	mg/1				28.46			32.4			28.56			29.81	32.4	28.5
26	Potassium	mg/1				2.94			2.6			2.84			2.79	2.9	2.6
27	T. Coliform	MPN/100ml				-			-			-					
28	F. Coliform	MPN/100ml				-			-						-	-	-
29	T.K.N.	mg/1				-			-						-	-	-
	Cat. A																

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2019-20

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Characteristic	Unit	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Stagnant water	Not in package	Not in package	05.07.19	Not in package	Not in package	14.10.19	Not in package	Not in package	03.01.20	Not in package	Not in package	-	-	-
2	Appearance					Clear			Clear			Clear			-	-	-
3	Temperature	oC				24			26			19			23	26	19
4	Turbidity	NTU				12.6			16.4			11.4			13.47	16.4	11.4
5	Colour	PCS				Colorless			Colorless			Colorless			-	-	-
6	Odour	T. No				Odourless			Odourless			Odourless			-	-	-
7	pH	pH Unit				7.07			8.04			7.88			7.66	8.04	7.07
8	Sp. Conductivity	µMhos/cm.				369.9			286.8			266.5			307.73	369.9	266.5
9	T. Solids	mg/1				268			202			188			219.33	268	188
10	D. Solids	mg/1				256			176			172			201.33	256	172
11	S. Solids	mg/1				12			26			16			18	26	12
12	Amm. Nitrogen	mg/1				0.18			BDL			BDL			0.18	0.18	0.18
13	Nitrite Nitrogen	mg/1				0.18			BDL			BDL			0.18	0.18	0.18
14	Nitrate Nitrogen	mg/1				2.6			1.24			1.14			1.66	2.60	1.14
15	Phosphate (PO_4)	mg/1				0.03			0.72			0.52			0.42	0.72	0.03
16	Chloride	mg/1				33.99			19.99			16.99			23.66	33.99	16.99
17	Sulphate (SO_4)	mg/1				15.42			8.4			7.86			10.56	15.42	7.86
18	T. Alkalinity	mg/1				102			168			160			143.33	168	102
19	T. Hardness	mg/1				110			170			152			144.00	170	110
20	CalciumHardness	mg/1				66			138			122			108.67	138	66
21	Magnesium H.	mg/1				44			32			30			35.33	44	30
22	D. Oxygen	mg/1				7.6			-			8			7.80	8	7.6
23	B.O.D.	mg/1				4.1			-			2.8			3.45	4.1	2.8
24	C.O.D.	mg/1				19.6			9.6			20			16.40	20	9.6
25	Sodium	mg/1				26.48			14.3			13.65			18.14	26.48	13.65
26	Potassium	mg/1				2.53			1.1			1.04			1.56	2.53	1.04
27	T. Coliform	MPN/100ml				-			-			-					
28	F. Coliform	MPN/100ml				-			-			-			-	-	-
29	T.K.N.	mg/1				-			-						-	-	-
	Cat. -A																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2020-2021

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal) **2137**

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.04.20	22.05.20	08.06.20	15.07.2020	06.08.20	03.09.20	28.10.20	27.11.20	09.12.20	04.01.21	23.2.2021	22.3.2021	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	31 ºC	30	30	30	30	30				20			28.33	30.0	20.0
4	Turbidity	NTU	7.48	7.9	8.1	13.2	14.2	6.6	5.1	8.1	7.7	5.1	3.5	1.1	7.34	14.2	1.1
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Colorless	Greenish	40	40	40	30	20	10	30.00	40.0	10.0
6	Odour	T. No	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	observed	Odourless	-	-	-
7	pH	pH Unit	8.77	8.02	7.93	8.66	8.65	8.18	7.56	7.96	7.9	7.72	7.48	7.7	8.04	8.8	7.5
8	Sp. Conductivity	µMhos/cm.	262.7	218.6	227.7	93.35	902.5	269.8	185.2	178	164.9	210.4	223.3	255.2	265.97	902.5	93.4
9	T. Solids	mg/1	184	158	170	144	227	210	156	154	154	174	198	210	178.25	227.0	144.0
10	D. Solids	mg/1	166	140	148	130	186	134	128	128	122	152	164	182	148.33	186.0	122.0
11	S. Solids	mg/1	18	18	22	14	41	76	28	26	32	22	34	28	29.92	76.0	14.0
12	Amm. Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	0.62	0.31	0.65	0.05	BDL	BDL	0.41	0.7	0.1
14	Nitrate Nitrogen	mg/1	1.0	1.08	1.13	1.10	1.08	2.97	3.2	3.4	3.34	2.48	2.5	2.89	2.18	3.4	1.0
15	Phosphate (PO_4)	mg/1	0.16	0.16	0.15	0.13	0.14	0.23	0.16	0.26	0.05	0.31	BDL	BDL	0.18	0.3	0.1
16	Chloride	mg/1	12.81	13.99	15.99	10.99	12.90	9.99	25.07	13.49	14.46	9.78	13.69	15.65	14.07	25.1	9.8
17	Sulphate (SO_4)	mg/1	8.75	8.87	9.0	10.47	11.5	12.6	14.2	16.2	16.45	5.88	2.13	3.72	9.98	16.5	2.1
18	T. Alkalinity	mg/1	156	160	164	88	90	400	172	118	114	96	94	102	146.17	400.0	88.0
19	T. Hardness	mg/1	156	158	160	82	160	80	164	128	120	108	104	102	126.83	164.0	80.0
20	CalciumHardness	mg/1	130	126	130	50	130	42	130	82	84	52	86	84	93.83	130.0	42.0
21	Magnesium H.	mg/1	26	32	30	32	30	38	34	46	36	56	18	18	33.00	56.0	18.0
22	D. Oxygen	mg/1	8.7	8.9	9	9.3	8.0	8.4	8.4	10.3	8.5	7.7	7.9	6.8	8.49	10.3	6.8
23	B.O.D.	mg/1	1.1	1.1	ns. disorder	1.6	1.4	1.6	2	1.2	1.0	1.5	1.4	1.3	1.38	2.0	1.0
24	C.O.D.	mg/1	10	10	10	19.36	19.0	29.4	19.04	28.8	19.2	20	19.36	19.36	18.63	29.4	10.0
25	Sodium	mg/1	12.6	14.73	14.85	16.7	19.6	12.33	6.84	16.7	7.02	7.37	5.75	5.89	11.70	19.6	5.8
26	Potassium	mg/1	1.3	1.42	1.56	1.38	1.4	1.16	1.34	2.8	1.19	3.14	0.78	0.62	1.51	3.1	0.6
27	T. Coliform	MPN/100ml	41	39	40	34	63	280	28000	28000	11000	54000	430	430	10196.42	54000.0	34.0
28	F. Coliform	MPN/100ml	1.8	1.8	1.8	1.8	1.8	12	370	180	1600	3500	22	1.8	474.57	3500.0	1.8
29	Streptococcus	MPN/100ml	-	-	-	-	-	-	1.8	1.8	1.8	1.8	1.8	1.8	1.80	1.8	1.8
30	Calcium	mg/1	-	-	-	-	-	-	54.16	32.83	33.64	20.82	34.44	33.64	34.92	54.2	20.8
31	Magnesium	mg/1	-	-	-	-	-	-	8.29	11.17	8.74	13.59	4.37	4.37	8.42	13.6	4.4
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2020-2021

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal) **2138**

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.04.20	22.05.20	08.06.20	15.07.2020	06.08.20	03.09.20	28.10.20	27.11.20	09.12.20	04.01.21	23.2.2021	22.3.2021	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	Clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	32 ºC	31	31	30	30	30				20			28.67	31	20
4	Turbidity	NTU	7.24	7.3	7.8	28.4	13	6.2	6.6	7.5	7.6	5.6	2.9	1.3	8.45	28.4	1.3
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Colorless	Sl.	50	40	40	40	10	20	33.33	50	10
6	Odour	T. No	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	No Odour	Odourless	-	-	-
7	pH	pH Unit	8.52	7.84	7.91	8.23	7.87	8.03	7.75	8.02	8.02	7.65	7.59	7.66	7.92	8.52	7.59
8	Sp. Conductivity	µMhos/cm.	276.4	284.1	297.3	324.5	905.6	290.8	186.6	184	160.6	212.3	234.1	256.8	301.09	905.6	160.6
9	T. Solids	mg/1	168	202	212	148	190	200	162	144	158	180	180	208	179.33	212	144
10	D. Solids	mg/1	152	186	194	124	146	128	134	122	128	162	152	186	151.17	194	122
11	S. Solids	mg/1	16	16	18	24	44	72	28	22	30	18	28	22	28.17	72	16
12	Amm. Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	0.26	0.38	0.28	0.03	BDL	BDL	0.24	0.38	0.03
14	Nitrate Nitrogen	mg/1	1.25	1.32	1.36	1.29	1.31	3.13	3.18	3.14	3.15	2.46	2.91	3.01	2.29	3.18	1.25
15	Phosphate (PO_4)	mg/1	0.19	0.19	0.24	0.22	0.24	0.36	0.24	0.46	0.04	0.19	0.01	3.92	0.53	3.92	0.01
16	Chloride	mg/1	13.79	14.99	15.99	10.99	12.9	9.99	17.35	11.57	13.49	8.81	12.71	13.69	13.02	17.35	8.81
17	Sulphate (SO_4)	mg/1	6.25	6.3	6.48	5.89	6.78	16.4	14.2	22.2	17.24	5.72	3.4	0.05	9.24	22.20	0.05
18	T. Alkalinity	mg/1	118	122	126	88	136	90	158	110	110	104	86	106	112.83	158	86
19	T. Hardness	mg/1	134	120	122	84	136	72	164	138	114	118	114	106	118.50	164	72
20	Cal.Hardness	mg/1	114	104	100	52	114	46	132	96	84	48	84	84	88.17	132	46
21	Magnesium H.	mg/1	20	16	22	32	22	26	32	42	30	70	30	22	30.33	70	16
22	D. Oxygen	mg/1	8.9	9	9.1	8.8	8.7	8.6	8.3	10.9	8.8	6.9	7.6	7.1	8.56	10.9	6.9
23	B.O.D.	mg/1	1.1	1.2	trument Disor	2.2	3.9	3.2	2	1.4	1.3	1.7	1.5	1.2	1.88	3.9	1.1
24	C.O.D.	mg/1	10	10	10	19.36	24.9	29.4	19.04	9.6	19.2	20	9.68	9.68	15.91	29.4	9.6
25	Sodium	mg/1	16.8	16.78	16.79	14.92	14.94	11.62	16.2	18.2	9.65	7.59	6.11	6.04	12.97	18.2	6.04
26	Potassium	mg/1	1.1	1.13	1.18	1.12	1.14	1.17	1.24	2.7	1.15	3.09	0.8	0.53	1.36	3.09	0.53
27	T. Coliform	MPN/100ml	58	48	47	48	79	350	28000	43000	28000	92000	350	350	16027.50	92000	47
28	F. Coliform	MPN/100ml	1.8	1.8	1.8	1.8	1.8	24	550	200	2400	2600	38	1.8	485.23	2600	1.8
29	Boron	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
30	Streptococcus	MPN/100ml	-	-	-	-	-	-	1.8	1.8	24	4.5	1.8	1.8	5.95	24	1.8
31	T.K.N.	meq/l	-	Instrument out of order				-			1.4	BDL	2.1	2.1	1.87	2.1	1.4
32	Calcium	mg/1						-	55	38.44	33.64	19.22	33.64	33.64	35.60	55	19.22
33	Magnesium	mg/1						-	7.8	10.19	7.28	16.99	7.28	5.34	9.15	16.99	5.34
Cat. B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2020-2021

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal) 2139

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.04.20	22.05.20	08.06.20	15.07.2020	06.08.20	03.09.20	28.10.20	27.11.20	09.12.20	04.01.21	23.02.2021	22.03.2021	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Sl. Greenish	Clear	Clear	SLIGHTLY TURBID	Clear	CLEAR	CLEAR	-	-	-
3	Temperature	oC	29 ºC	31	30	29	29	31				19			28.17	31.0	19.0
4	Turbidity	NTU	8.2	8.3	8.5	17.5	11.9	5.9	4.8	8.4	8.7	4.6	0.8	0.7	7.36	17.5	0.7
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Colorless	Colorless	40	40	70	30	20		40.00	70.0	20.0
6	Odour	T. No	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	No Odour observed		-	-	-
7	pH	pH Unit	8.62	7.77	7.84	8.42	8.00	8.04	7.86	8.03	8.02	7.69	8.01	7.44	7.98	8.6	7.4
8	Sp. Conductivity	µMhos/cm.	261.3	269.4	274.3	286.9	922.7	247.6	185.1	187.8	180.4	211.7	221.9	260.7	292.48	922.7	180.4
9	T. Solids	mg/1	182	198	204	180	206	200	158	172	174	168	160	210	184.33	210.0	158.0
10	D. Solids	mg/1	162	174	178	154	175	132	134	138	136	148	144	192	155.58	192.0	132.0
11	S. Solids	mg/1	20	24	26	26	31	68	24	34	38	20	16	18	28.75	68.0	16.0
12	Amm. Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	1.24	0.32	0.31	0.02	BDL	BDL	0.47	1.2	0.0
14	Nitrate Nitrogen	mg/1	1.75	1.79	1.82	1.78	1.34	4.76	6.24	3.7	3.28	2.72	2.75	2.75	2.89	6.2	1.3
15	Phosphate (PO_4)	mg/1	0.13	0.14	0.14	0.12	7.4	0.38	0.32	0.47	0.64	0.33	BDL	6.79	1.53	7.4	0.1
16	Chloride	mg/1	13.79	13.99	14.99	12.99	17.94	9.99	26.99	12.53	15.42	10.76	13.69	13.69	14.73	27.0	10.0
17	Sulphate (SO_4)	mg/1	7.5	7.62	7.77	8.12	0.24	20.5	22.7	20.4	17.85	6.2	0.67	1.85	10.12	22.7	0.2
18	T. Alkalinity	mg/1	132	136	144	88	146	84	172	96	116	96	76	110	116.33	172.0	76.0
19	T. Hardness	mg/1	156	158	160	80	168	72	178	128	126	104	86	104	126.67	178.0	72.0
20	Cal.Hardness	mg/1	130	134	134	46	138	46	142	80	92	62	72	76	96.00	142.0	46.0
21	Magnesium H.	mg/1	26	24	26	34	30	26	36	48	34	42	14	28	30.67	48.0	14.0
22	D. Oxygen	mg/1	8.5	8.3	8.7	8.5	10.0	8.8	7.9	10.2	8.0	6.7	7.9	6.6	8.34	10.2	6.6
23	B.O.D.	mg/1	1.3	1.4	ument Disc	2.6	3.4	2.7	1.8	1.6	1.9	1.2	0.8	0.9	1.78	3.4	0.8
24	C.O.D.	mg/1	20	10.0	20	19.36	20.6	29.4	28.56	28.8	19.2	20	9.68	9.68	19.61	29.4	9.7
25	Sodium	mg/1	14.7	14.83	14.96	17.61	16.67	13.53	36.4	16.7	11.64	7.63	6.05	7.12	14.82	36.4	6.1
26	Potassium	mg/1	1.4	1.47	1.52	1.75	1.19	1.34	3.4	2.9	1.28	3.22	0.65	0.42	1.71	3.4	0.4
27	T. Coliform	MPN/100ml	63	58	63	70	46	170	17000	17000	92000	92000	280	280	18252.50	92000.0	46.0
28	F. Coliform	MPN/100ml	1.8	1.8	1.8	1.8	1.8	20	400	180	17000	3500	27	1.8	1761.48	17000.0	1.8
29	Boron	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
30	Streptococus	MPN/100ml						-	1.8	1.8	13	1.8	1.8	1.8	3.67	13.0	1.8
31	Calcium	mg/1	-	-	-	-	-	-			36.84	24.83	28.83	30.43	30.23	36.8	24.8
32	Magnesium	mg/1	-	-	-	-	-	-			8.25	10.19	3.39	6.79	7.16	10.2	3.4
33	T.K.N.	mg/1	-	-	-	-	-	-			1.4	BDL	1.4	2.1	1.63	2.1	1.4
	Cat. B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2020-2021

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.04.20	22.05.20	08.06.20	15.07.2020	06.08.20	03.09.20	28.10.20	27.11.20	09.12.20	04.01.21	23.02.202	22.03.202	-	-	-
2	Appearance		Clear	Clear	Clear	Clear	Clear	Sl. Turbid	Clear	Clear	clear	Clear	Clear	Clear	-	-	-
3	Temperature	oC	32	30	31	29	30	31				19			28.86	32.0	19.0
4	Turbidity	NTU	6.2	6.6	6.9	10.3	16.9	8.2	4.9	8.4	6.8	4.2	1.8	1.4	6.88	16.9	1.4
5	Colour	PCS	Colorless	Colorless	Colorless	Colorless	Colorless	Sl.	40	50	30	40	20	20	33.33	50.0	20.0
6	Odour	T. No	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	Odourless	-	No odoure	Odourless	-	-	-
7	pH	pH Unit	8.42	7.93	8.00	8.94	7.97	8.99	7.26	8.04	8	7.68	7.66	7.5	8.04	9.0	7.3
8	Sp. Conductivity	µMhos/cm.	256.4	264.2	275.7	253.7	914.01	266.4	162.3	183.8	169.1	208.6	223.7	216.5	282.87	914.0	162.3
9	T. Solids	mg/1	172	194	198	158	194	216	150	156	152	160	194	172	176.33	216.0	150.0
10	D. Solids	mg/1	156	176	178	132	164	128	128	134	126	144	166	148	148.33	178.0	126.0
11	S. Solids	mg/1	16	18	20	26	30	88	22	22	26	16	28	24	28.00	88.0	16.0
12	Amm. Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	0.14	0.33	0.16	0.03	BDL	BDL	0.17	0.3	0.0
14	Nitrate Nitrogen	mg/1	1.1	1.02	1.08	1.13	1.14	2.46	3.46	2.04	3.55	2.6	2.5	2.85	2.08	3.6	1.0
15	Phosphate (PO_4)	mg/1	0.09	0.08	0.09	0.06	0.09	0.32	0.36	0.28	0.07	0.28	BDL	BDL	0.17	0.4	0.1
16	Chloride	mg/1	11.82	12.99	13.99	11.99	12.9	9.99	21.21	12.53	10.6	8.81	11.74	15.65	12.85	21.2	8.8
17	Sulphate (SO_4)	mg/1	5.75	6.84	7.0	7.56	7.90	13.3	16.2	15.8	15.24	5.24	2.13	3.86	8.90	16.2	2.1
18	T. Alkalinity	mg/1	106	132	136	86	132	92	162	112	114	102	98	106	114.83	162.0	86.0
19	T. Hardness	mg/1	128	116	126	94	128	90	156	116	116	116	96	104	115.50	156.0	90.0
20	CalciumHardness	mg/1	104	94	100	50	108	46	128	86	90	58	66	84	84.50	128.0	46.0
21	Magnesium H.	mg/1	24	22	26	44	20	44	28	30	26	58	30	20	31.00	58.0	20.0
22	D. Oxygen	mg/1	8.7	8.8	8.7	8.8	6.0	8.3	8.2	10.7	8.6	7.1	7.6	7.6	8.26	10.7	6.0
23	B.O.D.	mg/1	0.9	1	ment out of	1.8	2.5	2.2	2.1	1.4	0.9	2.1	1.2	1.1	1.56	2.5	0.9
24	C.O.D.	mg/1	10.0	10	10	19.36	27.6	29.4	19.04	19.2	9.6	30	9.68	9.68	16.96	30.0	9.6
25	Sodium	mg/1	10.8	11.84	11.97	10.26	11.98	8.37	7.64	14.6	6.45	6.91	5.75	6.42	9.42	14.6	5.8
26	Potassium	mg/1	1.3	1.33	1.42	1.33	1.41	1.28	1.14	2.7	1.13	3.04	0.78	0.76	1.47	3.0	0.8
27	T. Coliform	MPN/100ml	48	41	43	40	110	220	11000	22000	4600	43000	350	310	6813.50	43000.0	40.0
28	F. Coliform	MPN/100ml	1.8	1.8	1.8	1.8	1.8	14	360	180	280	4700	17	1.8	463.48	4700.0	1.8
29	Boron	mg/1	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
30	T.K.N.	mg/1	-	-	-	-	-				0.7	BDL	2.1	1.8	1.53	2.1	0.7
31	Streptococcus	MPN/100ml	-	-	-	-	-	-	1.8	1.8	4	1.8	1.8	2.8	2.33	4.0	1.8
Category--B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2020-2021

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal)

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.04.20	Not in package	Not in package	Not in package	Not in package	10.09.20	Not in package	Not in package	Not in package	Not in package	23.02.2021	Not in package	-	-	-
2	Appearance		Clear					clear					Slightly turbid		-	-	-
3	Temperature	oC	30					30							30.00	30.0	30.0
4	Turbidity	NTU	12.8					7.3					2.1		7.40	12.8	2.1
5	Colour	PCS	Colorless					colourless					50		-	-	-
6	Odour	T. No	Odourless					Odourless					-		-	-	-
7	pH	pH Unit	8.02					8.13					7.85		8.00	8.1	7.9
8	Sp. Conductivity	µMhos/cm.	302.2					159.1					214.9		225.40	302.2	159.1
9	T. Solids	mg/1	216					152					136		168.00	216.0	136.0
10	D. Solids	mg/1	194					124					118		145.33	194.0	118.0
11	S. Solids	mg/1	22					28					18		22.67	28.0	18.0
12	Amm. Nitrogen	mg/1	BDL					BDL					BDL				
13	Nitrite Nitrogen	mg/1	BDL					BDL					BDL				
14	Nitrate Nitrogen	mg/1	2.25					1.85					2.5		2.20	2.5	1.9
15	Phosphate (PO_4)	mg/1	0.6					0.05					BDL		0.33	0.6	0.1
16	Chloride	mg/1	16.75					22.99					13.69		17.81	23.0	13.7
17	Sulphate (SO_4)	mg/1	12.5					14.92					2.08		9.83	14.9	2.1
18	T. Alkalinity	mg/1	156					146					102		134.67	156.0	102.0
19	T. Hardness	mg/1	162					142					86		130.00	162.0	86.0
20	CalciumHardness	mg/1	130					118					52		100.00	130.0	52.0
21	Magnesium H.	mg/1	32					24					34		30.00	34.0	24.0
22	D. Oxygen	mg/1	8.1					8.6					6.8		7.83	8.6	6.8
23	B.O.D.	mg/1	2.1					3.3					0.9		2.10	3.3	0.9
24	C.O.D.	mg/1	10					19.04					10		13.01	19.0	10.0
25	Sodium	mg/1	24.8					16.85					6.51		16.05	24.8	6.5
26	Potassium	mg/1	1.9					1.68					0.76		1.45	1.9	0.8
27	T. Coliform	MPN/100ml	-					-					-				
28	F. Coliform	MPN/100ml	-					-					-		-	-	-
29	T.K.N.	mg/1	-					-					4.2		4.20	4.2	4.2
	Cat. A																

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2020-2021

Natural Water Resource : Upper Lake near Sehore Naka.

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		17.04.20	Not in package	Not in package	Not in package	Not in package	**10.09.20**		Not in package	Not in package	Not in package	23.02.2021	Not in package	-	-	-
2	Appearance		Clear					clear					Slightly		-	-	-
3	Temperature	oC	30					31							30.5	31	30
4	Turbidity	NTU	8.14					9.7					4		7.28	9.7	4
5	Colour	PCS	colourless					colourless					60		60	60	60
6	Odour	T. No	odourless					odourless					-		-	-	-
7	pH	pH Unit	8.32					8.99					7.73		8.35	8.99	7.73
8	Sp. Conductivity	µMhos/cm.	268.8					162.5					304.2		245.17	304.2	162.5
9	T. Solids	mg/1	200					154					190		181.33	200	154
10	D. Solids	mg/1	182					132					174		162.67	182	132
11	S. Solids	mg/1	18					22					16		18.67	22	16
12	Amm. Nitrogen	mg/1	BDL					BDL					BDL				
13	Nitrite Nitrogen	mg/1	BDL					BDL					BDL				
14	Nitrate Nitrogen	mg/1	0.3					1.05					4.51		1.95	4.51	0.3
15	Phosphate (PO_4)	mg/1	0.16					0.05					BDL		0.11	0.16	0.05
16	Chloride	mg/1	14.73					24.99					15.65		18.46	24.99	14.73
17	Sulphate (SO_4)	mg/1	8.75					7.62					8.58		8.32	8.75	7.62
18	T. Alkalinity	mg/1	134					150					90		124.67	150	90
19	T. Hardness	mg/1	148					146					80		124.67	148	80
20	CalciumHardness	mg/1	124					114					50		96.00	124	50
21	Magnesium H.	mg/1	24					32					30		28.67	32	24
22	D. Oxygen	mg/1	8.2					8.8					6.1		7.70	8.8	6.1
23	B.O.D.	mg/1	2.4					1.4					1.8		1.87	2.4	1.4
24	C.O.D.	mg/1	10.0					19.04					10		13.01	19.04	10
25	Sodium	mg/1	10.4					9.65					7.94		9.33	10.4	7.94
26	Potassium	mg/1	1.0					1.16					1.87		1.34	1.87	1
27	T. Coliform	MPN/100ml	-					-					-				
28	F. Coliform	MPN/100ml	-			-		-	-				-		-	-	-
29	T.K.N.	mg/1	-			-			-				BDL		-	-	-
	Cat. A																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2021-22

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal) **2137**

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake) **XGN ID: 22114**

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		4/23/2021	Sampling not done due to covid-19	7/6/2021		12/8/2021	13/9/2021	29/10/21	22/11/21	13/12/21	10/1/2022	24/02/22	10/3/2022	-	-	-
2	Appearance		Clear		clear		Clear								-	-	-
3	Temperature	oC	30		31		28		29	24	21	18	22	25	25.33	31.0	18.0
4	Turbidity	NTU	4		4.2		9.4								5.87	9.4	4.0
5	Colour	PCS	30		40		20								30.00	40.0	20.0
6	Odour	T. No	Odourless		NO O.		NO O.		40	20	30	20	30	30	28.33	40.0	20.0
7	pH	pH Unit	7.72		7.62		8.21	7.46	7.4	7.77	7.32	8.02	7.93	7.73	7.72	8.2	7.3
8	Sp. Conductivity	µMhos/cm.	226.5		212.3		288.9	244.2	248.9	267.5	219.8	239.5	275.8	281.4	250.48	288.9	212.3
9	T. Solids	mg/1	194		178		206								192.67	206.0	178.0
10	D. Solids	mg/1	168		158		184	188	178	188	162	166	198	180	177.00	198.0	158.0
11	S. Solids	mg/1	26		20		22	10	10	20	4	8	18	42	18.00	42.0	4.0
12	Amm. Nitrogen	mg/1	BDL		1.8		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	1.80	1.8	1.8
13	Nitrite Nitrogen	mg/1	0.05		0.03		0.03	BDL	0.02	0.05	0.01	BDL	BDL	0.01	0.03	0.1	0.0
14	Nitrate Nitrogen	mg/1	2.59		2.58		2.56	4.63	4.73	2.65	2.01	3.83	2.33	2.61	3.05	4.7	2.0
15	Phosphate (PO_4)	mg/1	0.09		0.35		0.28	1.61	1.78	0.27	0.26	1.12	2.12	BDL	0.88	2.1	0.1
16	Chloride	mg/1	9.99		10.76		13.69	15.65	16.63	12.81	12.99	31.3	19.56	18.58	16.20	31.3	10.0
17	Sulphate (SO_4)	mg/1	12.46		5.66		6.85	3.81	4.96	6.75	6.01	2.63	11.03	2.95	6.31	12.5	2.6
18	T. Alkalinity	mg/1	118		98		88	94	96	88	92	128	92	108	100	128.0	88.0
19	T. Hardness	mg/1	122		112		86	96	96	86	84	138	102	112	103	138.0	84.0
20	CalciumHardness	mg/1	68		70		56	64	62	58	62	96	72	66	67	96.0	56.0
21	Magnesium H.	mg/1	54		42		30	32	34	28	22	42	30	46	36	54.0	22.0
22	D. Oxygen	mg/1	8.6		7.2		7.7	6.4	6.8	7.1	4.7	5.7	6.4	6.3	6.69	8.6	4.7
23	B.O.D.	mg/1	1.2		1.3		3.1	1.3	1.1	1.2	3.6	1.1	1.4	1.2	1.65	3.6	1.1
24	C.O.D.	mg/1	10		19.6		20	10	10	19.36	29.64	9.8	9.52	9.52	14.74	29.6	9.5
25	Sodium	mg/1	8.53		6.38		5.51	6.72	7.14	4.15	5.83	6.71	5.77	4.97	6.17	8.5	4.2
26	Potassium	mg/1	2.08		2.14		0.37	0.68	1.23	2.32	2.4	2.8	0.62	0.64	1.53	2.8	0.4
27	T. Coliform	MPN/100ml	430		430		240	140	150	220	280	130	350	280	265	430.0	130.0
28	F. Coliform	MPN/100ml	11		3.6		1.8	6.4	6.8	7.1	4.7	5.7	6.4	6.3	5.98	11.0	1.8
29	Streptococcus	MPN/100ml	1.8		1.8		1.8						1.8	1.8	1.80	1.8	1.8
30	T.K.N.	mg/1	1.4		1.8		1.4	1.4	1.4	1.4	1.4	1.4	BDL	2.8	1.60	2.8	1.4
31	Calcium	mg/1	27.23		28.03		22.42		24.82	23.22	24.82	38.45	28.83	26.43	27.14	38.5	22.4
32	Magnesium	mg/1	13.11		10.19		7.28		8.25	6.79	5.34	10.19	7.28	11.17	8.84	13.1	5.3
33	Boron	mg/3	BDL		1.8		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	1.80	1.8	1.8
Category - B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2021-22

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal) **XGN ID: 22115**

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23/04/2021	Sampling not done due to covid-19	7/6/2021		8/12/2021	9/29/2021	10/7/2021	11/22/2021	12/9/2021	1/12/2022	2/9/2022	3/9/2022	-	-	-
2	Appearance		Clear		clear		Clear								-	-	-
3	Temperature	oC	29		32		29		30	24	24	19	23	24	26.00	32	19
4	Turbidity	NTU	3.9		4.6		11.4	3.1	1.7	2.8	8.4	6.1	3.2	1.4	4.66	11.4	1.4
5	Colour	PCS	20		40		30		50	30	40	20	20	30	31.11	50	20
6	Odour	T. No	Odourless		NO O.		NO O.								-	-	-
7	pH	pH Unit	7.69		7.62		8.08	7.74	7.44	7.58	7.29	8.2	7.62	7.99	7.73	8.2	7.29
8	Sp. Conductivity	µMhos/cm.	224.8		212.3		227.2	247.9	248.8	241.7	205.6	226.6	260.4	278.9	237.42	278.9	205.6
9	T. Solids	mg/1	178		178		174	192	180	198	154	174	202	202	183.20	202	154
10	D. Solids	mg/1	156		158		144	178	168	172	144	158	186	176	164.00	186	144
11	S. Solids	mg/1	22		20		30	14	12	26	10	16	16	26	19.20	30	10
12	Amm. Nitrogen	mg/1	BDL		BDL		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	0.05		0.03		0.04	BDL	BDL	0.04	0.06	BDL	BDL	0.01	0.04	0.06	0.01
14	Nitrate Nitrogen	mg/1	2.59		2.58		2.32	4.91	4.99	3.32	2.46	4.25	2.15	2.67	3.22	4.99	2.15
15	Phosphate (PO_4)	mg/1	0.09		0.35		0.12	BDL	BDL	0.11	0.21	BDL	1.08	BDL	0.33	1.08	0.09
16	Chloride	mg/1	8.99		10.76		15.65	13.69	14.67	12.81	12.99	20.54	20.54	19.56	15.02	20.54	8.99
17	Sulphate (SO_4)	mg/1	12.46		5.66		5.84	8.5	9.7	5.48	4.22	6.96	13.08	2.56	7.45	13.08	2.56
18	T. Alkalinity	mg/1	112		98		88	90	92	88	94	126	90	106	98.40	126	88
19	T. Hardness	mg/1	116		112		84	92	98	82	88	138	102	114	102.60	138	82
20	Cal.Hardness	mg/1	74		70		50	58	68	52	56	80	68	68	64.40	80	50
21	Magnesium H.	mg/1	42		42		34	34	30	30	32	58	34	46	38.20	58	30
22	D. Oxygen	mg/1	8.6		7.2		7.4	6.8	6.4	5.2	5.9	6.2	6.3	9.2	6.92	9.2	5.2
23	B.O.D.	mg/1	1.2		1.3		4	1.2	1.8	1.9	1.5	1.2	1.3	0.7	1.61	4	0.7
24	C.O.D.	mg/1	10		19.6		40	10	10	9.68	9.88	9.8	9.52	9.52	13.80	40	9.52
25	Sodium	mg/1	8.53		6.38		5.65	6.86	6.99	6.55	7.5	4.12	6.45	5.45	6.45	8.53	4.12
26	Potassium	mg/1	2.08		2.14		0.43	0.62	0.82	0.42	1.25	1.63	0.37	0.42	1.02	2.14	0.37
27	T. Coliform	MPN/100ml	430		430		170	170	170	350	430	150	140	240	268.00	430	140
28	F. Coliform	MPN/100ml	11		3.6		5.5	33	33	2	1.8	23	3.6	3.7	12.02	33	1.8
29	Boron	mg/1	BDL		BDL		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
30	Streptococcus	MPN/100ml	1.8		1.8		1.8						1.8	1.8	1.80	1.8	1.8
31	T.K.N.	meq/l	1.4		BDL		1.4	1.4	1.4	2.8	1.4	2.8	1.8	2.8	1.91	2.8	1.4
32	Calcium	mg/1	29.63		28.03		20.02		27.23	20.82	22.42	32.04	27.23	27.23	26.07	32.04	20.02
33	Magnesium	mg/1	10.19		10.19		8.25	-	7.28	7.28	7.77	14.08	8.25	11.17	9.38	14.08	7.28
Category -B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2021-22

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal) 2139

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the La **XGN ID: 22116**

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23/4/2021	Sampling not done due to covid-19	7/6/2021		8/12/2021	9/29/2021	10/7/2021	11/22/2021	12/9/2021	1/12/2022	2/9/2022	3/9/2022	-	-	-
2	Appearance		Clear		clear		Clear								-	-	-
3	Temperature	oC	28		32		27	29	28	,26	24	18	23	22	25.67	32.0	18.0
4	Turbidity	NTU	3.2		4.1		11.4	2.7	1.7	3.5	5.5	2.3	2.5	2.6	3.95	11.4	1.7
5	Colour	PCS	30		30		30	30	50	30	30	30	20	50	33.00	50.0	20.0
6	Odour	T. No	Odourless		NO O. OB.		NO O. OB.								-	-	-
7	pH	pH Unit	7.77		7.79		8.08	7.75	7.48	7.85	7.68	8.14	8.15	7.03	7.77	8.2	7.0
8	Sp. Conductivity	µMhos/cm.	219.9		214.6		227.2	258.6	285.5	254.2	231.3	230	259.4	310.4	249.11	310.4	214.6
9	T. Solids	mg/1	170		176		174	194	214	198	178	172	202	214	189.20	214.0	170.0
10	D. Solids	mg/1	146		158		144	182	204	184	172	162	184	176	171.20	204.0	144.0
11	S. Solids	mg/1	24		18		30	12	10	14	6	10	18	38	18.00	38.0	6.0
12	Amm. Nitrogen	mg/1	BDL		BDL		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	0.09		0.02		0.04	BDL	BDL	0.03	0.04	BDL	BDL	0.09	0.05	0.1	0.0
14	Nitrate Nitrogen	mg/1	3.12		2.27		3.05	4.96	5.01	3.06	2.38	4.42	BDL	4.27	3.62	5.0	2.3
15	Phosphate (PO_4)	mg/1	0.24		0.31		0.25	BDL	0.01	0.24	0.42	BDL	0.89	BDL	0.34	0.9	0.0
16	Chloride	mg/1	11.99		11.74		15.65	12.71	13.69	14.78	18.99	21.52	21.52	21.52	16.41	21.5	11.7
17	Sulphate (SO_4)	mg/1	18.42		6.1		6.32	7.84	8.12	6.23	6.55	6.82	11.7	5.16	8.33	18.4	5.2
18	T. Alkalinity	mg/1	124		86		116	94	96	110	116	136	90	112	108.00	136.0	86.0
19	T. Hardness	mg/1	122		106		118	96	100	112	118	140	102	126	114.00	140.0	96.0
20	Cal.Hardness	mg/1	64		64		76	66	68	78	80	82	72	94	74.40	94.0	64.0
21	Magnesium H.	mg/1	58		42		42		32	34	38	58	30	32	40.67	58.0	30.0
22	D. Oxygen	mg/1	7.2		7.6		4.1	6.7	8	6.9	6	6.1	6.4	4.9	6.39	8.0	4.1
23	B.O.D.	mg/1	1.7		1.2		1.5	1.1	1.2	1.2	1.2	1.4	1.2	1.5	1.32	1.7	1.1
24	C.O.D.	mg/1	20		29.4		30	10	10	9.68	9.88	19.6	9.52	19.04	16.71	30.0	9.5
25	Sodium	mg/1	9.45		7.36		5.72	0.62	1.12	0.46	1.5	1.14	0.97	0.86	2.92	9.5	0.5
26	Potassium	mg/1	3.16		3.11		0.45	5.87	6.24	5.27	6.4	5.24	5.6	4.97	4.63	6.4	0.5
27	T. Coliform	MPN/100ml	920		430		350	280	240	240	280	240	240	350	357.00	920.0	240.0
28	F. Coliform	MPN/100ml	24		3.7		3.6	25	24	1.8	1.8	27	1.8	5.6	11.83	27.0	1.8
29	Boron	mg/1	BDL		BDL		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
30	Streptococus	MPN/100ml	1.8		1.8		1.8								1.80	1.8	1.8
31	Calcium	mg/1	25.63		25.63		30.43		27.23	31.23	32.03	32.84	28.83	37.64	30.17	37.6	25.6
32	Magnesium	mg/1	14.08		10.19		10.19	7.28	7.77	8.25	9.22	14.08	7.28	7.77	9.61	14.1	7.3
33	T.K.N.	mg/1	3.5		BDL		1.4	BDL	1.4	2.8	1.4	1.4	1.8	2.8	2.06	3.5	1.4
	Cat. -B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2021-22

XGN ID: 22025

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		23/4/2021	Sampling not done due to covid-19	7/6/2021		8/12/2021	9/29/2021	10/25/2021	11/22/2021	12/21/2021	1/28/2022	2/17/2022	3/13/2022	-	-	-
2	Appearance		Clear		clear		Clear								-	-	-
3	Temperature	oC	29		32		28		30	25	23	19	21	23	25.56	32.0	19.0
4	Turbidity	NTU	3.6		4.6		11.9	2.4	1.7	4.2	7.4	6.7	2.6	3.9	4.90	11.9	1.7
5	Colour	PCS	20		30		20	20	50	20	40	20	20	20	-	-	-
6	Odour	T. No	Odourless		NO O. OB.		NO O. OB.								-	-	-
7	pH	pH Unit	7.66		7.38		7.73	7.2	7.37	7.18	7.28	7.42	7.69	7.46	7.44	7.7	7.2
8	Sp. Conductivity	µMhos/cm.	229.3		218.5		246.6	226.4	235.9	292.2	305.1	237.4	387.1	274.1	265.26	387.1	218.5
9	T. Solids	mg/1	170		178		182	180	186	226	210	180	306	246	206.40	306.0	170.0
10	D. Solids	mg/1	152		160		162	166	172	208	200	168	286	198	187.20	286.0	152.0
11	S. Solids	mg/1	18		18		20	14	14	18	10	12	20	48	19.20	48.0	10.0
12	Amm. Nitrogen	mg/1	BDL		BDL		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
13	Nitrite Nitrogen	mg/1	0.03		0.04		0.05	BDL	BDL	0.04	0.02	BDL	BDL	0.02	0.03	0.1	0.0
14	Nitrate Nitrogen	mg/1	2.47		3.58		3.26	3.78	3.88	2.63	3.05	4.16	2.31	3.52	3.26	4.2	2.3
15	Phosphate (PO_4)	mg/1	0.06		0.38		0.32	BDL	BDL	0.23	0.27	BDL	0.47	BDL	0.29	0.5	0.1
16	Chloride	mg/1	6.99		11.74		14.67	13.69	14.67	11.82	13.99	19.56	19.56	20.54	14.72	20.5	7.0
17	Sulphate (SO_4)	mg/1	9.82		6.42		6.86	11.07	13.11	6.86	6.82	10.72	10.53	2.17	8.44	13.1	2.2
18	T. Alkalinity	mg/1	106		114		82	90	94	82	86	138	100	102	99.40	138.0	82.0
19	T. Hardness	mg/1	110		118		80	94	98	80	88	144	108	112	103.20	144.0	80.0
20	CalciumHardness	mg/1	76		78		52	56	64	54	56	98	72	80	68.60	98.0	52.0
21	Magnesium H.	mg/1	34		40		28	38	34	26	32	46	36	32	34.60	46.0	26.0
22	D. Oxygen	mg/1	8.1		7.6		7.4	6.3	6	7.8	5	6	6.6	4.8	6.56	8.1	4.8
23	B.O.D.	mg/1	1.1		1.9		4.2	1.2	1.6	1.3	2.3	1.3	1	1	1.69	4.2	1.0
24	C.O.D.	mg/1	10		29.4		50	10	10	9.68	19.76	9.8	9.52	9.52	16.77	50.0	9.5
25	Sodium	mg/1	6.42		5.82		5.57	7.86	8.31	4.57	4.81	7.16	6.75	5.32	6.26	8.3	4.6
26	Potassium	mg/1	2.01		2.06		0.38								1.48	2.1	0.4
27	T. Coliform	MPN/100ml	350		220		220	150	170	170	220	140	430	350	242.00	430.0	140.0
28	F. Coliform	MPN/100ml	6		1.8		1.8	23	26	1.8	1.8	27	3.7	6.1	9.90	27.0	1.8
29	Boron	mg/1	BDL		BDL		BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
30	T.K.N.	mg/1	1.4		BDL		2.80	1.4	1.4	2.8	2.8	1.4	BDL	2.8	2.10	2.8	1.4
31	Streptococcus	MPN/100ml	1.8		1.8		1.80	-					1.8	1.8	1.80	1.8	1.8
Category -B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2021-22

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal) XGN ID: 22158

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Sampling not done due to covid-19	Not in package	Not in package	Not in package	Not in package	9/29/2021	Not in package	Not in package	Not in package	Not in package	2/17/2022	Not in package	-	-	-
2	Appearance														-	-	-
3	Temperature	oC											22		22.00	22.0	22.0
4	Turbidity	NTU						9.8					8.1		8.95	9.8	8.1
5	Colour	PCS													-	-	-
6	Odour	T. No											50		50.00	50.0	50.0
7	pH	pH Unit						8.16					8.32		8.24	8.3	8.2
8	Sp. Conductivity	µMhos/cm.						275.4					242.1		258.75	275.4	242.1
9	T. Solids	mg/1						224					204		214.00	224.0	204.0
10	D. Solids	mg/1						186					178		182.00	186.0	178.0
11	S. Solids	mg/1						38					26		32.00	38.0	26.0
12	Amm. Nitrogen	mg/1						BDL					BDL				
13	Nitrite Nitrogen	mg/1						BDL					BDL				
14	Nitrate Nitrogen	mg/1						5.64					BDL		5.64	5.6	5.6
15	Phosphate (PO_4)	mg/1						BDL					0.3		0.30	0.3	0.3
16	Chloride	mg/1						40					9.52		24.76	40.0	9.5
17	Sulphate (SO_4)	mg/1						12.52					14.06		13.29	14.1	12.5
18	T. Alkalinity	mg/1						88					68		78.00	88.0	68.0
19	T. Hardness	mg/1						98					94		96.00	98.0	94.0
20	CalciumHardness	mg/1						44					62		53.00	62.0	44.0
21	Magnesium H.	mg/1						54					32		43.00	54.0	32.0
22	D. Oxygen	mg/1						6.9					7.2		7.05	7.2	6.9
23	B.O.D.	mg/1						4.9					1.4		3.15	4.9	1.4
24	C.O.D.	mg/1						40					9.52		24.76	40.0	9.5
25	Sodium	mg/1						7.28					8.69		7.99	8.7	7.3
26	Potassium	mg/1						2.02					1.12		1.57	2.0	1.1
27	T. Coliform	MPN/100ml						0					0		0.00	0.0	0.0
28	F. Coliform	MPN/100ml						0					0		0.00	0.0	0.0
29	T.K.N.	mg/1	-					7					1.8		4.40	7.0	1.8
	Cat. -A																

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2021-22

Natural Water Resource : Upper Lake near Sehore Naka. XGN ID: 22159

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		Sampling not done due to covid-19	Not in package	Not in package	Not in package	Not in package	9/29/2021	Not in package	Not in package	Not in package	Not in package	2/17/2022	Not in package	-	-	-
2	Appearance														-	-	-
3	Temperature	oC											22		22	22	22
4	Turbidity	NTU						2.8					2.7		2.75	2.8	2.7
5	Colour	PCS											50		-	-	-
6	Odour	T. No													-	-	-
7	pH	pH Unit						7.51					7.74		7.63	7.74	7.51
8	Sp. Conductivity	µMhos/cm.						496.2					342.8		419.50	496.2	342.8
9	T. Solids	mg/1						334					296		315.00	334	296
10	D. Solids	mg/1						310					252		281.00	310	252
11	S. Solids	mg/1						24					44		34.00	44	24
12	Amm. Nitrogen	mg/1						BDL					BDL				
13	Nitrite Nitrogen	mg/1						0.11					0.75		0.43	0.75	0.11
14	Nitrate Nitrogen	mg/1						5.63					3.64		4.64	5.63	3.64
15	Phosphate (PO_4)	mg/1						0.6					0.71		0.66	0.71	0.6
16	Chloride	mg/1						28.37					21.52		24.95	28.37	21.52
17	Sulphate (SO_4)	mg/1						7.76					11.36		9.56	11.36	7.76
18	T. Alkalinity	mg/1						160					90		125.00	160	90
19	T. Hardness	mg/1						172					102		137.00	172	102
20	CalciumHardness	mg/1						110					80		95.00	110	80
21	Magnesium H.	mg/1						62					22		42.00	62	22
22	D. Oxygen	mg/1						4					3.8		3.90	4	3.8
23	B.O.D.	mg/1						2.5					1.8		2.15	2.5	1.8
24	C.O.D.	mg/1						20					9.52		14.76	20	9.52
25	Sodium	mg/1						7.36					6.69		7.03	7.36	6.69
26	Potassium	mg/1						1.78					0.86		1.32	1.78	0.86
27	T. Coliform	MPN/100ml															
28	F. Coliform	MPN/100ml				-			-								
29	T.K.N.	mg/1	-			-		7					2.8		4.90	7	2.8
30	Calcium	mg/1											32.03		32.03	32.03	32.03
31	Magnesium	mg/1											5.35		5.35	5.35	5.35
	Cat. -B																

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2022-23

Natural Water Resource : Upper Lake , Bhopal. (Distt.-Bhopal) **2137**

Description of Sampling Station: Near w/s intake well at Yatch Club (South side of the Lake) **XGN ID: 22114**

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.2022	19.5.2022	20.6.2022	26.7.2022	16.8.2022	8.9.2022	10.10.2022	23.11.2022	27.12.2023	18.1.2023	8.02.2023	1.03.2023	-	-	-
2	Alkalinity	mg/l	138	136	108	106	108	96	108	122	114	114	120	136	117.17	138.0	96.0
3	Amm. Nitrogen	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
4	B.O.D	mg/l	1.4	2.8	1.1	2	0.9	1	1.9	1	1.1	1.3	1.4	4.4	1.69	4.4	0.9
5	Boron	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
6	Cadmium	mg/l	BDL														
7	Calcium	mg/l	35.24	32.03	26.43	32.83	32.83	25.63	31.23	30.43	38.44	27.24	30.43	38.44	31.77	38.4	25.6
8	Calcium Hardness	Mg/Lts	88	80	66	82	82	64	78	76	96	68	76	96	79.33	96.0	64.0
9	C.O.D.	mg/l	20	19.6	9.68	9.88	10	9.4	10	10	9.88	10	9.52	28.8	13.06	28.8	9.4
10	Chloride	mg/l	21.68	24.88	12.99	19.56	21.99	16.75	18.99	14.67	25.62	15.65	16.63	29.56	19.91	29.6	13.0
11	Colour	Pt.Co.Scale	40	40	30	170	30	30	20	30	40	20	50	30	44.17	170.0	20.0
12	Conductivity	micromho/cm	306.6	310.7	355.5	270.1	257.8	172	161.7	218.2	240.1	261.4	231.1	238.7	251.99	355.5	161.7
13	Dissolved Oxygen	mg/l	5	7.7	5.4	6	8.6	5.6	6.2	7.1	7.2	7	7.2	9.8	6.90	9.8	5.0
14	Fecal Coliform	MPN/100 ml	31	22	3.6	3.6	6.1	21	3.7	1.8	27	11	38	49	18.15	49.0	1.8
15	Magnesium	mg/l	13.59	14.57	10.68	7.28	7.77	8.74	7.77	10.68	6.79	13.6	12.62	13.59	10.64	14.6	6.8
16	Magnesium Hard.	Mg/Lts	56	60	44	30	32	36	32	44	28	56	52	56	44	60.0	28.0
17	Nitrate	mg/l	2.93	3.43	2.32	3.25	3.25	1.05	3.52	2.13	2.05	3.12	3.29	6.04	3	6.0	1.1
18	Nitrite	mg/l	BDL	BDL	0.02	BDL	0.04	BDL	0.06	0.09	BDL	BDL	BDL	0.11	0	0.1	0.0
19	pH	pH Units	7.92	8.06	7.35	7.05	7.14	7.8	7.88	7.52	7.85	7.78	8.04	8.02	8	8.1	7.1
20	Phosphate	mg/l	1.02	BDL	BDL	0.19	BDL	1.02	0.02	BDL	0.09	BDL	BDL	0.06	0.40	1.0	0.0
21	Potassium	mg/l	2.12	0.24	0.71	0.56	0.34	1.96	0.43	0.51	1.58	0.92	1.73	1.02	1.01	2.1	0.2
22	Sodium	mg/l	5.86	5.11	2.97	5.23	5.67	6.12	5.76	5.43	7.34	3.24	4.34	6.1	5.26	7.3	3.0
23	(SAR)	SAR	0.21	0.18	0.12	0.21	0.23	0.26	0.23	0.21	0.22	0.1	0.16	0.21	0.20	0.3	0.1
24	streptococci	MPN/100 ml	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.80	1.8	1.8
25	Sulphate	mg/l	5.16	5.66	4.41	2.71	2.71	4.19	3.71	10.34	9.21	10.35	8.87	5.52	6	10.4	2.7
26	Suspended Solids	mg/l	26	8	38	36	76	12	28	22	22	22	32	30	29.33	76.0	8.0
27	Temperature	Centigrade	31	31	27	26	24	26	24	17	17	14	21	19	23.08	31.0	14.0
28	Total coliform	MPN/100 ml	140	280	280	280	120	110	210	280	280	210	120	280	215.83	280.0	110.0
29	Total Dissolved Solid	mg/l	224	182	260	198	186	126	118	114	162	188	172	172	175.17	260.0	114.0
30	Total Hardness as $CaCO_3$	mg/l	144	140	110	112	114	100	110	120	124	124	128	152	123.17	152.0	100.0
31	T.K.N.	mg/l	2.8	2.8	1.4	1.4	1.4	Instrument out of order		1.4	1.4	1.4	1.4	2.8	1.82	2.8	1.4
32	Total Solids	mg/l	250	190	298	234	262	138	146	136	184	210	204	202	204.50	298.0	136.0
33	Turbidity	N.T.U.	3.4	4.8	6.6	25	5.2	2.7	1.4	0.9	18.8	3.9	4.3	7.8	7.07	25.0	0.9
Category - B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2022-23

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal) **XGN ID: 22115**

Description of Sampling Station : Near w/s intake well at Karbala (North side of the Lake)

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.2022	19.5.2022	20.6.2022	26.7.2022	16.8.2022	8.9.2022	10.10.2022	23.11.2022	27.12.2023	18.1.2023	8.02.2023	1.03.2023	-	-	-
2	Alkalinity	mg/l	128	134	106	110	112	84	110	116	124	84	112	140	113.33	140.0	84.0
3	Amm. Nitrogen	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
4	B.O.D	mg/l	1.1	1.5	0.8	2.2	1.1	1.4	1.6	1	2.8	1.4	0.9	2.3	1.51	2.8	0.8
5	Boron	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
7	Calcium	mg/l	33.64	34.44	27.23	33.64	28.83	21.62	28.83	28.03	35.24	24.03	29.63	33.64	29.90	35.24	21.62
8	Calcium Hardness	Mg/Lts	84	86	68	84	72	54	72	70	88	60	74	84	74.67	88	54
9	C.O.D.	mg/l	10	9.8	9.68	9.88	10	18.8	10	10	39.52	10	19.04	19.2	14.66	39.52	9.68
10	Chloride	mg/l	20.69	26.8	18.99	19.56	21.99	15.77	22.99	16.63	21.68	18.59	14.67	31.53	20.82	31.53	14.67
11	Colour	Pt.Co.Scale	50	50	20	110	50	40	20	20	30	30	70	30	43.33	110	20
12	Conductivity	micromho/c	337.7	314.2	298.5	266.2	231	174.8	154.8	218.6	234.3	280.3	225.9	229.1	247.12	337.7	154.8
13	Dissolved Oxygen	mg/l	5.3	5.8	6.7	5	6.8	7.1	7	5	6.9	6.5	8	9.6	6.64	9.6	5
14	Fecal Coliform	MPN/100 ml	26	13	3.6	5.6	3.7	21	6.1	3.6	23	17	17	48	15.63	48.00	3.6
15	Fixed Dissolved S.	mg/l															
16	Fluoride	mg/l	0.12	0.1	BDL	BDL	BDL	0.14	BDL	BDL	0.14	0.12	0.27	BDL	0.15	0.27	0.1
17	Magnesium	mg/l	13.59	13.59	10.68	7.28	10.68	8.74	10.68	13.11	10.68	8.74	10.68	18.45	11.41	18.45	7.28
18	Magnesium Ha.	Mg/Lts	56	56	44	30	44	36	44	54	44	36	44	76	47.00	76	30
19	Nitrate	mg/l	4.18	2.31	2.65	2.64	2.76	0.44	3.67	2.37	3.26	2.68	3.75	5.66	3.03	5.66	0.44
20	Nitrite	mg/l	BDL	BDL	0.03	BDL	0.02	BDL	0.04	0.06	0.06	BDL	BDL	BDL	0.04	0.06	0.02
21	pH	pH Units	7.29	7.68	7.78	7.14	7.38	7.94	8.25	7.5	7.57	8.06	8.16	8.03	7.73	8.25	7.14
22	Phosphate	mg/l	BDL	BDL	BDL	0.21	BDL	BDL	0.03	BDL	0.1	BDL	BDL	0.01	0.09	0.21	0.01
23	Potassium	mg/l	1.21	0.91	0.43	0.24	0.35	1.18	0.53	0.5	1.89	1.1	1.2	1.12	0.89	1.89	0.24
24	Sodium	mg/l	4.71	4.92	4.45	5.45	6.54	3.26	6.45	5.23	6.67	4.24	4.5	5.1	5.13	6.67	3.26
25	(SAR)	SAR	0.17	0.17	0.18	0.22	0.26	0.15	0.25	0.2	0.25	0.18	0.17	0.17	0.20	0.26	0.15
26	streptococci	MPN/100 ml	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.80	1.8	1.8
27	Sulphate	mg/l	5.82	4.95	2.48	2.55	2.05	5.87	2.06	9.76	11.45	8.57	9.67	7.51	6.06	11.45	2.05
28	Suspended Solids	mg/l	32	20	28	12	74	16	18	32	26	20	34	34	28.83	74	12
29	Temperature	Centigrade	30	33	28	24	25	27	24	17	16	14	22	20	23.33	33	14
30	Total coliform	MPN/100 ml	170	210	240	240	210	140	120	210	280	220	110	240	199.17	280	110
31	Total Dissolved	mg/l	238	194	218	188	170	126	112	112	144	204	166	164	169.67	238	112
32	Total Hardness as	mg/l	140	142	112	114	116	90	116	124	132	96	118	160	121.67	160	90
33	T.K.N.	mg/l	2.8	4.2	2.8	1.4	2.8	Instrument out of order			2.8	2.8	1.4	2.8	2.64	4.2	1.4
34	Total Solids	mg/l	270	214	246	200	244	142	130	144	170	224	200	198	198.50	270	130
35	Turbidity	N.T.U.	9.6	8.7	9.7	7	5.4	5.6	1.8	0.8	6.5	3.5	3.9	6.2	5.73	9.7	0.8
Category - B																	

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2022-23

Natural Water Resource : Upper Lake , Bhopal(Distt.-Bhopal) 2139

Description of Sampling Station : Near w/s intake well at Bairagarh (West side of the La **XGN ID: 22116**

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.2022	19.5.2022	20.6.2022	26.7.2022	16.8.2022	8.9.2022	10.10.2022	23.11.2022	27.12.2023	18.1.2023	8.02.2023	1.03.2023	-	-	-
2	Alkalinity	mg/l	108	90	120	114	120	76	122	118	104	106	116	134	110.67	134.0	76.0
3	Ammo. Nitrogen	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
4	B.O.D	mg/l	1.3	1.2	1.3	2.5	1	1.3	1.7	0.4	1	1.4	0.9	2.3	1.36	2.5	0.4
5	Boron	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
6	Calcium	mg/l	27.23	31.23	37.64	37.64	38.44	22.42	37.64	28.03	33.64	20.82	32.83	30.43	31.50	38.4	20.8
7	Calcium Ha.	Mg/Lts	68	78	94	94	96	56	94	70	84	52	82	76	78.67	96.0	52.0
8	C.O.D.	mg/l	20	9.6	19.36	19.76	20	9.4	10	20	29.64	10	19.04	19.2	17.17	29.6	9.4
9	Chloride	mg/l	21.68	23.92	18.99	19.56	22.99	14.78	21.99	16.63	21.68	17.61	14.67	29.56	20.34	29.6	14.7
10	Colour	Pt.Co.Scale	70	20	50	110	30	50	30	30	20	40	150	20	51.67	150.0	20.0
11	Conductivity	micromho/cm	341.6	304.6	296.5	269.7	228.8	174.7	195	214.5	214	194.5	227.8	232.4	241.18	341.6	174.7
12	Dissolved Oxygen	mg/l	5.2	6.6	4.9	6	7.6	5.3	5.3	7.4	7.5	8.1	8.5	8.9	6.78	8.9	4.9
13	Fecal Coliform	MPN/100 ml	33	17	5.6	3.7	5.6	23	5.6	6.1	21	24	48	40	19.38	48.0	3.7
14	Fixed Dissolved S.	mg/l	**Instrument out of order**														
15	Fluoride	mg/l	0.42	BDL	BDL	BDL	BDL	0.26	BDL	BDL	0.16	0.33	0.41	BDL	0.32	0.4	0.2
16	Magnesium	mg/l	12.63	7.77	7.77	7.77	7.77	6.79	7.77	13.59	6.79	15.06	9.71	19.42	10.24	19.4	6.8
17	Magnesium Ha.	Mg/Lts	52	32	32	32	32	28	32	56	28	62	40	80	42.17	80.0	28.0
19	Nitrate	mg/l	4.06	4.13	4.22	4.17	4.72	2.55	4.27	2.15	1.26	2.74	3.05	5.72	3.59	5.7	1.3
20	Nitrite	mg/l	BDL	0.01	0.08	0.68	0.08	BDL	0.1	0.07	0.04	BDL	BDL	BDL	0.15	0.7	0.0
21	pH	pH Units	7.24	7.89	7.96	7.04	7.48	8.06	8.34	7.71	7.89	8.09	8.13	8.13	7.83	8.3	7.0
22	Phosphate	mg/l	BDL	0.08	BDL	0.2	BDL	BDL	0.02	BDL	BDL	BDL	BDL	0.02	0.08	0.2	0.0
23	Potassium	mg/l	1.02	1.01	0.96	0.68	0.89	1.04	0.89	0.45	1.69	1.08	1.11	1.02	0.99	1.7	0.5
24	Sodium	mg/l	4.96	3.91	4.96	3.97	5.76	3.18	7.67	5.05	7.1	4.06	4.47	3.1	4.85	7.7	3.1
25	SAR	SAR	0.2	0.16	0.19	0.15	0.22	0.14	0.29	0.19	0.28	0.16	0.2	0.1	0.19	0.3	0.1
26	streptococci	MPN/100 ml	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.80	1.8	1.8
27	Sulphate	mg/l	6.24	7.31	5.16	5.61	1.61	4.47	2.61	11.16	9.15	8.61	8.94	7.14	6.50	11.2	1.6
28	Suspended Solids	mg/l	56	12	36	14	66	18	20	40	12	22	40	18	29.50	66.0	12.0
29	Temperature	Centigrade	29	32	29	25	23	26	22	16	17	16	20	19	22.83	32.0	16.0
30	Total coliform	MPN/100 ml	220	280	350	350	220	210	220	350	220	350	220	430	285.00	430.0	210.0
31	T. Dissolved Solids	mg/l	242	218	220	194	166	128	138	156	124	142	168	166	171.83	242.0	124.0
32	Total Ha.	mg/l	120	110	126	126	128	84	126	126	112	114	122	156	120.83	156.0	84.0
33	T.K.N.	mg/l	2.8	1.4	2.8	1.4	1.4	Instrument out of order			1.4	1.4	4.2	1.4	1.4	4.2	1.4
34	Total Solids	mg/l	298	230	256	208	232	146	158	196	196	136	164	208	184	298.0	136.0
35	Turbidity	N.T.U.	9.3	5.8	11.8	7	5.6	5.3	2.4	0.4	0.4	8	3.7	4.6	6.5	11.8	0.4

Category - B

MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2022-23

XGN ID: 22025

Natural Water Resource : Upper Lake,at Kamla Park Near w/s.intake well Bhopal.

Description of Sampling Station :

S.No.	Characteristic	Unit	*April*	May	June	*July*	Aug.	Sept.	*Oct.*	Nov.	Dec.	*Jan.*	Feb	March	Av.	Maxi.	Mini.
1	Date of Sampling		7.4.2022	19.5.2022	20.6.2022	26.7.2022	16.8.2022	8.9.2022	10.10.2022	23.11.2022	27.12.2023	18.1.2023	8.02.2023	1.03.2023	-	-	-
2	Alkalinity	mg/l	120	124	108	108	110	82	106	116	118	96	114	142	112.00	142.0	82.0
3	Ammo.Nitrogen	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL			
4	B.O.D	mg/l	1.2	2.8	1	2.1	1	0.9	1.5	0.8	0.9	1.1	1	2.2	2.3	2.8	0.8
5	Boron	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL		-	-
6	Calcium	mg/l	31.24	30.43	31.23	33.64	28.03	23.22	29.63	29.63	33.64	24.83	28.83	35.24	29.97	35.2	23.2
7	Calcium Hardness	Mg/Lts	78	76	78	84	70	58	74	74	84	62	72	88	76	88.0	58.0
8	C. O. D.	mg/l	20	19.6	9.68	9.88	10	9.4	10	10	9.88	10	9.52	9.6	19.2	20.0	9.4
9	Chloride	mg/l	22.66	24.88	12.99	18.58	19.99	14.78	20.99	16.63	21.68	16.63	16.63	27.59	29.56	29.6	13.0
10	Colour	Pt.Co.Scale	40	40	30	200	50	30	10	20	30	20	60	30	20	200.0	10.0
11	Conductivity	micromho/c	295.9	323.1	301.8	267.9	230.2	179.4	161.3	226.1	219.8	226.5	224.6	234.7	232.4	323.1	161.3
12	Copper	mg/l	0.027												0.03	0.0	0.0
13	Dissolved Oxygen	mg/l	5.8	5.9	5.8	5.1	7.2	5	6.9	4.7	7.6	6.7	8.8	7.8	6.44	8.8	4.7
14	Fecal Coliform	MPN/100 ml	23	26	6.1	6.1	4	23	4	5.5	23	17	22	120	23.31	120.0	4.0
15	Fluoride	mg/l	0.18	0.09	BDL	BDL	BDL	0.21	BDL	BDL	0.1	0.26	0.63	BDL	0.25	0.6	0.1
16	Iron	mg/l	0.2												0.20	0.2	0.2
17	Magnesium	mg/l	13.11	15.05	8.25	6.79	10.68	7.28	9.71	9.22	10.68	10.2	12.14	17.48	10.88	17.5	6.8
18	Magnesium H.	Mg/Lts	54	62	34	28	44	30	40	38	44	42	50	72	44.83	72.0	28.0
19	Nitrate	mg/l	5.15	2.33	2.61	2.16	2.16	2.76	2.61	2.02	2.16	2.96	3.88	5.86	3.06	5.9	2.0
20	Nitrite	mg/l	BDL	BDL	0.01	BDL	0.02	BDL	0.04	0.04	BDL	BDL	BDL	BDL	0.03	0.0	0.0
21	pH	pH Units	7.88	7.65	7.97	7.1	7.21	7.6	8.14	7.59	7.78	7.61	8.14	7.88	7.71	8.1	7.1
22	Phosphate	mg/l	BDL	BDL	BDL	0.17	BDL	BDL	0.01	BDL	0.07	BDL	0.12	0.03	0.08	0.2	0.0
23	Potassium	mg/l	1.98	1.01	0.55	0.46	0.61	1.82	0.16	0.25	1.29	1.04	1.13	1.1	0.95	2.0	0.2
24	Sodium	mg/l	6.82	4.19	2.32	4.79	5.77	4.23	5.77	5.64	5.17	3.96	4.93	3.01	4.72	6.8	2.3
25	Sodium (SAR)	SAR	0.26	0.15	0.09	0.19	0.23	0.19	0.23	0.23	0.33	0.16	0.21	0.1	0.20	0.3	0.1
26	streptococci	MPN/100 ml	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.80	1.8	1.8
27	Sulphate	mg/l	6.06	6.26	4.18	2.59	2.59	4.31	3.29	9.06	9.41	8.82	8.28	6.29	5.93	9.4	2.6
28	Suspended Solids	mg/l	34	16	42	32	82	14	24	30	16	28	30	40	32.33	82.0	14.0
29	Temperature	Centigrade	29	32	29	25	25	25	23	16	15	15	22	20	23.00	32.0	15.0
30	Total coliform	MPN/100 ml	130	280	350	350	110	130	110	170	170	240	170	350	213.33	350.0	110.0
31	Dissolved Solids	mg/l	218	210	224	196	168	136	116	96	142	164	176	168	167.83	224.0	96.0
32	Total Hardness	mg/l	132	138	112	112	114	88	114	122	128	104	122	160	120.50	160.0	88.0
33	T.K.N.	mg/l	1.4	2.8	1.4	1.4	1.4	Instrument out of order		1.4	1.4	2.8	1.4	2.8	1.82	2.8	1.4
34	Total Solids	mg/l	252	226	266	228	250	150	140	126	158	192	206	208	200.17	266.0	126.0
35	Turbidity	N.T.U.	3.5	4.5	8.3	21	5.9	2.3	1.4	0.8	8.9	5.3	4.2	7.2	6.11	21.0	0.8
Category - B																	

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2022-23

Natural Water Resource Upper Lake , Bhopal. (Distt.-Bhopal) XGN ID: 22158

Description of Sampling Station : Near Khanugaon,Bhopal.

S.No.	Par	Unit	April(27-04-2022)	May()	June()	July()	Aug()	Sep(22-09-2022)	Oct()	Nov()	Dec()	Jan()	Feb(08-02-2023)	March()	Av.	Maxi.	Mini.
1	Alkalinity	mg/l	86					90					112		96.00	112.0	86.0
2	Ammo. Nitrogen	mg/l	BDL					BDL					BDL				
3	B.O.D	mg/l	1.2					0.5					0.6		0.77	1.2	0.5
4	Boron	mg/l	BDL					BDL					BDL				
5	Calcium	mg/l	26.43					25.63					25.63		25.90	26.4	25.6
6	Calcium Hard.	Mg/Lts	66					64					64		64.67	66.0	64.0
7	C.O. D.	mg/l	9.52					10					9.52		9.68	10.0	9.5
8	Chloride	mg/l	24.63					13.49					17.61		18.58	24.6	13.5
9	Colour	Pt.Co.Scale	30					20					60		36.67	60.0	20.0
10	Conductivity	micromho/c	335.7					174.1					224.6		244.80	335.7	174.1
11	Dissolved Oxygen	mg/l	7.9					5.8					10		7.90	10.0	5.8
12	Fecal Coliform	MPN/100 ml											58		58.00	58.0	58.0
13	FDS	mg/l	NR					Instrumen					Instrument				
14	Fluoride	mg/l	1.56					BDL					BDL		1.56	1.6	1.6
15	Magnesium	mg/l	9.22					7.77					13.11		10.03	13.1	7.8
16	Magnesium Hard	Mg/Lts	38					32					54		41.33	54.0	32.0
17	Nitrate	mg/l	5.16					2.64					3.96		3.92	5.2	2.6
18	Nitrite	mg/l	0.01					0.62					BDL		0.32	0.6	0.0
19	P-Alkanity	--	NIL					NIL					NIL				
20	pH	pH Units	7.8					7.46					8.03		7.76	8.0	7.5
21	Phosphate	mg/l	0.21					0.22					BDL		0.22	0.2	0.2
22	Potassium	mg/l	3.29					0.74					1.39		1.81	3.3	0.7
23	Sodium	mg/l	20.1					8.96					6.12		11.73	20.1	6.1
24	SAR	SAR											0.24		0.24	0.2	0.2
25	streptococci	MPN/100 ml											1.8		1.80	1.8	1.8
26	Sulphate	mg/l	1.21					14.06					8.27		7.85	14.1	1.2
27	Suspended Solids	mg/l	54					22					42		39.33	54.0	22.0
28	Temperature	Centigrade	30					23					17		23.33	30.0	17.0
29	Total coliform	MPN/100 ml											280		280	280.0	280.0
30	T.D. Solids	mg/l	118					118					178		138	178.0	118.0
31	Total Hard	mg/l	104					96					118		106	118.0	96.0
32	T.K.N.	mg/l	2.8					Instrumen					1.4		2.10	2.8	1.4
33	Total Solids	mg/l	172					140					220		177.33	220.0	140.0
34	Turbidity	N.T.U.	29.5					1.2					6.1		12.27	29.5	1.2
	Category - A																

REGIONAL OFFICE,MADHYA PRADESH POLLUTION CONTROL BOARD, BHOPAL.

Water Quality during Year - 2022-23

Natural Water Resource : Upper Lake near Sehore Naka. XGN ID: 22159

Description of Sampling Station : Near Sehore naka, Bhopal.

S.No.	Par	Unit	April(27-04-2022)	May()	June()	July()	Aug()	Sep(22-09-2022)	Oct()	Nov()	Dec()	Jan()	Feb(08-02-2023)	March()	Av.	Maxi.	Mini.
1	Alkalinity	mg/l	114					94					132		113.33	132	94
2	Ammo. Nitrogen	mg/l	BDL					BDL					BDL				
3	B.O.D	mg/l	0.9					0.4					0.9		0.73	0.9	0.4
4	Boron	mg/l	BDL					BDL					BDL				
5	Calcium	mg/l	40.84					32.83					33.64		35.77	40.84	32.83
6	Calcium Hard.	Mg/Lts	102					82					84		89.33	102	82
7	C.O. D.	mg/l	9.52					10					19.04		12.85	19.04	9.52
8	Chloride	mg/l	44.34					15.42					21.52		27.09	44.34	15.42
9	Colour	Pt.Co.Scale	30					10					50		30.00	50	10
10	Conductivity	micromho/c	470.1					193.5					287.2		316.93	470.1	193.5
11	Dissolved Oxygen	mg/l	7.5					4.6					9		7.03	9	4.6
12	Fecal Coliform	MPN/100 ml											34		34.00	34	34
13	Fixed Dissolved	mg/l	NOT					Instrumen					Instrument				
14	Fluoride	mg/l	1.47					BDL					BDL		1.47	1.47	1.47
15	Magnesium	mg/l	11.65					4.85					12.62		9.71	12.62	4.85
16	Magnesium Hard.	Mg/Lts	48					20					52		40.00	52	20
17	Nitrate	mg/l	7.15					3.46					5.93		5.51	7.15	3.46
18	Nitrite	mg/l	0.23					0.58					BDL		0.41	0.58	0.23
19	pH	pH Units	7.86					7.77					7.37		7.67	7.86	7.37
20	Phosphate	mg/l	0.98					0.46					BDL		0.72	0.98	0.46
21	Potassium	mg/l	5.07					0.87					1.93		2.62	5.07	0.87
22	Sodium	mg/l	32.96					6.71					6.93		15.53	32.96	6.71
23	SAR	SAR											0.25		0.25	0.25	0.25
24	streptococci	MPN/100 ml											1.8		1.80	1.8	1.8
25	Sulphate	mg/l	3.28					11.38					8.12		7.59	11.38	3.28
26	Suspended Solids	mg/l	78					24					28		43.33	78	24
27	Temperature	Centigrade	31					22					17		23.33	31	17
28	Total coliform	MPN/100 ml											140		140.00	140	140
29	T. Dissolved	mg/l	204					136					208		182.67	208	136
30	Total Hard.	mg/l	150					102					136		129.33	150	102
31	T.K. N.	mg/l	2.8					Instrumen					1.4		2.10	2.8	1.4
32	Total Solids	mg/l	282					160					236		226.00	282	160
33	Turbidity	N.T.U.	40.2					1.4					3.6		15.07	40.2	1.4
	Cat. - A																